

European Utilities & Clean Energy

Regulated Networks Primer series: Part 3: Growth and Capex Outlooks in European Power networks



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Regulated utilities are one of the most dynamic subsectors within European Utilities. Given investor interest, we have launched the Regulated Networks Primer Series - in [Part 1](#) we analysed the attractiveness of regulation across jurisdictions, in [Part 2](#) we focused on valuation, and in this Part 3 we look at the underlying drivers of the capex growth for electricity networks in more detail.

European regulated power grids are currently riding 5 key megatrends: we identify **1)** growing power demand, **2)** decarbonisation of power production, **3)** a push towards energy sovereignty and power markets integrity, **4)** networks digitalisation and cybersecurity and **5)** infrastructure resilience and environmental standards as key megatrends for European regulated power grids. These megatrends each provide strong tailwinds for investment and asset growth, with high visibility.

This is driving a structural improvement in earnings quality as sector investment increasingly flows into regulated assets: We see significant value in utility earning streams under supportive regulatory frameworks, with particular appeal in the valuable macroeconomic protections offered and the high level of revenue visibility. As such, we look favorably on the ongoing earnings mix shift in the sector, with companies such as SSE and Iberdrola seeing regulated earnings grow to 60% plus of the total by 2030.

Whilst the megatrends driving grid investment are well understood, the potential supercycle duration is not: we do not believe that the market currently appreciates the duration of an investment runway which we see extending for at least a decade as European energy grids modernize, digitize and decarbonize. In our view, these drivers underpin an investment runway until at least the mid-2030s, even assuming only limited uptake of low carbon technologies such as electric vehicles and heat pumps, and modest data center capacity growth.

There is potential for a further supercycle extension if power demand develops as anticipated: The key uncertainty, in our view, surrounds the potential for growing power demand to take over the baton as the current electrification-and-decarbonisation-driven supercycle reaches maturity from the mid-2030s onwards. In the event that key drivers such as EV and heat pump uptake, data centers and even green hydrogen play out as anticipated, power demand will grow significantly during the 2030s, adding another leg to the investment runway as new generation capacity is added to serve the additional demand, requiring accompanying network investment.

The debate matters for valuations given the importance of RAB growth in driving the terminal RAB premium: The single most important parameter in our regulated utility valuation models is the terminal premium to RAB assumption, which in turn primarily reflects **1)** the level of outperformance expected across both financial and operational metrics, and **2)** the growth outlook. Our analysis in this note is highly supportive for the second element, with potential further upside in a favorable power demand scenario.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	29 Sep 2026	Price Target	TTM	Adjusted EPS				Adjusted P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
ELI.BB (Elia)	O	EUR	118.00	145.00	12.5%	EUR	5.51	6.91	7.84	21.4	17.1	15.1
NG/.LN (National Grid)	O	GBp	1,123.00	1,310.00	(2.7)%	GBP	0.78	0.89	1.02	14.5	12.6	11.0
EOAN.GR (E.ON)	CS	EUR	16.91	0.000	(4.4)%	EUR	1.16			14.6		
IBE.SM (Iberdrola)	M	EUR	20.49	20.00	18.0%	EUR	0.95	1.02	1.12	13.4	12.5	11.7
ELE.SM (Endesa)	M	EUR	41.99	41.00	44.4%	EUR	2.22	2.46	2.60	18.9	17.1	16.1
ENGI.FP (Engie)	O	EUR	23.02	27.70	16.3%	EUR	1.96	2.09	2.14	11.8	11.0	10.7
TRN.IM (Terna)	O	EUR	9.19	10.30	(3.3)%	EUR	0.52	0.51	0.54	17.7	18.0	17.0
RED.SQ (Redeia)	O	EUR	14.85	18.60	(20.2)%	EUR	0.90	0.92	0.93	16.4	16.2	15.9
RENE.PL (Redes Energéticas)	O	EUR	3.56	4.35	6.2%	EUR	0.18	0.22	0.21	19.7	16.3	16.7
EDME			1,554.34									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

IBE.SM valuation is EV/Adj EBITDA (x); NG/.LN base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

VALUATION COMPS TABLE

EXHIBIT 1: Bernstein European Utilities comparative valuation sheet

	BERN Rating	Curr.	Close Price	Market Cap (m)	P/E			EV/EBITDA			ND/EBITDA			DY (%)		
					2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E
Generation																
Fortum	-	EUR	23.8	21,328	23.4x	22.1x	20.6x	15.3x	14.5x	13.7x	1.3x	1.1x	1.0x	3.8%	4.0%	4.2%
Verbund	U	EUR	64.4	22,374	22.8x	16.8x	19.8x	12.7x	10.1x	8.9x	2.0x	1.9x	0.0x	2.2%	3.0%	2.5%
Average					23.1x	19.5x	20.2x	14.0x	12.3x	11.3x	1.6x	1.5x	0.5x	3.0%	3.5%	3.4%
Integrated																
Iberdrola	M	EUR	20.5	136,733	20.0x	18.3x	16.8x	12.1x	11.7x	11.5x	4.1x	4.2x	4.3x	3.6%	3.8%	4.0%
Enel	M	EUR	8.8	89,253	12.5x	11.4x	11.1x	7.7x	7.6x	7.5x	3.3x	3.5x	3.5x	5.9%	6.2%	6.4%
Engie	O	EUR	23.0	58,527	11.0x	10.7x	10.9x	8.0x	7.3x	7.5x	3.7x	3.3x	3.5x	6.4%	6.6%	6.8%
Endesa	M	EUR	42.0	43,743	17.1x	16.1x	15.3x	9.1x	9.4x	9.1x	1.9x	2.3x	2.3x	4.4%	4.6%	4.8%
Naturgy	M	EUR	29.0	28,119	13.2x	13.5x	14.2x	8.2x	8.2x	8.3x	3.0x	3.0x	2.9x	6.4%	6.6%	5.6%
SSE	O	GBP	2,437	29,546	13.4x	11.3x	10.2x	11.7x	10.6x	9.8x	3.0x	3.3x	3.4x	3.0%	3.2%	3.5%
EDP	O	EUR	4.8	19,991	14.4x	15.0x	14.1x	8.3x	8.5x	8.2x	2.9x	3.0x	2.9x	4.4%	4.5%	4.7%
Centrica	M	GBP	147	6,772	11.6x	10.2x	9.1x	6.5x	6.6x	5.7x	-0.8x	-0.6x	-0.4x	4.1%	4.5%	5.5%
Average					14.1x	13.3x	12.7x	9.0x	8.8x	8.4x	2.7x	2.7x	2.8x	4.8%	5.0%	5.2%
Regulated																
National Grid	O	GBP	1,123	56,451	12.6x	11.0x	9.8x	11.7x	10.9x	10.4x	5.6x	5.5x	5.5x	4.4%	4.5%	4.6%
E.ON	-	EUR	16.9	44,665	15.7x	13.8x	12.9x	10.2x	9.4x	9.1x	4.8x	4.7x	4.7x	3.4%	3.6%	3.7%
Terna	O	EUR	9.2	18,480	18.0x	17.0x	15.9x	11.9x	11.7x	11.8x	5.6x	5.9x	6.3x	4.3%	4.3%	4.5%
Snam	M	EUR	5.6	18,861	13.1x	13.1x	12.7x	12.5x	12.3x	12.4x	6.4x	6.4x	6.8x	5.6%	5.8%	6.1%
Elia	O	EUR	118.0	12,871	17.1x	15.1x	12.6x	15.1x	14.7x	14.0x	8.9x	9.6x	9.7x	1.8%	1.8%	1.8%
Redeia	O	EUR	14.9	8,035	16.2x	15.9x	14.9x	10.8x	10.9x	10.7x	4.6x	4.7x	5.0x	5.5%	5.6%	5.7%
Severn Trent	O	GBP	3,068	9,267	15.2x	11.9x	10.8x	13.3x	12.1x	11.7x	7.4x	7.1x	7.1x	4.3%	4.3%	4.4%
United Utilities	O	GBP	1,407	10,452	13.5x	12.4x	12.7x	12.6x	12.1x	12.4x	6.2x	6.4x	7.0x	3.9%	4.1%	4.2%
Italgas	M	EUR	8.4	8,533	11.4x	10.4x	9.6x	9.2x	8.7x	8.5x	5.0x	4.8x	4.8x	5.7%	6.3%	6.8%
Enagas	O	EUR	16.7	4,365	18.1x	14.6x	14.8x	11.3x	9.4x	9.2x	4.6x	3.7x	3.4x	6.0%	6.0%	6.0%
Pennon	M	GBP	442	2,086	16.9x	14.2x	13.0x	12.6x	12.4x	12.2x	8.8x	8.9x	9.0x	6.9%	7.0%	7.1%
REN	O	EUR	3.6	2,375	16.3x	16.7x	15.7x	8.4x	8.3x	8.1x	4.5x	4.6x	4.5x	4.6%	4.8%	4.9%
Average					15.3x	13.8x	13.0x	11.6x	11.1x	10.9x	6.0x	6.0x	6.1x	4.7%	4.8%	5.0%
Renewable																
RWE	-	EUR	59.1	46,128	19.5x	17.6x	16.4x	10.8x	9.7x	9.4x	2.2x	2.2x	2.5x	2.2%	2.4%	2.7%
Orsted	-	DKK	136.1	179,815	17.0x	13.7x	13.0x	7.7x	7.4x	5.9x	0.9x	1.0x	0.3x	2.0%	2.4%	2.5%
EDPR	O	EUR	12.8	13,532	26.2x	25.3x	22.7x	10.0x	9.7x	9.3x	4.0x	3.5x	3.3x	1.5%	1.6%	1.8%
Acciona Energia	M	EUR	22.0	7,158	50.2x	31.8x	33.4x	8.4x	9.7x	9.6x	2.8x	3.2x	3.2x	1.2%	1.1%	1.1%
Grenergy	O	EUR	74.9	2,141	19.7x	17.5x	10.0x	16.2x	14.2x	11.8x	6.6x	7.2x	7.4x	0.0%	0.0%	0.0%
Solaria	O	EUR	16.1	2,135	11.2x	12.3x	14.2x	9.9x	9.9x	10.0x	4.7x	5.4x	5.8x	0.0%	0.0%	0.0%
Average					24.0x	19.7x	18.3x	10.5x	10.1x	9.3x	3.5x	3.8x	3.7x	1.2%	1.2%	1.3%
Environment																
Veolia	O	EUR	30.9	22,965	12.8x	11.6x	10.8x	7.8x	7.3x	7.1x	3.8x	3.6x	3.5x	5.2%	5.7%	6.1%
Utilities Sector Average					16.7x	14.7x	13.9x	10.3x	9.9x	9.5x	4.1x	4.1x	4.1x	3.8%	3.9%	4.1%
Other																
Mettlen	M	EUR	49.1	6,992	11.7x	8.9x	6.8x	9.4x	7.5x	5.8x	3.1x	2.4x	1.6x	3.0%	3.9%	5.2%

Bloomberg pricing data as of market close 29 September 2026

Source: Company data, Bloomberg, Bernstein analysis and estimates

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DETAILS

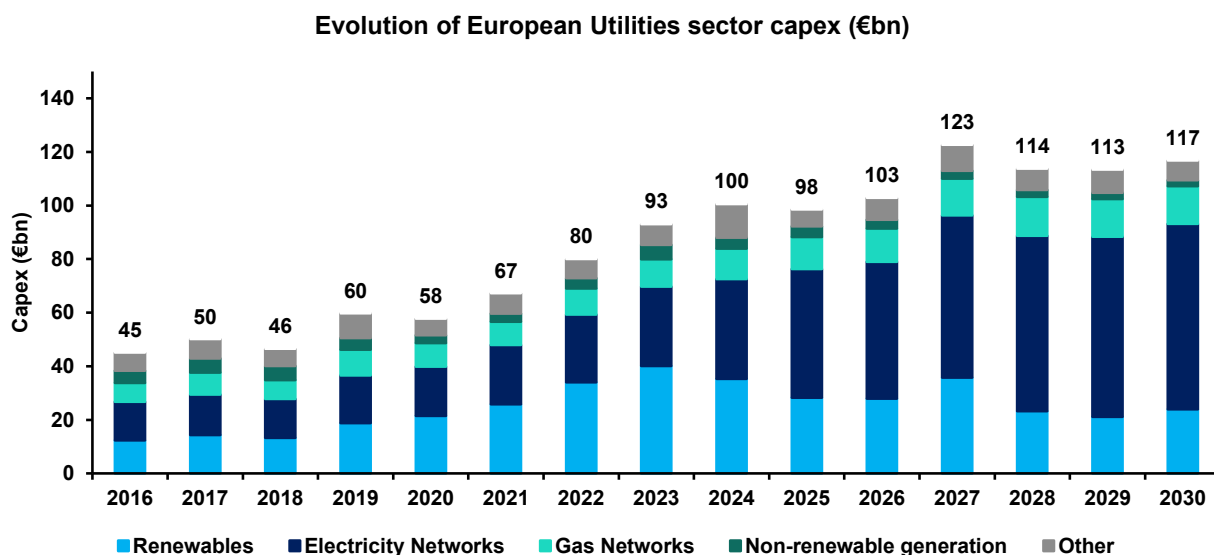
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SIGNIFICANT RESEARCH CONCLUSIONS

The most important conclusion from our work in this note is that the current investment supercycle in European electricity grids is underpinned by very high quality growth drivers with good visibility into the mid-2030s, longer than the market currently appreciates, with potential further upside beyond this point in a favorable power demand scenario.

EUROPEAN UTILITIES ARE INCREASING THEIR INVESTMENT IN REGULATED ELECTRICITY NETWORKS AS THEY RIDE THE GRID SUPERCYCLE

European regulated power grids are currently riding 5 key megatrends, including **1)** growing power demand; **2)** decarbonisation of power production; **3)** a push towards energy sovereignty and power markets integrity; **4)** networks digitalisation and cybersecurity; and **5)** infrastructure resilience and environmental standards. These megatrends each provide strong tailwinds for investment and asset growth, with high visibility. As set out in Exhibit 2, the net impact of these drivers is a significant increase in electricity network capex within our coverage universe.

EXHIBIT 2: **Evolution of European Utility capex, 2016-2030**

Shows illustrative cohort

Source: Company data, Bernstein analysis and estimates

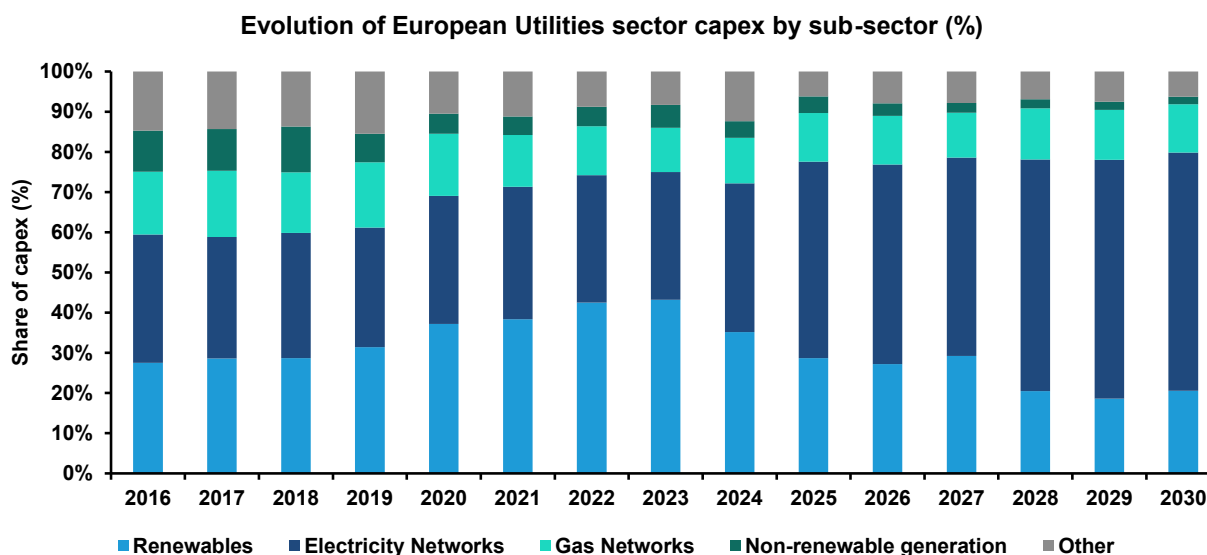
At the company level, the significant investment runway is reflected in individual RAB growth expectations, e.g. SSE at a ~30% CAGR YE25A-YE30E, RWE's Amprion at >18% CAGR from YE25A-YE35E, and Elia at >20% from YE24A-YE28E. We expect continuing strong RAB and earnings growth for each of these companies beyond the guidance period.

...THIS IS REBALANCING THEIR TOTAL INVESTMENTS INTO REGULATED ACTIVITIES AND IMPROVING EARNINGS VISIBILITY

One important implication of this ongoing investment wave in electricity networks is a material improvement in earnings quality for some utility companies. We see significant value in regulated earnings streams under supportive regulatory frameworks, with particular appeal in the valuable macroeconomic protections offered and the high level of revenue visibility, and as such we look favorably on the ongoing earnings mix shift in the sector, with companies such as SSE and Iberdrola seeing regulated earnings grow to >60% of the total by 2030.

On an aggregate basis, as we set out in Exhibit 3, we see electricity network capex rising from ~40% of our coverage universe total in 2024 to ~60% by 2030.

EXHIBIT 3: Evolution of European Utility shares of capex, 2016-2030

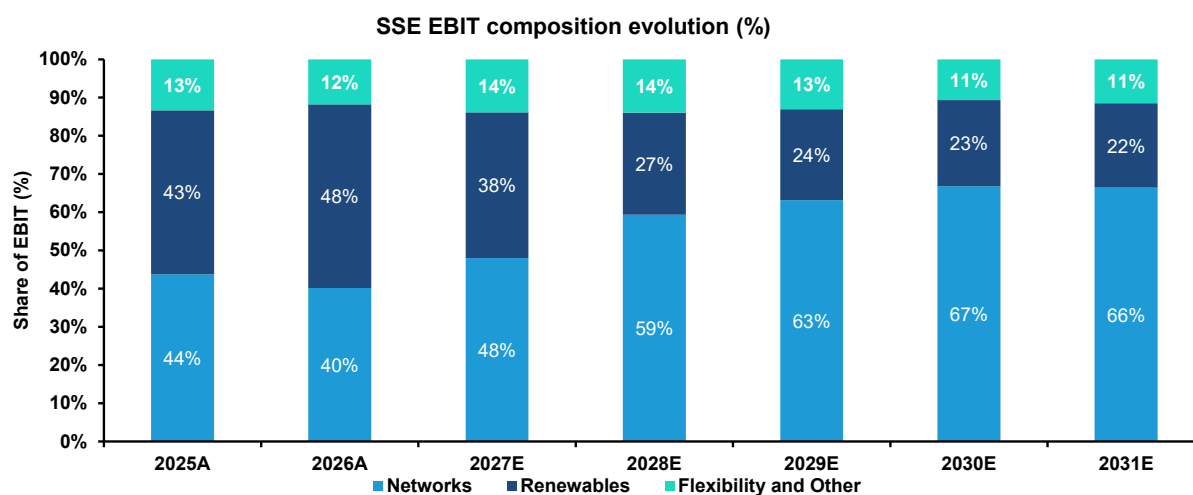


Shows illustrative cohort

Source: Company data, Bernstein analysis and estimates

We forecast that UK integrated utility SSE will increase its share of regulated network earnings from ~40% in FY26A to ~66% in FY31E, as set out in Exhibit 4. The increasing earnings quality shown by SSE is a key part of our thesis for the company and other players in the sector, such as Iberdrola, where the EBITDA from regulated networks will account for 60% of the total vs 38% in 2024.

EXHIBIT 4: We forecast that SSE will grow the share of regulated networks earnings in its mix from ~40% in FY26A to >65% in FY31E



Source: SSE, Bernstein analysis and estimates

THE INVESTMENT RUNWAY IN ELECTRICITY GRIDS IS LONGER THAN THE MARKET CURRENTLY APPRECIATES

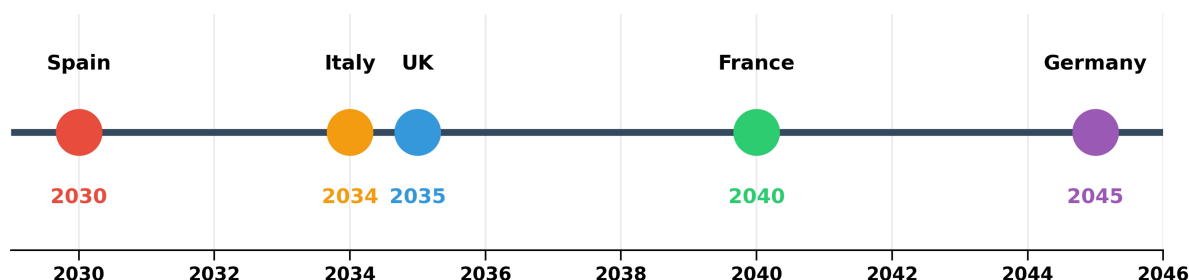
We do not believe that the market currently appreciates the duration of an investment runway which we see extending for at least a decade as European energy grids modernize, digitize and decarbonize. In our view, these drivers underpin an investment runway until at least the mid-2030s, even assuming only limited uptake of low carbon technologies such as electric vehicles and

heat pumps, and modest data center capacity growth.

We set out specific details of the grid planning horizons for which specific investment forecasts are available in Exhibit 5. We apply a rigorous qualification threshold, requiring official recognition of an investment need, for example via a national grid planning document, in order for visibility to be recognised. Germany emerges as the market with the greatest formal visibility, with planning scenarios stretching into the 2040s.

EXHIBIT 5: Major European electricity markets have good grid investment visibility well into the 2030s and beyond

Electricity grid investment visibility horizons in major European electricity markets

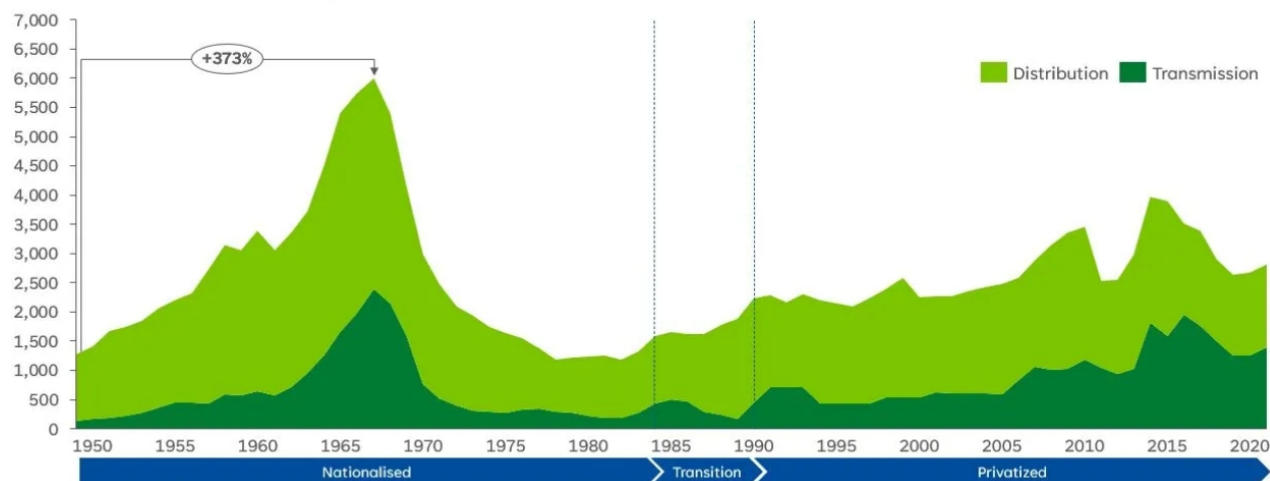


Source: Individual country national grid planning documentation, Bernstein analysis

THE TRAJECTORY OF POWER DEMAND WILL DETERMINE WHEN THE CURRENT SUPERCYCLE ENDS

The key uncertainty, in our view, surrounds the potential for growing power demand to take over the baton as the current decarbonisation-dominated supercycle reaches maturity from the mid-2030s onwards. In the event that key drivers such as EV and heat pump uptake, data centers and even green hydrogen play out as anticipated, power demand will grow significantly during the 2030s, adding a further leg to the investment runway as new generation capacity is added to serve the additional demand, requiring accompanying network investment. However, as is clear from Exhibit 6, we concede that historical precedent does point to a significant growth slowdown once the current investment wave does mature.

EXHIBIT 6: Evolution of annual electricity network capex in the UK, 1950 to 2020



Source: Arthur Downing via Substack

THE DEBATE MATTERS FOR VALUATIONS GIVEN THE IMPORTANCE OF RAB GROWTH IN DRIVING THE TERMINAL RAB PREMIUM

The single most important parameter in our regulated utility valuation models is the terminal premium to RAB assumption, which in turn primarily reflects **1)** the level of outperformance expected across both financial and operational metrics, and **2)** the growth outlook. Our analysis in this note is highly supportive for the second element, with potential further upside in a favorable

power demand scenario.

As we set out in Exhibit 7, for a given level of returns spread, the RAB growth outlook is a very significant driver of valuation.

EXHIBIT 7: **Theoretical RAB premium under a range of growth and return (relative to cost of capital) scenarios**

		Return/WACC-1 (WACC = 5%)						
		0%	5%	10%	15%	20%	25%	30%
10 year RAB CAGR *	0%	0%	5%	10%	15%	20%	25%	30%
	5%	0%	9%	17%	24%	32%	39%	47%
	10%	0%	15%	26%	38%	50%	61%	73%
	15%	0%	24%	41%	59%	77%	94%	112%

*Assumes 0% RAB growth after 10 years

Source: Bernstein analysis and estimates

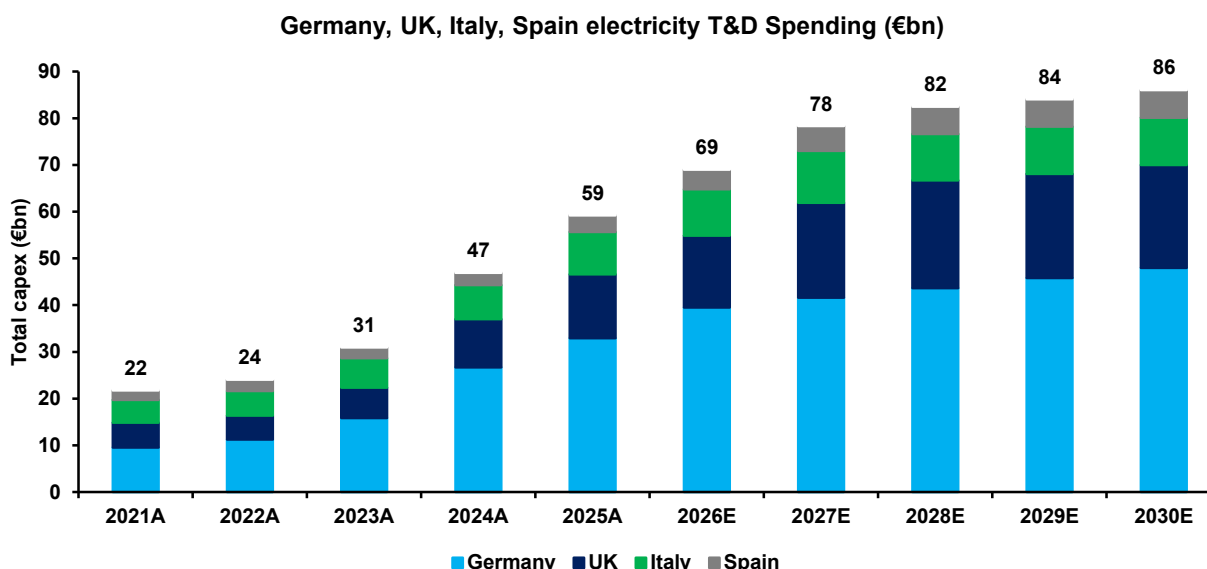
EUROPEAN REGULATED POWER GRIDS ARE CURRENTLY RIDING 5 KEY MEGATRENDS

European electricity grids are currently in a period of very significant growth. The European Commission estimates that [~€1.2 trillion in electricity grid investment is required by 2040](#) across the EU alone. We identify the following as key megatrends driving growth in European regulated power grids:

- 1) Growing power demand:** Following years of stagnant power demand, optimism is returning as growth drivers such as electrification and the data center capacity roll-out begin to mature. In the UK, transmission network operator National Grid expects >30% demand growth to 2035.
- 2) Decarbonisation of power production:** Many European countries have set ambitious renewable generation targets for 2030, including Spain, the UK and Germany at 80% or more of total generation. Integrating the resulting new renewable generation capacity into the broader system will require significant grid investment.
- 3) A push towards energy sovereignty and power markets integrity:** Europe is heavily reliant on energy imports, which in 2024 accounted for over 55% of the EU's energy needs. Electrification is a key lever for ameliorating this major strategic weakness, and grids are a key enabler.
- 4) Networks digitalisation and cybersecurity:** Modern electricity grids are increasingly complex, with intermittent renewable capacity and more volatile demand patterns putting increasing strain on grid operators. Digitalisation is a key enabler of improved performance and increased cybersecurity spending is a clear corollary of greater digitalisation.
- 5) Infrastructure resilience and environmental standards:** Alongside new capacity growth, European grids have significant asset maintenance requirements. Given that around 40% of European electricity distribution grids are over 40 years old, and that electrification is making reliable power supply increasingly vital, maintenance capex is a major spending driver by itself.

These megatrends each provide strong tailwinds for investment and asset growth, with high visibility.

We set out an overview of electricity network capex in selected major European countries in Exhibit 8. Capex on electricity networks began to ramp up in 2024 and by 2027 is expected to be more than triple the levels seen in 2022. As we set out in more detail later in this note, we expect the megacycle to last significantly beyond 2030.

EXHIBIT 8: **Evolution of electricity network capex in selected major European countries, 2021A-2030E**

Source: Regulatory reports, TSO and DSO annual reports, Bernstein analysis and estimates

In connection with the ongoing supercycle, we would note the regulatory frameworks underpinning the investment across Europe, as covered in [Regulated Networks Primer series: Part 1: How do regulated return frameworks compare across jurisdictions?](#) and as set out in Exhibit 9.

EXHIBIT 9: **Bernstein European utilities regulatory framework ranking**

Jurisdiction	Headline Returns on new investments	Inflation Protection	Returns Protection	Treatment of WIP	Operational incentive opportunities	Overall attractiveness	Rank
Italy	5.0	5.0	4.0	3.0	4.5	4.3	1
UK	3.5	4.0	4.5	5.0	4.0	4.2	2
Belgium	2.0	3.0	5.0	4.0	5.0	3.8	3
Portugal	3.5	3.0	4.5	3.0	3.0	3.4	4
Germany	2.0	3.0	3.0	4.0	2.0	2.8	5
Sweden	4.0	3.0	1.0	1.0	2.0	2.2	6
Spain	3.0	1.0	1.0	2.0	3.5	2.1	7
France	2.0	3.0	1.0	2.0	2.0	2.0	8
Average	3.1	3.1	3.0	3.0	3.3	3.1	

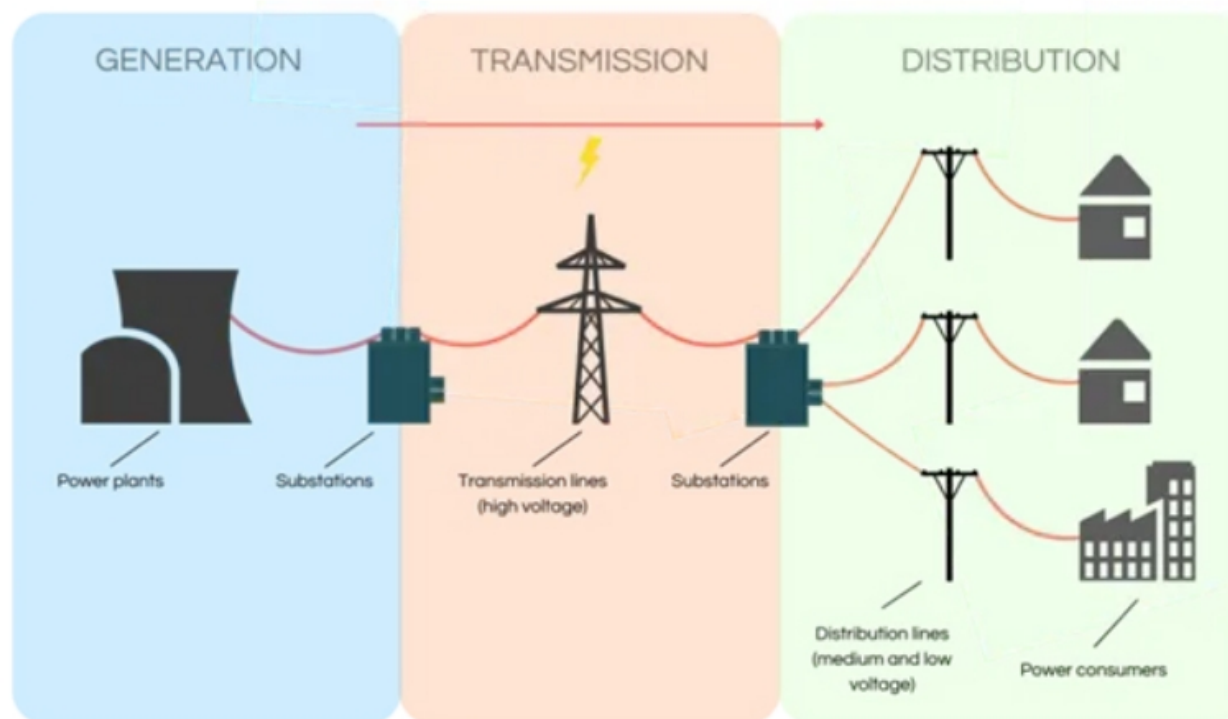
Source: Regulatory data, Bernstein analysis

THESE TRENDS APPLY ACROSS TRANSMISSION AND DISTRIBUTION

The drivers we identify apply to both electricity transmission and electricity distribution networks; whilst electricity distribution networks are less prominent, they are nonetheless vital enablers of renewable penetration and decarbonization. On the supply side, distribution-connected generation is a vital complement to the larger assets which connect to the transmission network; in the UK, distribution-connected generation currently accounts for ~23% of total capacity, whilst industry experts in Italy [estimate](#) that 70% of the additional renewable generation capacity to be added to the system between 2024 and 2030 will be added to the distribution system. On the demand side, distribution networks facilitate uptake of vital decarbonisation technologies such as electric vehicles and heat pumps. As such, as the decarbonization megatrend continues to play out, electricity distribution network operators are well-placed to benefit.

Alongside the discrete supply and demand angles, we would additionally highlight the role of the electricity distribution grid in facilitating “prosumers”, e.g. via two-way flows of power from houses with domestic solar panels and battery assets. These new flows of power will place new demands on the distribution network, further bolstering the investment runway for these assets.

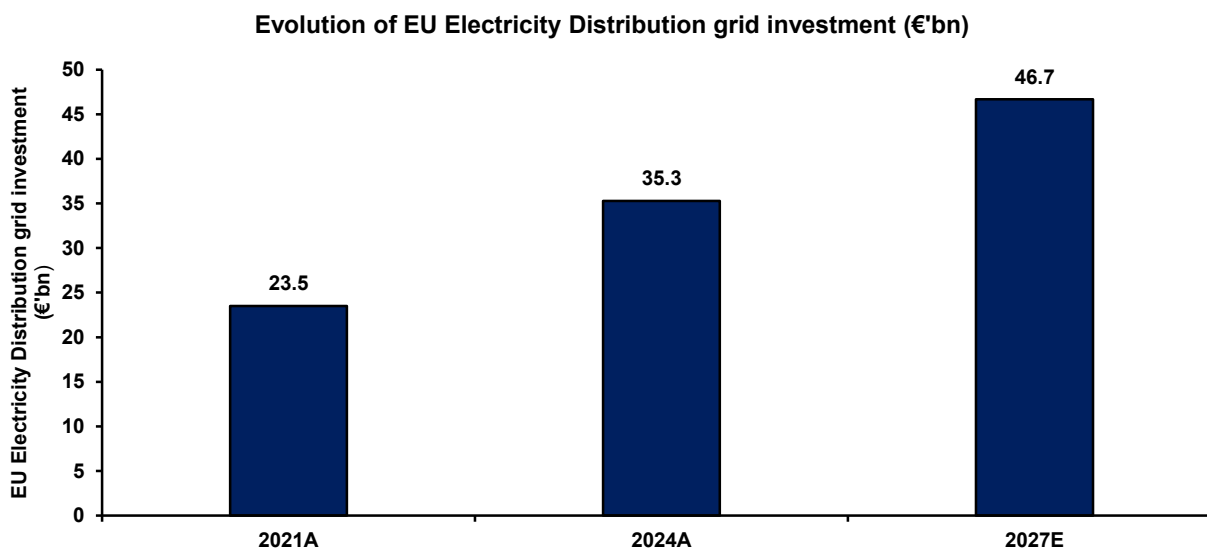
EXHIBIT 10: **The distribution network is a vital component of a decarbonised electricity system**



Source: Eurelectric

The importance of electricity distribution grids to the renewable build-out in Europe was illustrated by a [June 2026 study](#) by Afry on behalf of the German non-profit [Beyond Fossil Fuels](#). The study found that “across the 8 European markets analyzed (Bulgaria, Czechia, Great Britain, Germany, Greece, Italy, Poland and Spain), distribution-level queues have reached volumes that can directly influence system costs and market outcomes, making grid access a binding constraint on decarbonisation pathways”. As such, it is clear that distribution networks can be a critical bottleneck to the energy transition and that they are therefore a key focus for investment alongside the more prominent transmission networks.

EXHIBIT 11: Annual evolution of EU electricity distribution grid investment spending

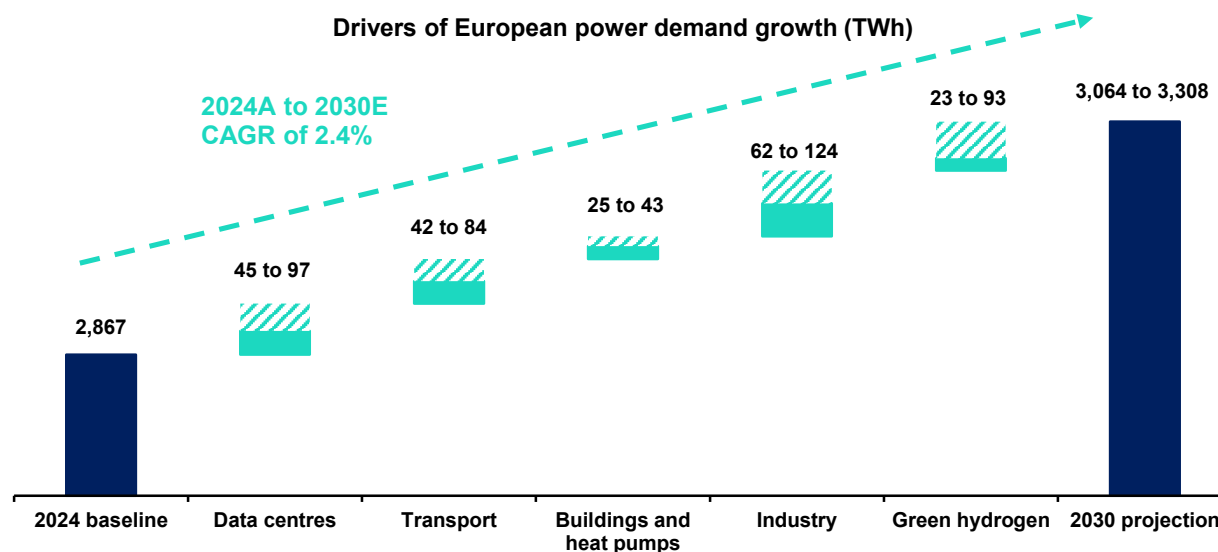


Data presented is for the EU27 less Bulgaria and Denmark plus Norway. Includes data from 191 DSOs with over 100,000 customers
 Source: European Union Agency for the Cooperation of Energy Regulators, Bernstein analysis

1) GROWING POWER DEMAND

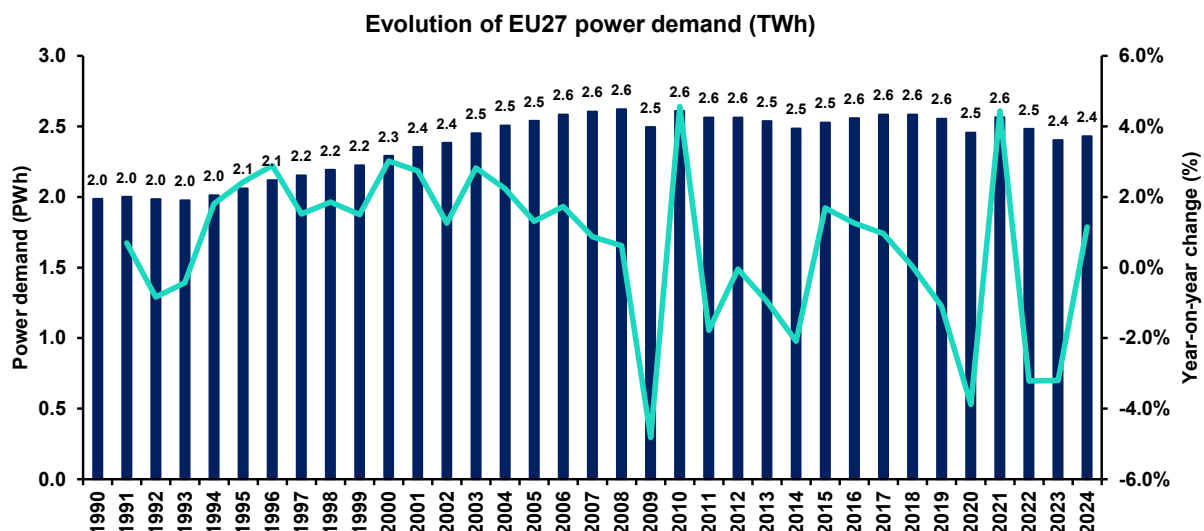
One of the most important investment drivers for European regulated electricity networks is a very strong outlook for power demand growth. In the UK, for example, electricity transmission system operator National Grid expects [>30% electricity demand growth](#) to 2035. We set out in Exhibit 12 an illustrative decomposition of potential demand drivers, using ranges derived from our own and 3rd-party forecasts.

EXHIBIT 12: We see a return to power demand growth in Europe following years of stagnation



Source: McKinsey, BNEF, IEA, Hydrogen Europe, Eurelectric, European Heat Pump Association, Bernstein analysis and estimates

As we set out in Exhibit 13, this outlook for demand growth in Europe is a strong contrast to the recent history; power demand in Europe has been stagnant since the early 2000s.

EXHIBIT 13: **European power demand has been stagnant since the early 2000s**

Source: Eurostat, Ember, Bernstein analysis

One primary theme in growing power demand is **electrification**, which refers to the process of substituting a non-electricity energy source with electricity, for example at the domestic level by replacing a gas boiler with a heat pump, or an internal combustion engine vehicle with an electric vehicle. Electrification stands to be one of the most significant drivers of growth for European electricity networks given the naturally accompanying increase in electricity demand. The European Commission on 17 July 2026 unveiled Europe's [Electrification Action Plan](#), setting an indicative target to increase the share of electricity in final energy consumption to 46% by 2040. The commission estimates that reaching the target could support the deployment of approximately 120 million battery electric vehicles (BEVs), compared with around 8 million today, and 100 million heat pumps, versus roughly 30 million currently installed. Additionally, achieving the 2040 target could reduce the EU's fossil fuel import bill by up to €260bn annually, with BEVs already contributing to an estimated €4.1bn in avoided oil imports in 2025.

This expected increase in electricity demand would require significant growth in both generation capacity and the accompanying grid infrastructure to accommodate that capacity within the electricity system. For example, in France, the transmission grid operator has [highlighted](#) that "while its current architecture is a considerable asset, the French grid is only well suited to a France where electricity supplies just over a quarter of the country's energy needs. Raising the share of electricity in the country's energy mix will make investment in the public transmission grid mandatory".

Increasing electrification of the European economy is also a strategic priority for European governments seeking greater energy independence and lower carbon emissions. In May 2022, in response to the energy shock that followed Russia's invasion of Ukraine, the European Commission implemented its [REPowerEU](#) plan, seeking to rapidly reduce the EU's dependence on fossil fuel imports from Russia.

Progress on electrification to date has, however, proven to be somewhat muted. As of 2024, electricity accounted for ~23% of Europe's total energy consumption, broadly unchanged vs the preceding 5 years. Under the EU's REPowerEU plans, this is targeted at 35% by 2030 and 61% by 2050, implying significant acceleration in coming years.

In the following sections we examine electrification and other drivers of growing power demand in more detail.

a) Electrification of transport

The electrification of transport refers to the replacement of internal combustion engine vehicles with electric vehicles, primarily battery electric vehicles (BEVs) but also including hybrid vehicles. Alongside other benefits, electrification of transport is a key decarbonisation lever; transport typically accounts for a material share of the carbon dioxide emissions of European countries - the IEA [reports](#) that ~45% of Spain's CO₂ emissions in 2023, for example, related to transport.

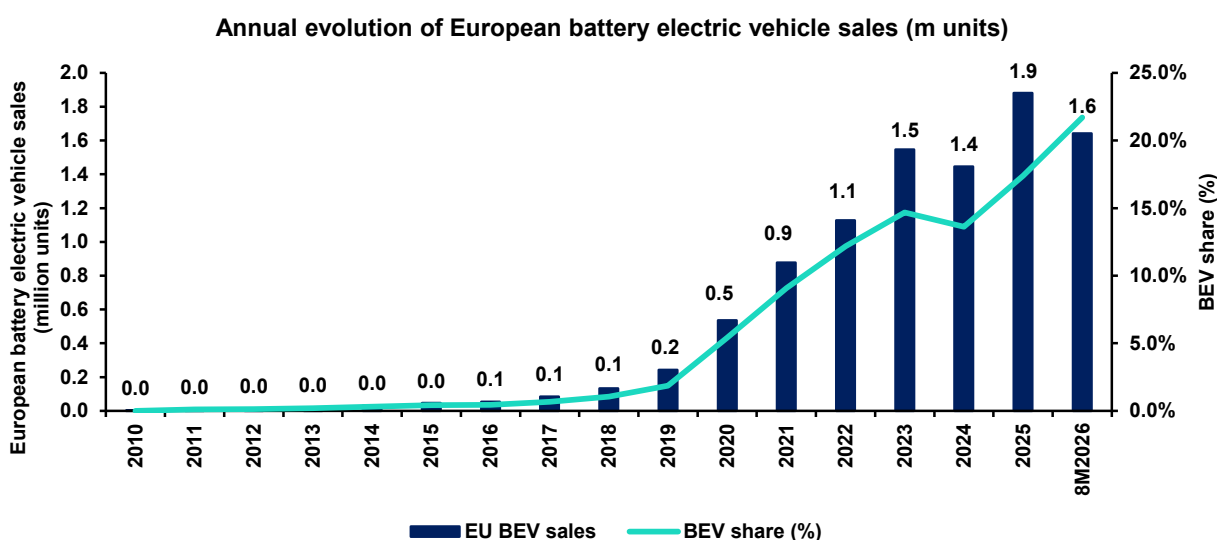
Electrification of transport: Power demand impact

We estimate that growing uptake of electric vehicles could have a modestly positive impact on overall European power demand. The average annual mileage of a passenger car in France is [11,600km](#), whilst the average EV fuel economy is [0.179KWh per kilometer](#), implying that a typical household using an electric vehicle will use an additional ~2,100 KWh per year. Given that the average annual electricity consumption of a French household is [~4,000kWh](#), this would imply that a French household's EV adoption raises its annual electricity demand by close to 50%. The key near-term sensitivity therefore relates to the pace of uptake, where every million electric vehicles bought increases annual power demand by ~2TWh, or ~0.1% of total European power demand in 2025 ([~2,775 TWh](#)). Total battery electric vehicle sales in the EU in 2025 were 1.9m units, implying a demand uplift of ~0.2% from new sales. A naive extrapolation of sales from the first 8 months of 2026 would imply total full year sales of ~2.5m units. As such, it is clear that we will need to see a significant acceleration in sales in order to materially impact total power consumption in the near term.

Looking further out, the EU's [46% electrification target by 2040](#), as set out in its [Electrification Action Plan](#), is consistent with **the uptake of over 100m additional EVs from the current car parc of ~8 million, representing a demand boost of ~10% from current levels**. The Commission notes that BEVs already generated an estimated €4.1bn in avoided oil imports in 2025.

We set out the annual evolution of EU battery electric vehicle sales and market shares in Exhibit 14 below.

EXHIBIT 14: Battery electric vehicles are taking significant market share in the EU, but will need further increases to make a material impact on power demand

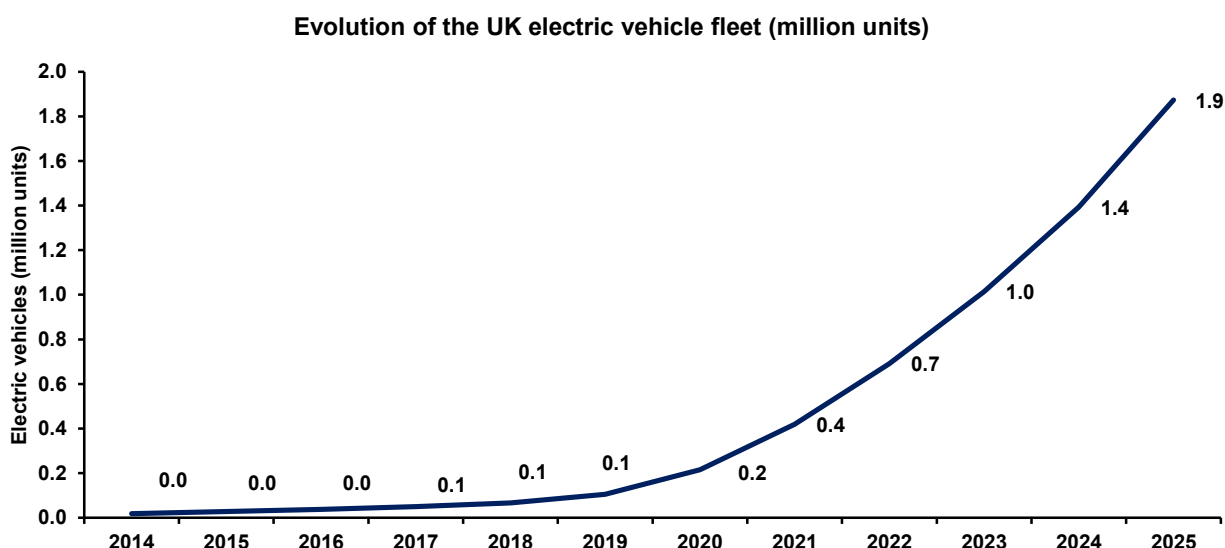


Source: ACEA, Bernstein analysis

Electrification of transport: Network load impact

Alongside electrification of industrial and commercial transport, **greater consumer adoption of electric vehicles has the potential to very significantly increase electricity network loads in European countries, particularly in the distribution network** (see [Electric Revolution 2026: Resilience - EV charging infrastructure and the durability of the e-mobility transition](#)). Industry data points to a [typical peak load level of around 2kW](#), so a typical domestic EV charger, which has a capacity of ~7kW in the UK, would therefore represent a significant increase in both network load and power consumption.

As set out in Exhibit 15, the UK EV fleet saw significant growth between 2014 and 2025, reaching ~1.9m units, or c.5% of the combined petrol and diesel total. Continued growth in electric vehicle adoption in European countries such as the UK is expected to be an important driver of distribution network investment.

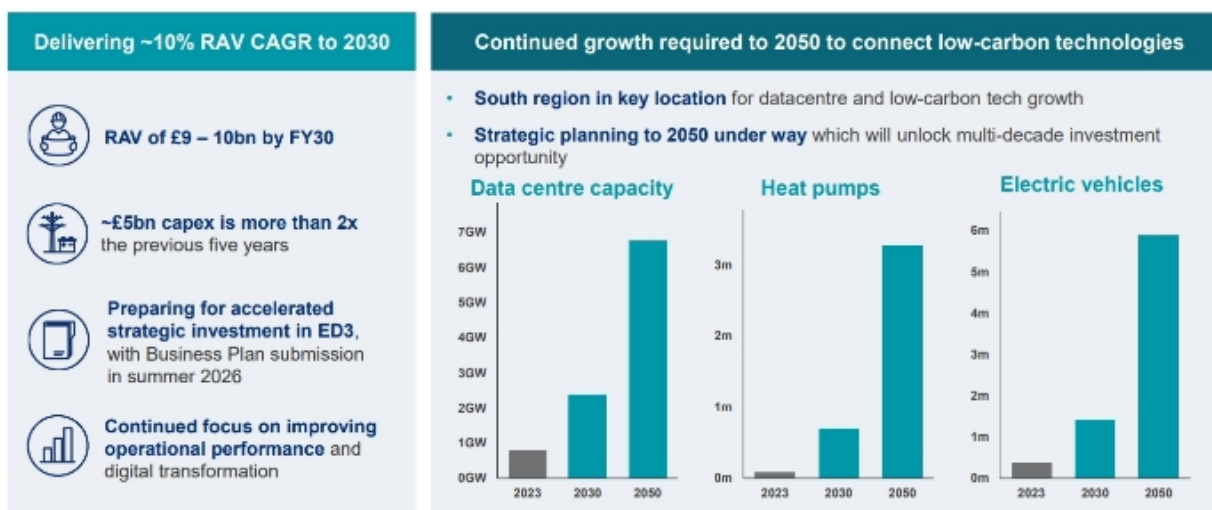
EXHIBIT 15: **Annual evolution of the UK electric vehicle fleet**

Source: UK Government Statistics, Bernstein analysis

As set out in Exhibit 16, SSE identify the electrification of domestic heating (via heat pumps) and of transport as key drivers of a 10% RAB CAGR in their distribution business to 2030.

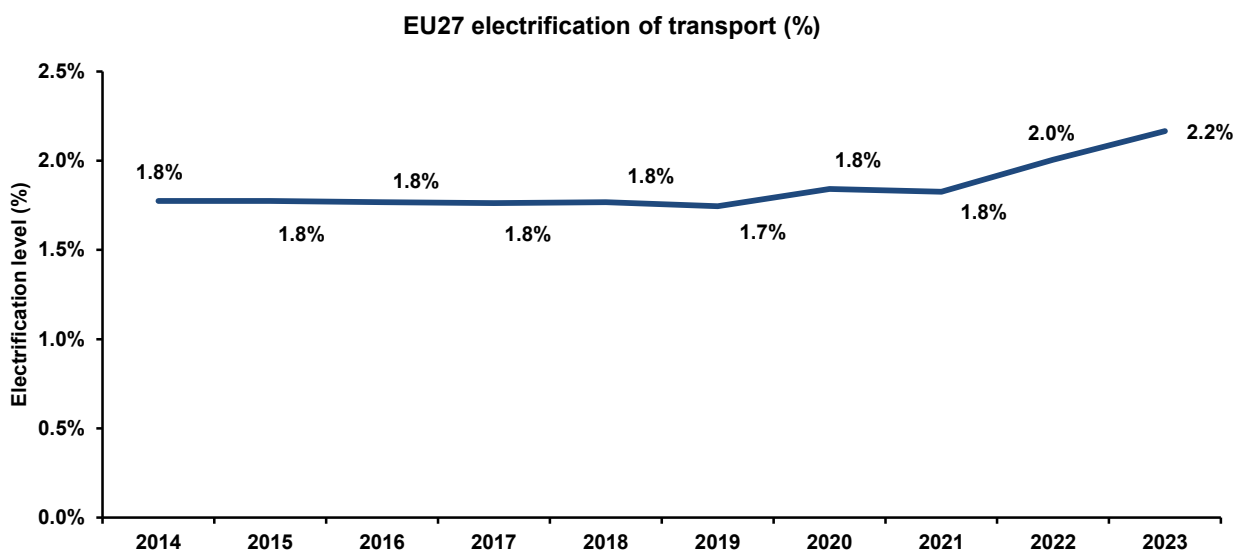
EXHIBIT 16: **SSE identify the electrification of domestic heating (via heat pumps) and of transport as key drivers of a ~10% RAB CAGR for their distribution business**

Distribution: RAV to grow at ~10% CAGR driven by long term electrification trends



Source: SSE, Bernstein analysis

Despite its promise, electrification of transport has so far progressed at a very modest pace, increasing by only ~40bps between 2014 and 2023. We set out the recent evolution of electrification in EU27 transport in Exhibit 17. 2023 is the most recent year for which data is available.

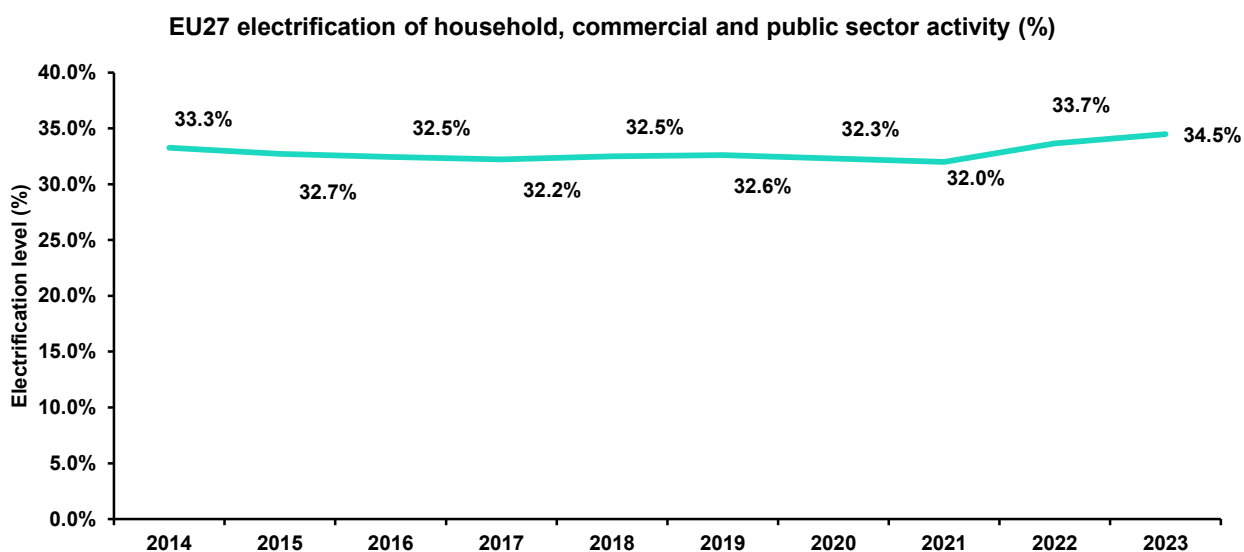
EXHIBIT 17: **EU27 electrification of transport, 2014-2023**

Source: Eurostat, Bernstein analysis

b) Electrification of heat

Alongside transport, heat generation is a major source of carbon dioxide emissions in European countries, and therefore a key focus for decarbonisation, often via electrification. Greater domestic uptake of low-carbon, electric heating technologies such as heat pumps, for example, has the potential to materially grow power demand. As explored in more detail below, **the UK's National Energy System operator expects heat pumps to increase UK power demand by at least 20 TWh in its most optimistic planning scenarios for 2035**, vs [total demand of ~320 TWh in 2025](#) (+6%).

As with the electrification of transport, electrification of heating remains at an early stage - we set out the recent evolution of electrification in the EU27 household, commercial and public sector in Exhibit 18 showing little development over the last 10 years. As we set out more detail below

EXHIBIT 18: **EU27 electrification of household, commercial and public sector**

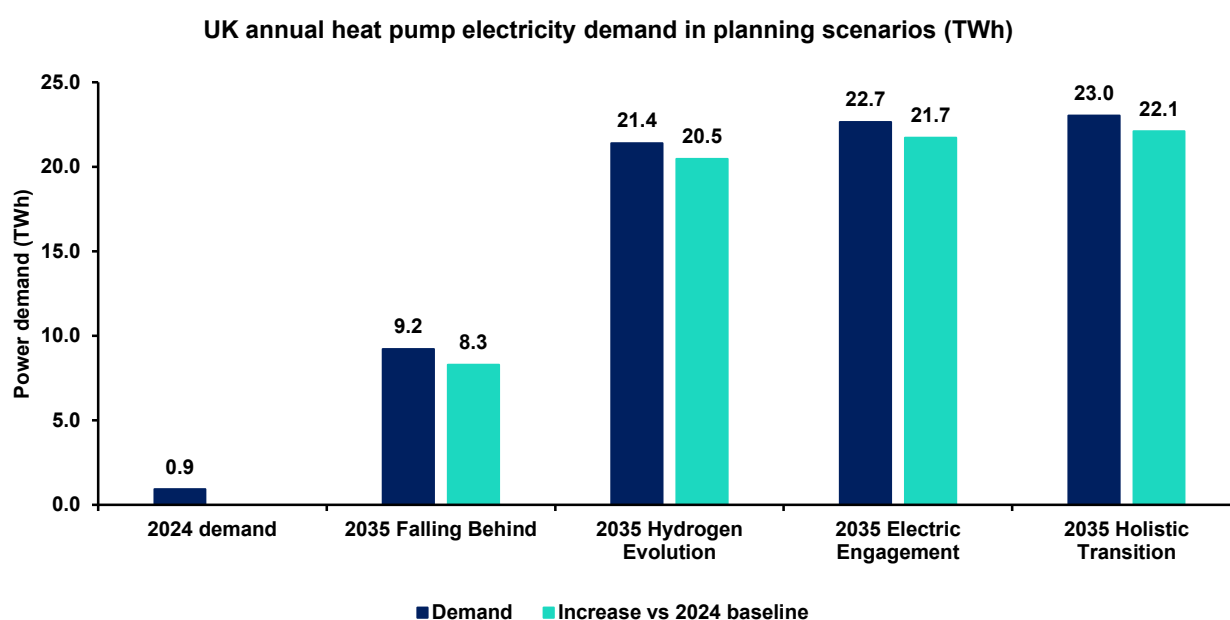
Source: Eurostat, Bernstein analysis

As previously noted, a primary focus of expectation for the electrification of heating is adoption of **heat pumps** (see [Bernstein Energy & Power: Heat pumps - an underappreciated lever of power demand growth?](#)). Heat pumps are electrically-powered heating devices that use a combination of ambient heat and compression technology to heat a building or other structure. The most commonly deployed version of the technology, the air source heat pump (ASHP), absorbs ambient heat from the air.

Heat pumps are at a lower state of maturity than electric vehicles, principally because of their lower cost competitiveness vs the equivalent fossil-fuelled technology. Nonetheless, we see significant growth potential over the long term. The UK's [National Infrastructure Commission](#) explains that between now and 2050 “electrification of heating and transport is expected to have the biggest impact on demand. Modeling by Regen and EA Technology for the Commission estimated that in 2050, heat electrification would account for 40-47GW of peak demand...and electric vehicle charging demand would account for 22-28GW....there will be...the greatest number of interventions and the largest investment in the low voltage levels of the network.”

We set out the illustrative potential impact of heat pump adoption on UK power demand in Exhibit 19 below.

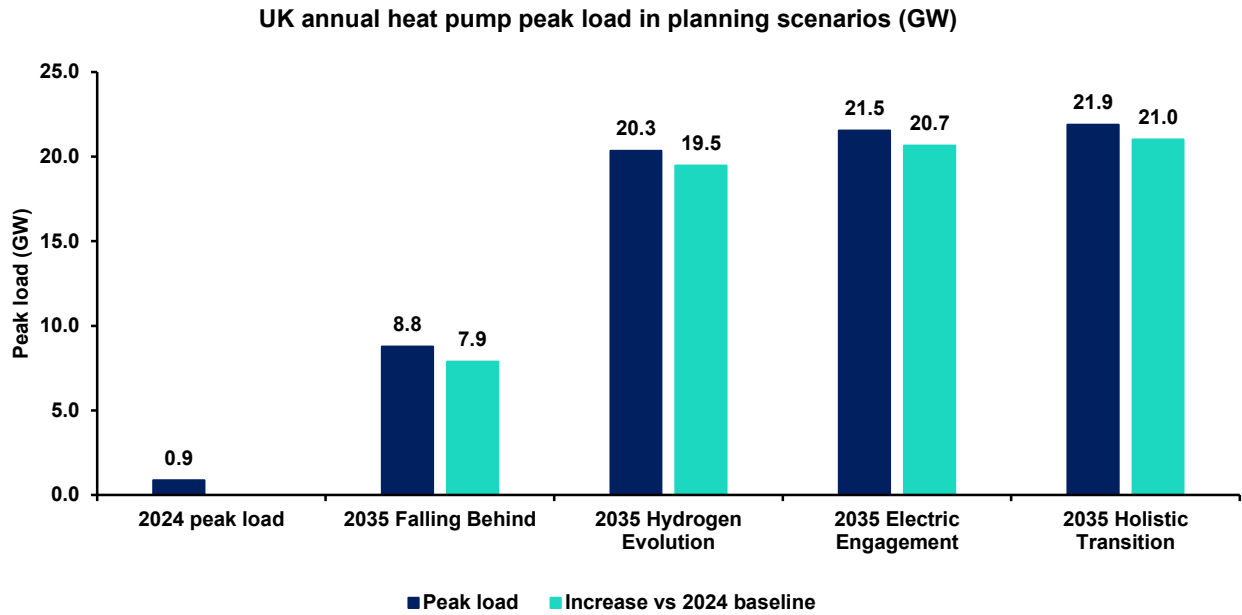
EXHIBIT 19: **Illustrative potential impact of heat pump adoption on UK power demand**



Source: National Energy System Operator, Bernstein analysis and estimates

Alongside their implications for power demand, heat pumps also have potentially significant implications for grid capacity, and hence grid investment needs. We set out the potential evolution of the distribution network load from heat pumps in Exhibit 20.

EXHIBIT 20: Illustrative potential impact of heat pump adoption on UK distribution network load



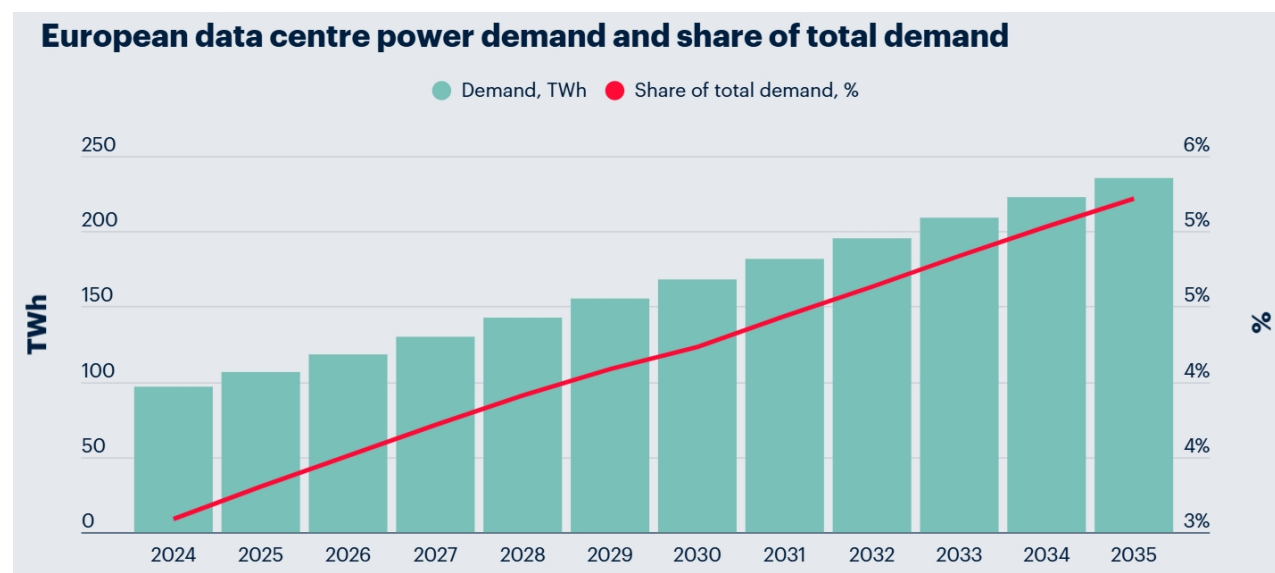
Source: National Energy System Operator, Bernstein analysis and estimates

c) AI and data centers

Artificial intelligence is one of the most widely recognised drivers of power demand growth in Europe over the coming decade. ICIS [estimate](#) that power demand from data centers will grow from ~96TWh in 2024 to ~168TWh by 2030 and ~236TWh by 2035, implying growth in data centers' share of European power demand from ~3.1% in 2024 to ~5.7% by 2035, notwithstanding significant growth in demand from other drivers outlined elsewhere in this report. Other forecasters place power demand from data centers in 2030 [as high as 287 TWh](#). Taking ICIS's contemplated ~72TWh increase in power demand between 2024 and 2030, vs the ~2,867TWh of total European power demand in 2024 (across the EU27, the UK and Norway), data centers emerge as a material factor, driving ~3% incremental demand vs the 2024 baseline.

We set out the ICIS forecasts in Exhibit 21 below.

EXHIBIT 21: ICIS estimate that European data center electricity demand will account for ~6% of the total by 2035

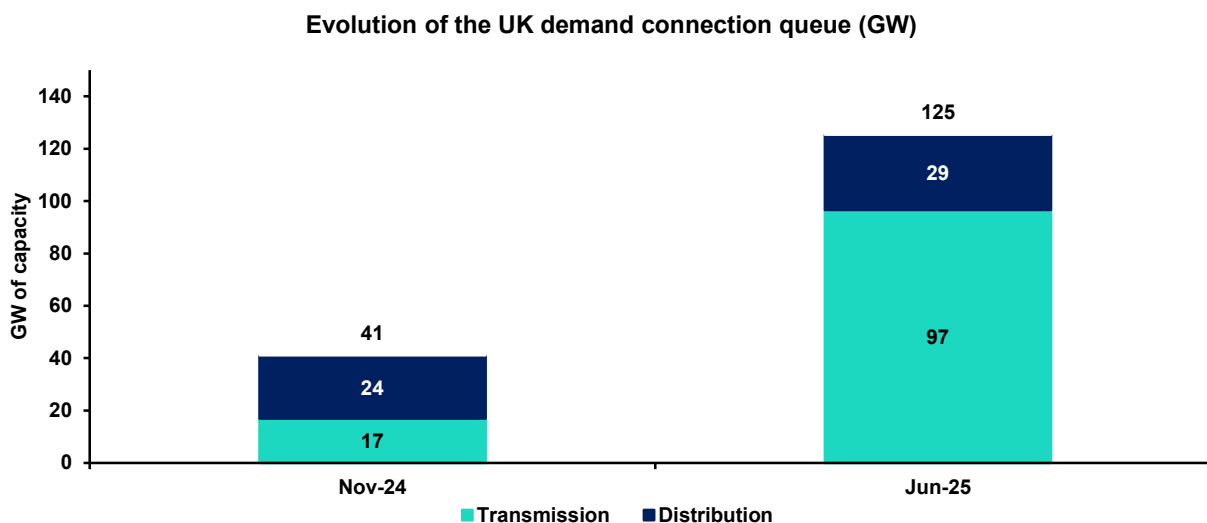


Source: ICIS

Alongside the power demand impact, data centers are also having a significant impact on grid connection queues.

As set out in Exhibit 22, the UK saw a ~470% increase in its electricity transmission grid demand connection queue between November 2024 and June 2025, largely driven by data centers. In Italy, transmission system operator Terna reported 30GW of connection requests, primarily from data centers, at end-FY24, but by 31st August 2026 the grid operator was reporting [~101GW of data center demand capacity](#) in the grid connection queue.

EXHIBIT 22: The UK saw an ~470% increase in its electricity transmission grid demand connection queue between November 2024 and June 2025, largely driven by data centers



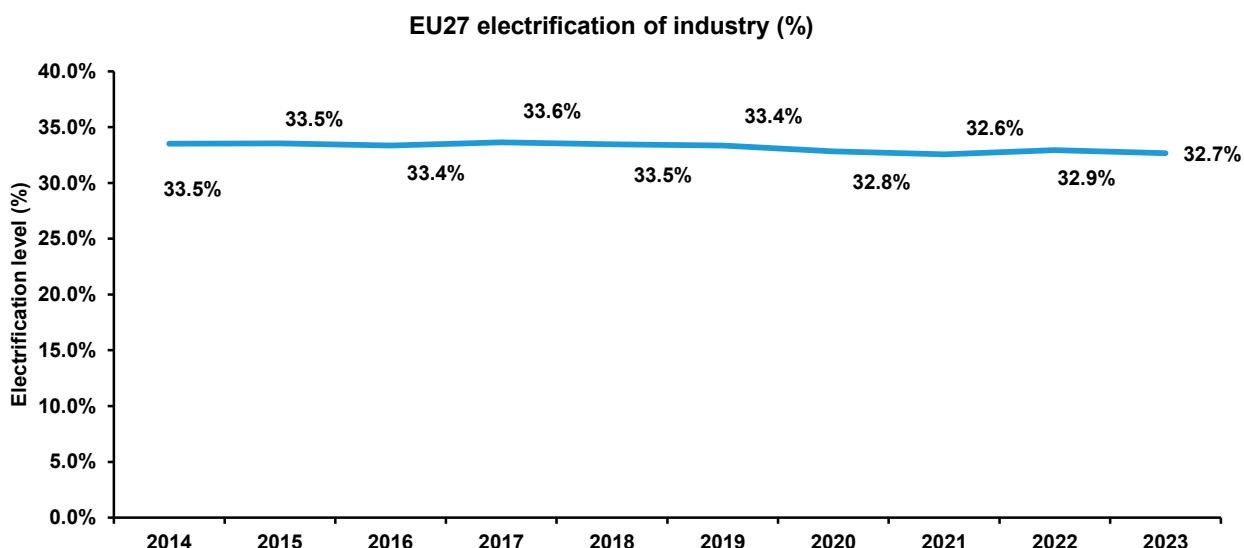
Source: Ofgem, Bernstein analysis

d) Electrification of industrial processes

A final potential electrification driver is the **electrification of industry in Europe**. We set out the recent development of electrification in EU27 industry in Exhibit 23. Given the potential cost competitiveness implications of electrification for industry,

this is an area where we are comparatively less optimistic, although we note the potential for changes in e.g. EU carbon pricing to shift the economics. As shown in Exhibit 23, in line with our more cautious view, electrification moved in reverse over the period 2014-23, falling by ~80bps.

EXHIBIT 23: **EU27 electrification of industry, 2014-23**



Source: Eurostat, Bernstein analysis

The cost-competitiveness of electrified industrial processes remains uncertain vs fossil-fuel-powered alternatives.

It remains uncertain whether industrial processes can be widely decarbonized whilst remaining cost-competitive, particularly in basic materials with strong price-based competition, e.g. steel, cement and chemicals. The EU [accounts for ~7% of global steel output](#), producing around 130 Mt of steel per year. EU steel is primarily made in one of two ways, either primary production using a blast furnace, or by recycling scrap steel using an electric arc furnace. In 2024, 55% of steel production in the EU used a blast furnace, but the average EU blast furnace is 50 years old, and almost half of EU blast furnaces are set to be retired by 2035. Steel production accounts for over 25% of the EU's industrial emissions.

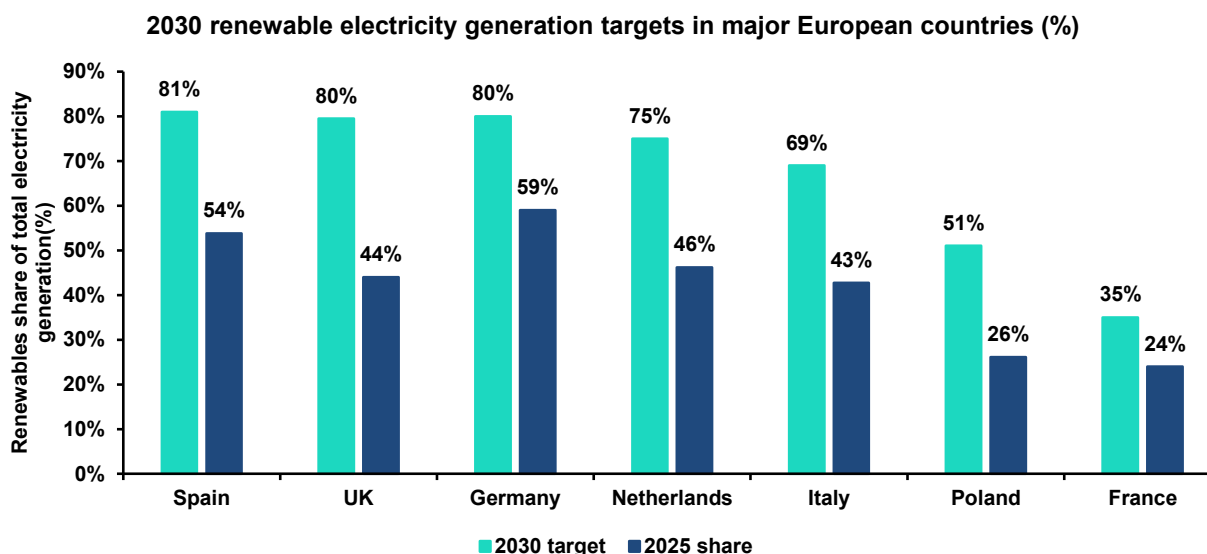
The leading technological process to decarbonize primary steel production is the [H2-DRI-EAF process](#), which uses hydrogen to produce direct reduced iron, which can then be processed in an electric arc furnace to produce steel. The cost evolution of hydrogen production will be a key driver of whether electrified steel production can gain broader traction, and whether this process can grow to become a more material contributor to power demand over time.

2) DECARBONISATION OF POWER PRODUCTION

The second megatrend and electricity network investment tailwind that we identify is an ongoing push towards decarbonization, which we expect to primarily manifest as higher renewable penetration in European electricity systems.

As we set out in Exhibit 24, major European countries have set ambitious renewable generation targets which will require significant grid investment to achieve.

EXHIBIT 24: Many European countries have set ambitious renewable generation targets which will require significant grid investment to achieve

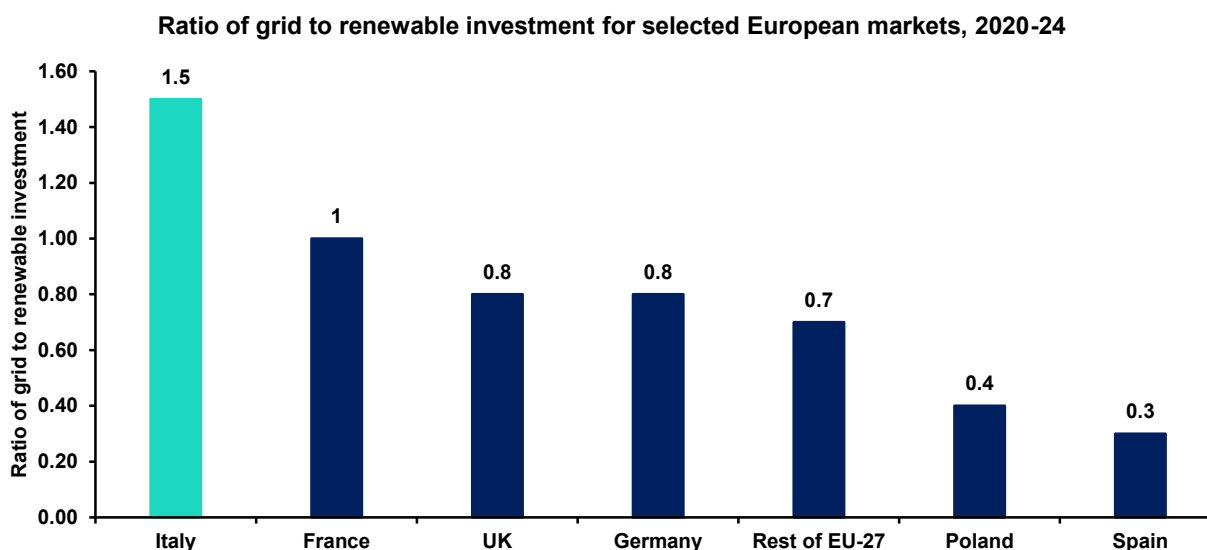


Source: Ember, Bernstein analysis

Higher renewable penetration in European electricity systems is inherently linked to the roll-out of electricity transmission capacity given the differences between renewable generation technologies and the fossil-fuel-based technologies which they are replacing. Fossil fuel generation assets tend to be located relatively close to demand centers, and as such require comparatively limited supporting network infrastructure. By contrast, renewable generation assets must be located in areas where conditions are most suitable, often in remote areas far away from where their electricity will be used. This location mismatch drives a corresponding need for greater supporting network infrastructure. Alongside raw capacity needs, ongoing increases in renewable penetration may also necessitate further investment in ancillary services, e.g. inertia, to compensate for the declining physical contribution from thermal generation.

As set out in more detail in the following country-specific sections, Europe has significant volumes of potential new renewable generation capacity stuck in grid connection queues because of a lack of transmission network capacity. As such, investing in networks is a key energy transition enabler and the energy transition is a powerful tailwind for regulated electricity networks.

As well as future decarbonisation requirements, European electricity grid operators also face a significant investment need for accommodating past renewable capacity investment given that many European countries have seen renewable investment significantly outpace grid investment in recent years. We set out details of the 2020-24 investment gap between electricity grids and renewable capacity in Exhibit 25. Spain, Germany and the UK all stand out as major European countries where significant further grid investment is required to accommodate past investment.

EXHIBIT 25: **Ratio of grid to renewable investment for selected European markets, 2020-24**

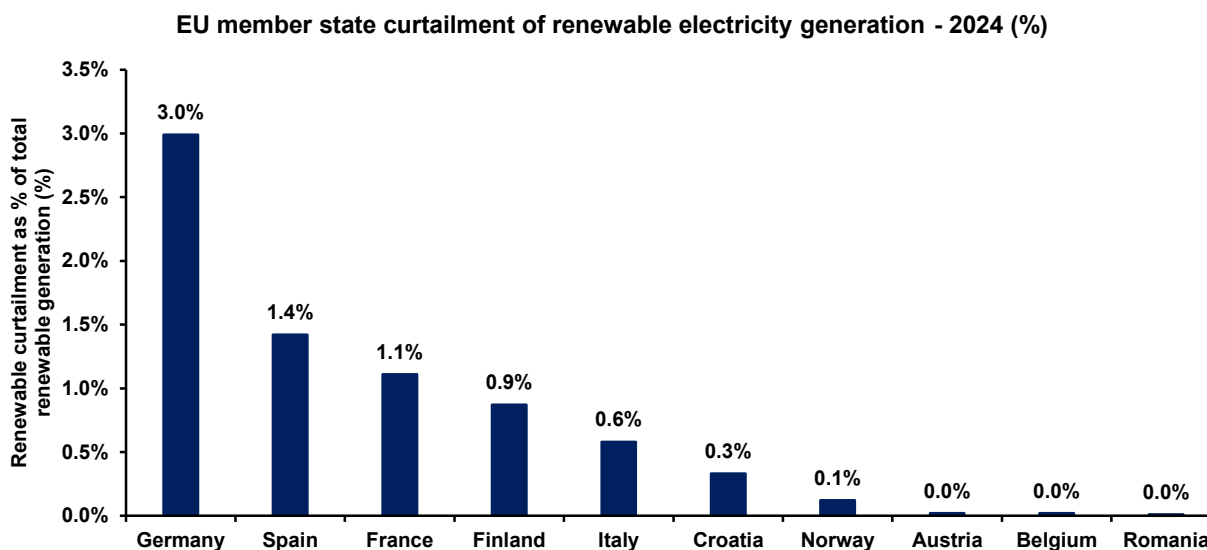
Source: Bloomberg New Energy Finance, Bernstein analysis

Underlining the importance of catch-up investment alongside future capacity roll-out, the IEA's [World Energy Investment 2026](#) noted that "previous editions of the World Energy Investment have highlighted how grid investments were lagging those for generation, creating electricity security risks." The organisation also stated that "in contrast with recent trends, investments in electricity supply are now learning towards grids and storage. Global spending on networks is expected at round \$550bn in 2026, up nearly 20% year-on-year."

The drivers of the prior electricity network investment deficit are complex and varied, but include the fact that regulators have historically been more focused on minimising the risk of stranded grid assets than preparing the grid for growth (described by Cambridge/Griffith University's [Paul Simshauser](#) as a "least cost, just-in-time framework"), with investment in renewable electricity production significantly outpacing investment in grids over the past decade.

Reinforcing the catch-up need, as set out in Exhibit 26, EU member states are already facing significant curtailment volumes as a result of the mismatch between generation and network capacity, with many expecting significant further increases in the absence of additional network investment. As such, the investment case for new networks assets is already substantially underpinned and does not rely on speculation regarding future need.

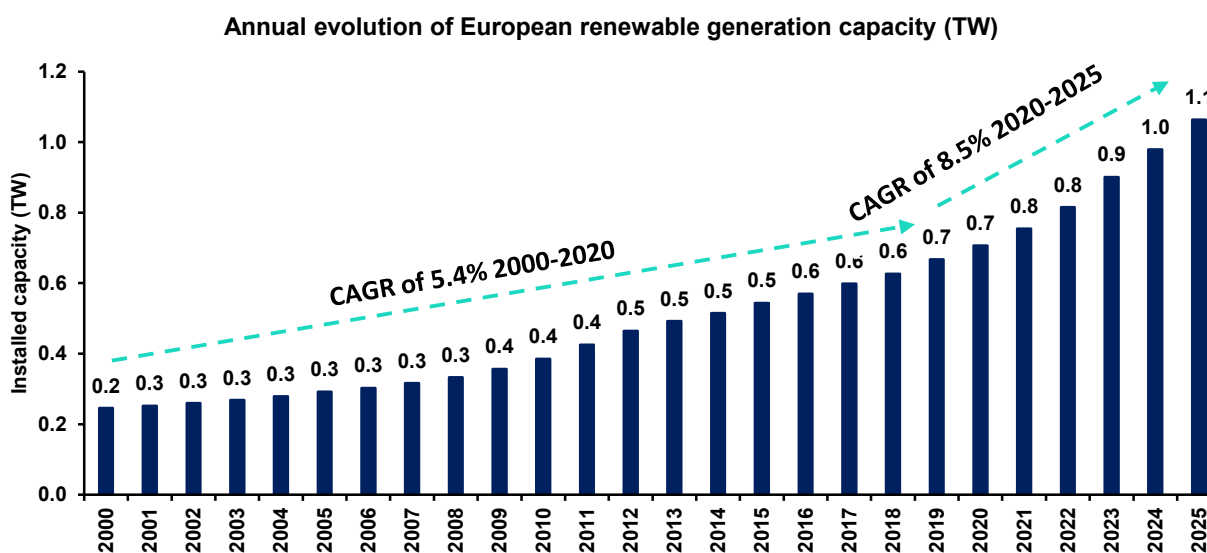
EXHIBIT 26: EU member states are already facing significant curtailment volumes as a result of a mismatch between generation and network capacity



Source: ACER, Bernstein analysis

We set out the evolution of total European renewable electricity generation capacity in Exhibit 27, clearly showing the acceleration in the pace of the renewables' capacity roll-out in the 2030s.

EXHIBIT 27: Annual evolution of European renewable electricity generation capacity



Source: Ember, Bernstein analysis

In the following sections we analyze the specific investment requirements and drivers in individual European countries.

A) GERMANY

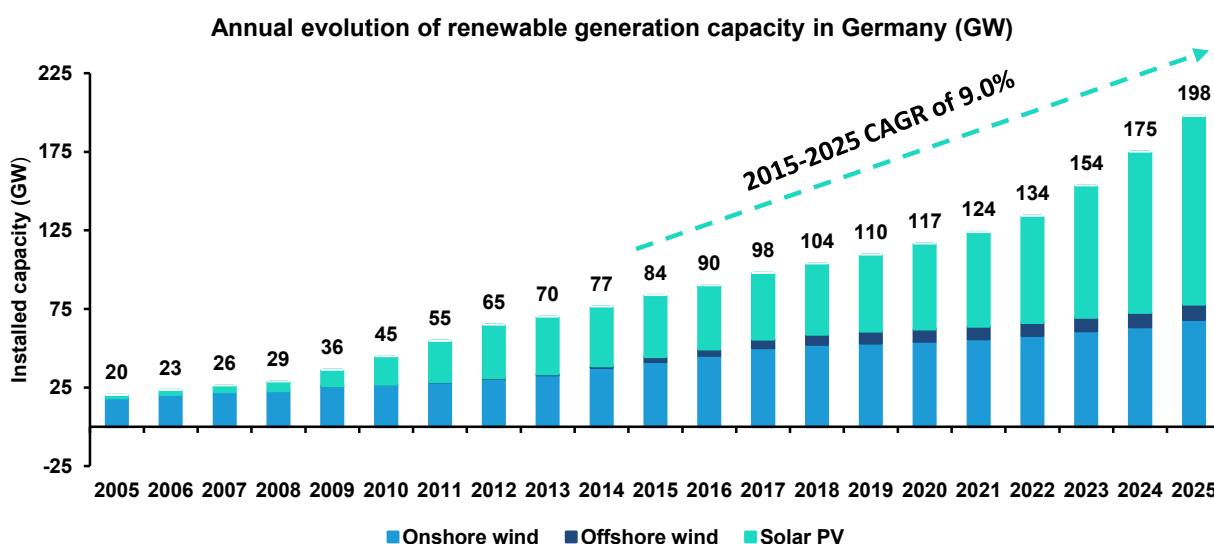
Germany has very ambitious climate ambitions. Under the Climate Change Act 2021, Germany seeks to achieve net zero by 2045, five years ahead of the EU target.

Key medium-term decarbonisation targets in Germany include:

- 80% renewable generation share of electricity consumption by 2030 (vs ~54% achieved in 2024)
- A complete phase out of coal generation by 2038
- 65% CO2 emissions reduction by 2030 compared to 1990 levels (vs ~48% reduction achieved by 2024)

As set out in Exhibit 28, Germany has very significantly increased its installed base of renewable electricity generation capacity, which has increased almost 10-fold in 2 decades. Renewable generation capacity grew at a CAGR of ~9% between 2015 and 2025.

EXHIBIT 28: **Annual evolution of renewable electricity generation capacity in Germany**

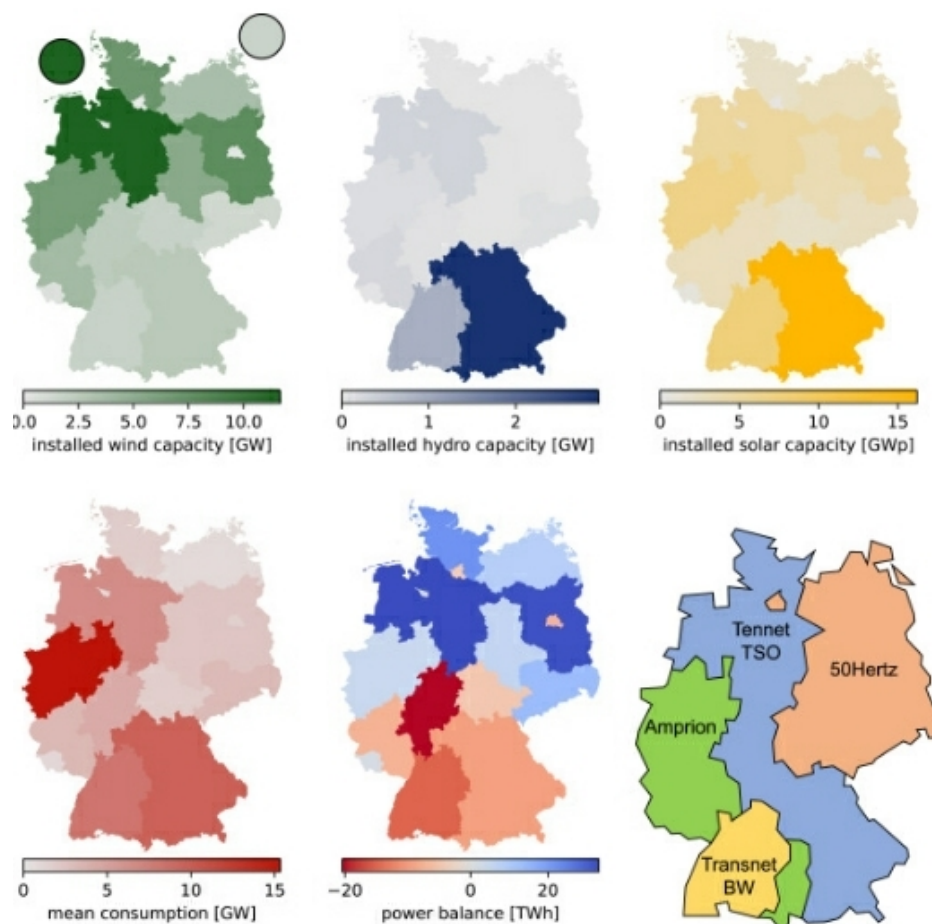


Source: Bundesministerium für Wirtschaft und Energie, Bernstein analysis

Germany's decarbonisation ambitions are a significant driver of electricity network investment

As we show in Exhibit 29, **Germany's ongoing push into renewables, particularly wind power, is a significant driver of electricity network investment.** Favorable locations for wind power are primarily located in the north of the country (onshore wind capacity is primarily concentrated in the federal states of Lower Saxony, Schleswig-Holstein and Brandenburg, with offshore wind located in the North and Baltic seas), while many densely populated areas and industrial centers, such as Munich and Stuttgart, are located in the south. This is an important contributor to a capex need which is [estimated at ~€365bn to €392bn](#) in total from 2025-2045, given that the new power generated must often be transported long distances. In comparison, the distribution of solar capacity is much more regionally balanced, whilst hydropower capacity is concentrated in the south.

EXHIBIT 29: Illustrative configuration of Germany's power system



Source: Maurizio Titz, Sebastian Pütz, Dirk Witthaut (Identifying drivers and mitigators for congestion and redispatch in the German electric power system with explainable AI - ScienceDirect)

The importance of the ongoing renewables roll-out in Germany to the significant levels of grid investment contemplated in that country can be seen in the national grid capacity planning documents. Electricity transmission grid planning documents are jointly prepared by the four German TSOs (TenneT Germany, Amprion, 50Hertz and TransnetBW) on the basis of a set of assumptions including the future evolution of demand and the level and profile of supply. The [latest draft version](#) contemplates the following developments:

- Onshore wind power capacity increasing from ~63.5GW in 2024 to between ~127GW and ~158GW by 2037 and between ~144GW and 176GW by 2045.
- Offshore wind capacity increasing from ~9GW in 2024 to between 50 and 56GW by 2037 and to between 60GW and 70GW by 2045.
- Solar PV capacity increasing from ~100GW in 2024 to between 270GW and ~380GW by 2037, and to between 315GW and 440GW by 2045.

In aggregate, according to the German [TSOs](#), approximately 33,000 km of new transmission infrastructure will be required by 2045. Total investment needs to that point are estimated at >€360bn.

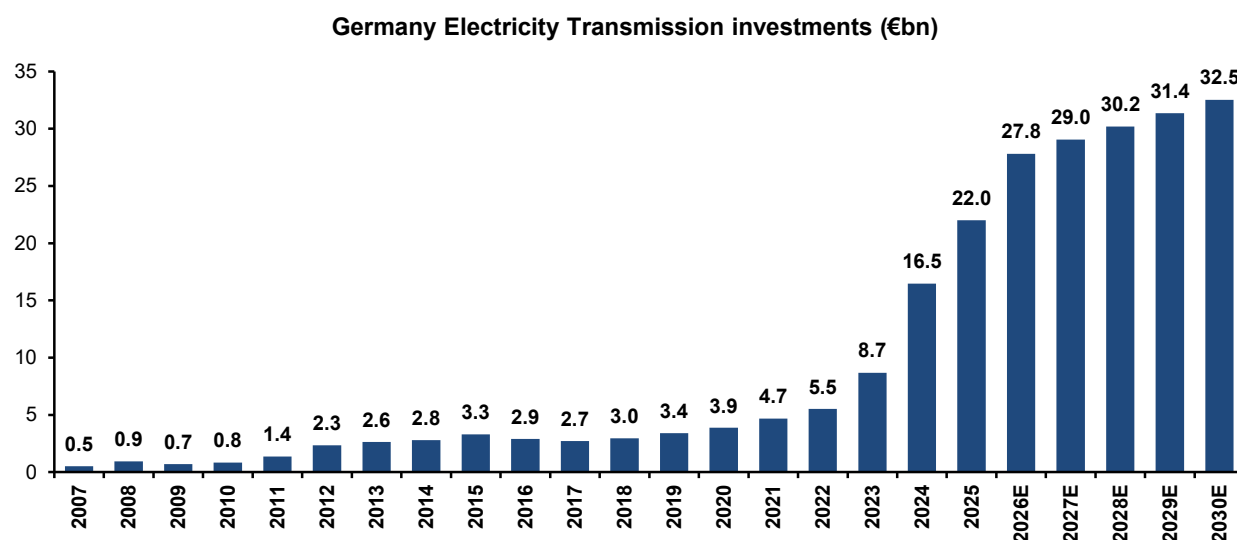
At the company level, as of summer 2025, the four German TSOs expected to invest ~€193bn cumulatively between 2020-2030:

- TenneT Germany planned a ~€65bn investment program from 2025-2029 and ~€10bn per annum from 2030-34.

- Amprion planned to invest €36.4bn between 2025-2029, of which €21.9bn was in onshore networks, which were expected to peak in 2029, while offshore investments were €14.6bn – these were expected to decrease towards the end of 2029.
- 50Hertz (80% owned by Elia group) planned to invest €19.3bn between 2025-2028.
- The Grid Development Plan 2035 (2021) identified around €10bn in investment needed for TransnetBW primarily for the major SuedLink and Ultranet projects.

We set out the historical and forecast evolution of electricity transmission grid investments in Germany since 2007 in Exhibit 30.

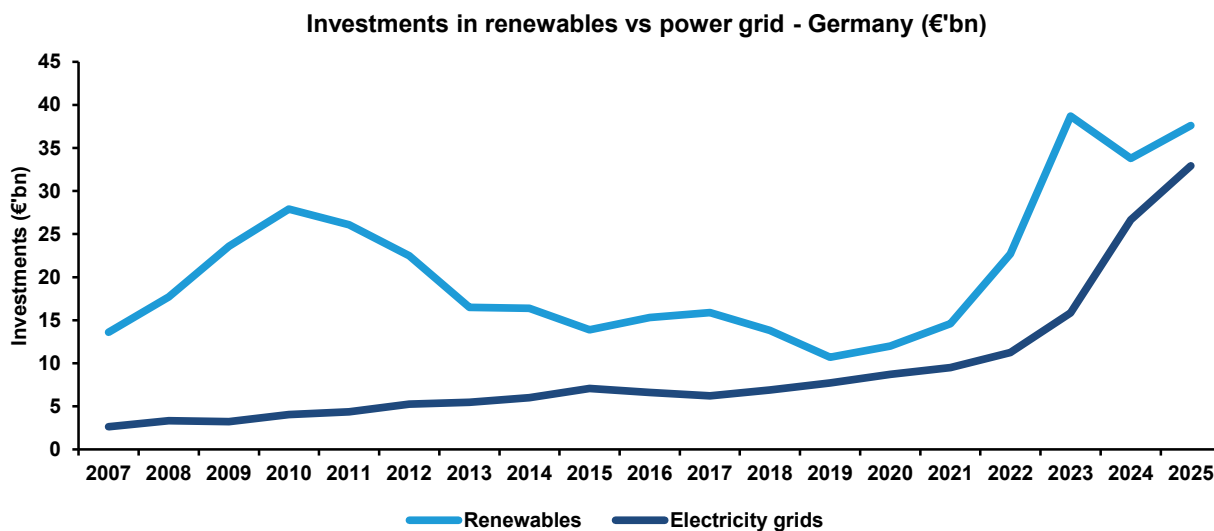
EXHIBIT 30: **Electricity grid investments in Germany have started to pick up in pace since 2023**



Source: BNetzA (historic investments), Company reports, Bernstein analysis and estimates

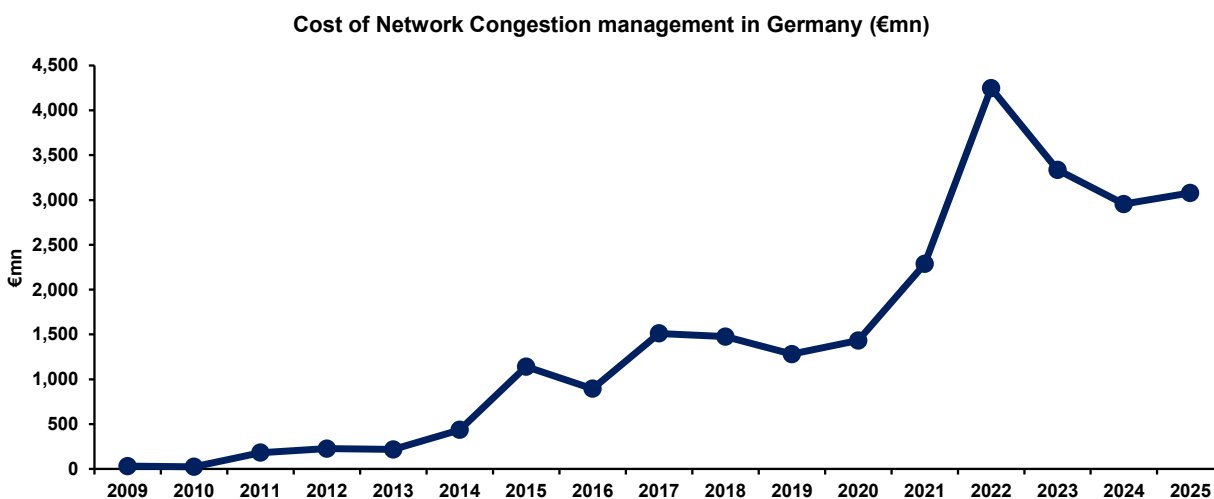
The catch-up electricity network investment need in Germany is also very significant

Germany is also a prominent example of a European country with a major pressing need for catch-up electricity network investment. As we set out in Exhibit 31, renewable investment has outpaced grid investment spend in each of the last 18 years, accelerating since 2021. Whilst the gap has started to close since 2023, the electricity network investment runway in Germany is substantially underpinned even in the event of a renewable build-out slowdown - a significant further network roll-out is not predicated on further renewable capacity expansion.

EXHIBIT 31: Grid investments in Germany have lagged renewable investments and are only now starting to catch up

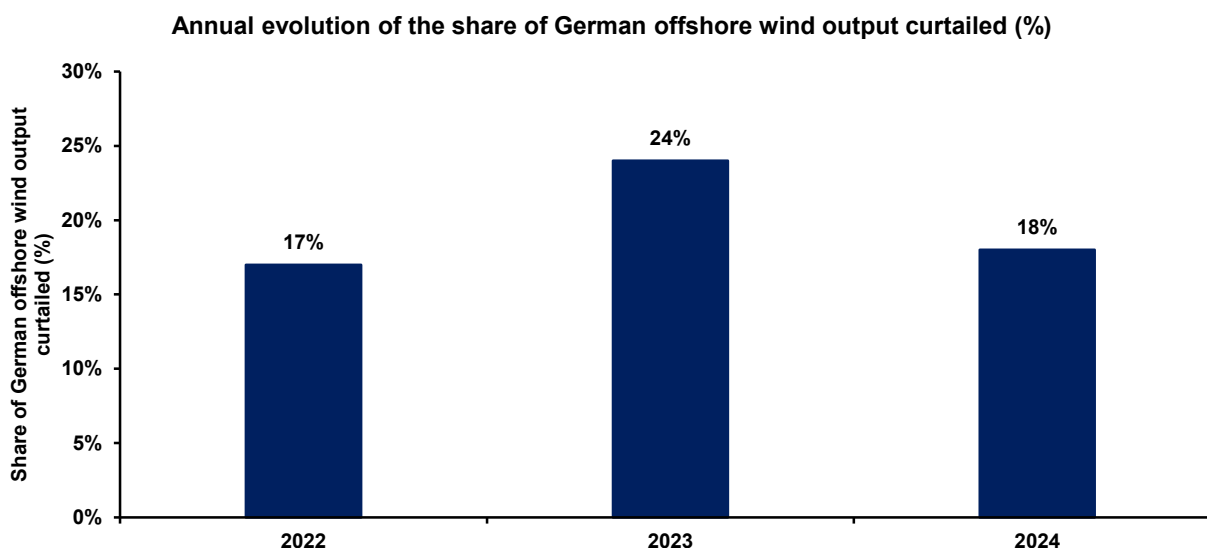
Source: ZSW (renewable investments), BNetzA (power grid investments - distribution plus transmission), Bernstein analysis

The disconnect in grid and renewables expansion has also been apparent in the rising cost of renewable curtailment and re-dispatch costs, which have been rising significantly since 2021, as shown in Exhibit 32 and as we discuss later, feeding into network tariffs (which also include impact of congestion measures apart from remuneration of costs and returns to grid operators). As [per Eurelectric](#), Germany has the highest grid management costs in Europe, with over 50% of total European costs in the past four years.

EXHIBIT 32: Network charges in Germany have increased significantly in recent years as a result of the mismatch between renewable deployment and grid investments

Source: BNetzA (2009-2022), SMARD.DE (2023,2024,2025), Bernstein analysis

A final indication of the scale of the catch-up investment need in Germany is the volume of offshore wind output which is currently being curtailed, as set out in Exhibit 33; curtailment figures in the double digits underline the need for grid investment to properly integrate the new output into the system.

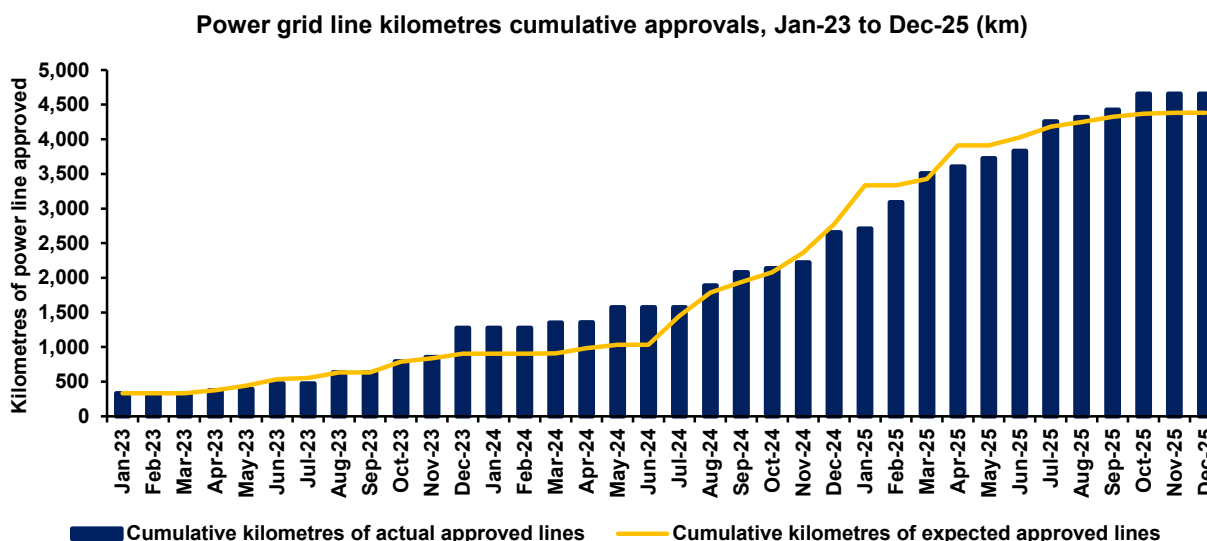
EXHIBIT 33: **Annual evolution of the share of German offshore wind output curtailed**

Source: BNEF, Bernstein analysis

The German government is pushing to simplify the electricity network roll-out

In response to the pressing need for further electricity grid capacity to enable the country's energy transition, the German government has taken significant action. Over the course of 2022, following Russia's invasion of Ukraine, Germany implemented a simplification of permitting procedures.

As shown in Exhibit 34, following the reforms, the cadence of new power line approvals in Germany has increased significantly, underlining the potential effectiveness of government focus on addressing the bottleneck.

EXHIBIT 34: **Monthly new German power line approvals in Germany have increased significantly since January 2023**

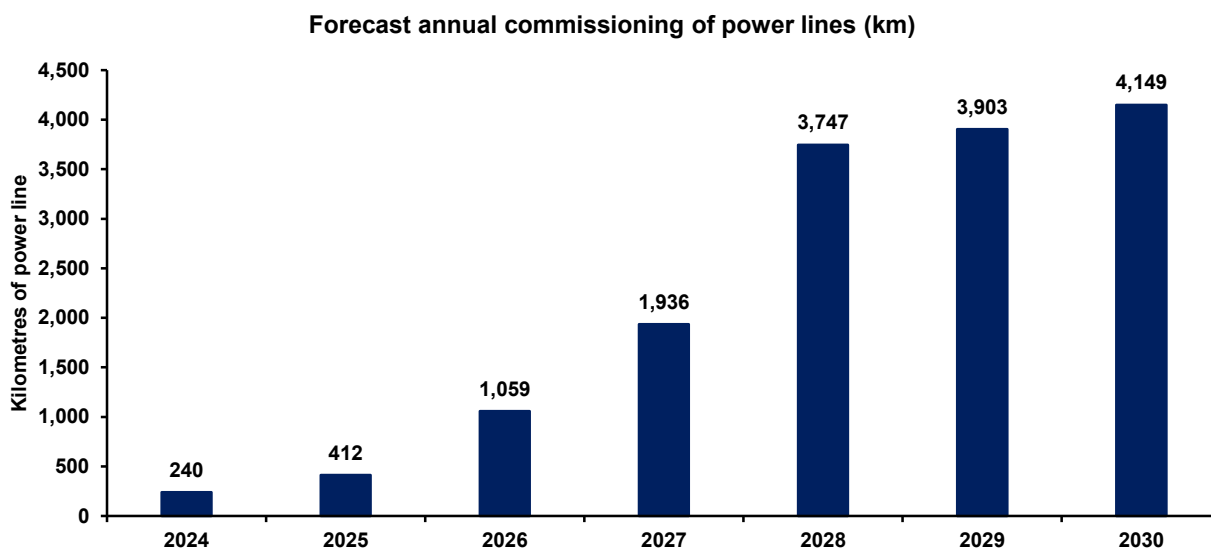
Source: Bundesnetzagentur, Bernstein analysis

German legislation (the Federal Requirements Plan Act and the Power Grid Expansion Act) calls for around 16,800km of new power lines in Germany. The Bundesnetzagentur is responsible for the examination and approval of ~9,600km of these lines, with the remainder processed by the respective federal states. The planned network expansion is equivalent to ~50% of the

existing network.

We set out the historical and forecast evolution of new power line commissioning in Germany on an annual basis from 2024-30 in Exhibit 35 below.

EXHIBIT 35: **Historical and forecast annual evolution of new power line commissioning in Germany, 2024-30**



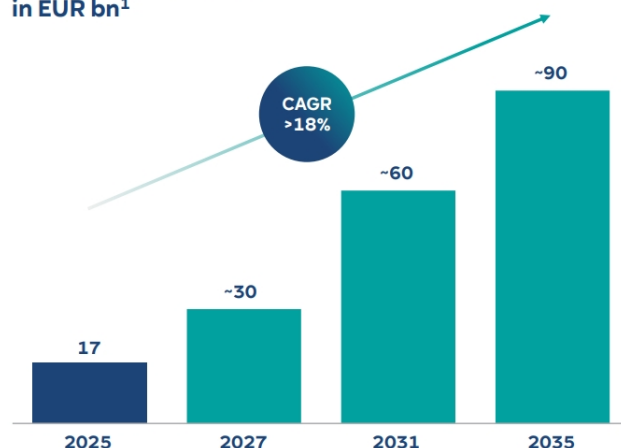
Source: Bundesnetzagentur, Bernstein analysis

Given the above, it is unsurprising that Germany has a very strong outlook for electricity transmission network investment at the company level. RWE [recently announced](#) its €3.6bn acquisition of an additional 35% stake in Amprion, a German TSO with an asset base in the west of the country, citing a RAB CAGR of >18% from 2025 to 2035 - see Exhibit 36.

EXHIBIT 36: RWE forecasts a >18% RAB CAGR for its German network assets

Amprion offers extraordinary RAB growth in the next ten years

Regulated Asset Base in EUR bn¹



Key highlights

- **System-relevant infrastructure** with 11,000 km transmission grid in Germany
- **Located along Germany's economic heartland** with more than 29 m people living in Amprion's core area
- **Strongly growing regulated asset base (RAB)** driven by high investment needs in energy transition based on grid development plan
- **Robust and predictable regulatory framework**
- **Proven track record** of strong financial performance

¹ 100% view

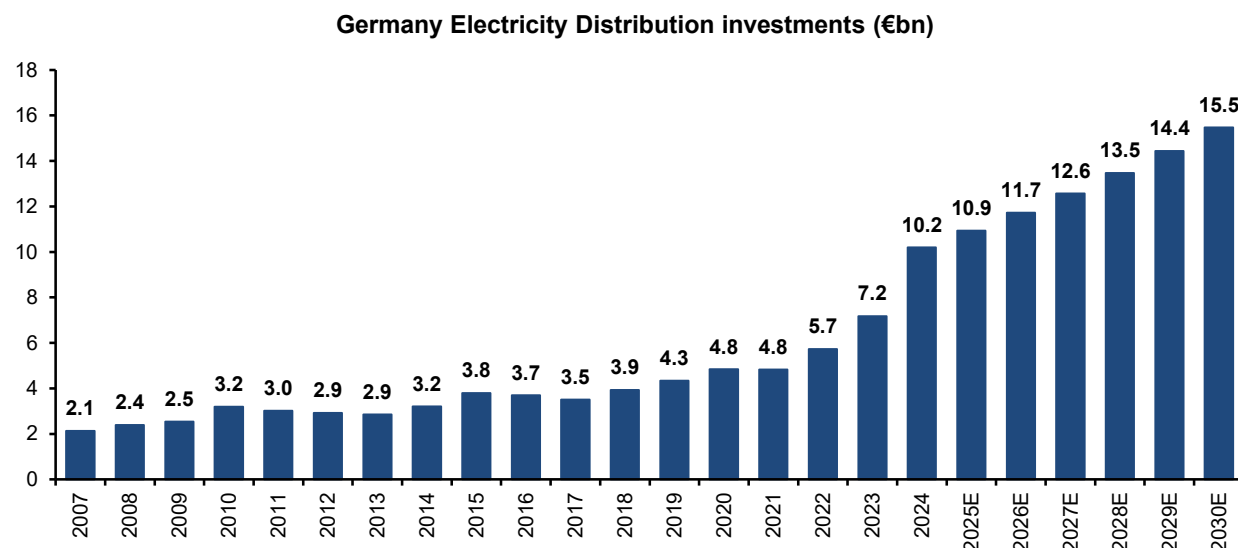
RWE 22 June 2026 RWE expands into Regulated Grid Infrastructure

Source: RWE

Germany's distribution network will also benefit from the decarbonisation push

Germany's electricity distribution system is highly fragmented in terms of ownership. There are [approximately 866 DSOs](#) in the country. As of December 2025, there were 43GW of wind capacity in distribution network connection queues (~37% of the targeted 2030 total) and 130GW of battery capacity (vs the 50GW target).

We set out the annual evolution of electricity distribution network investments in Germany in Exhibit 37. As with electricity transmission we see a significant growth runway.

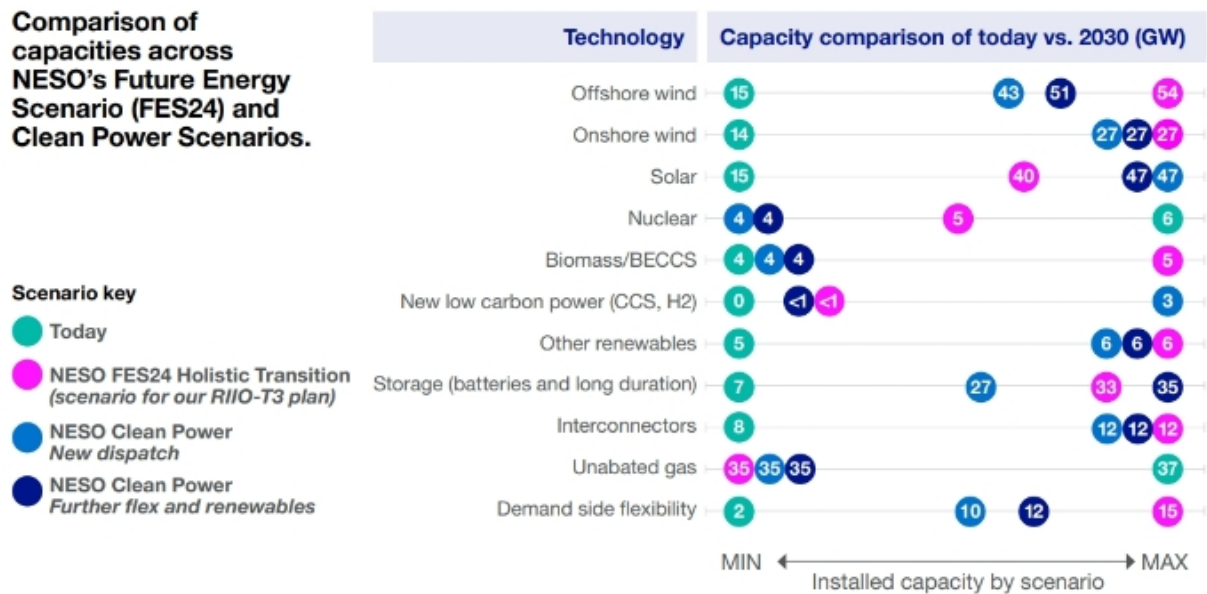
EXHIBIT 37: **Annual evolution of electricity distribution network investments in Germany**

Source: BNetzA, BDEW, Bernstein analysis and estimates

B) UK

As with Germany, the UK is contemplating a very significant roll-out of renewable generation capacity between 2026 and 2030 and further out, as set out in Exhibit 38. This is spearheaded by offshore wind, where capacity is seen close to tripling in even the least optimistic scenario. The Labour government in the UK has set out the [Clean Power 2030 Action Plan](#), with the aim that “in a typical weather year, the 2030 power system will see clean sources produce at least as much power as Great Britain consumes in total, over the whole year, and at least 95% of Great Britain’s generation.” The centrality of electricity network decarbonisation to the government’s political strategy and credibility has given decarbonisation significant momentum in the UK, as seen in the blockbuster AR7 subsidy auction (see [Quick Take: AR7 offshore wind auctions - Positive for RWE, Vestas, SSE with 8+GW clearing at £91/MWh](#)).

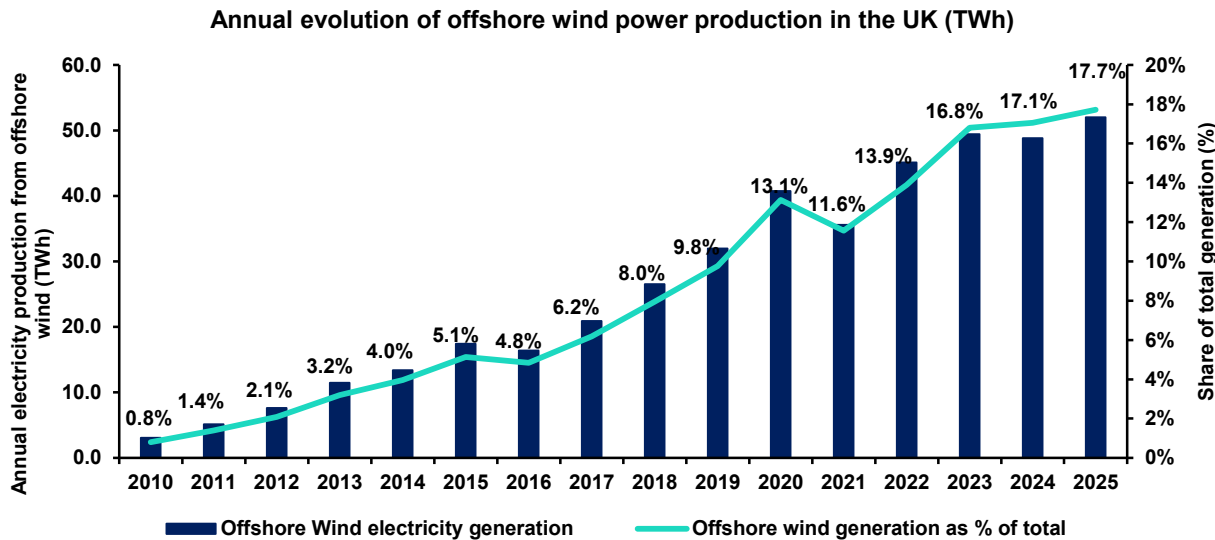
EXHIBIT 38: The UK is contemplating a very significant roll-out of renewable generation between 2026 and 2030



Source: National Grid

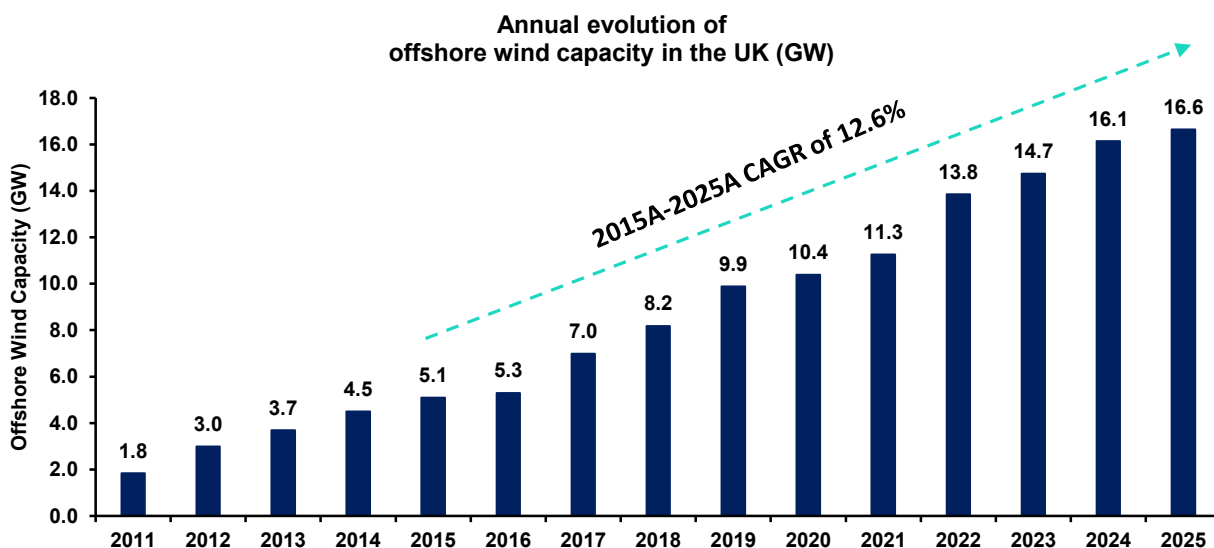
As set out in Exhibit 39, the UK has already seen very significant growth in offshore wind generation since 2010. The growth of this technology has driven significant demand for additional grid capacity given its typical location very far from demand centres.

EXHIBIT 39: Offshore wind, a highly distributed form of electricity generation, has taken significant share in the UK since 2010



Source: DUKES, Bernstein analysis

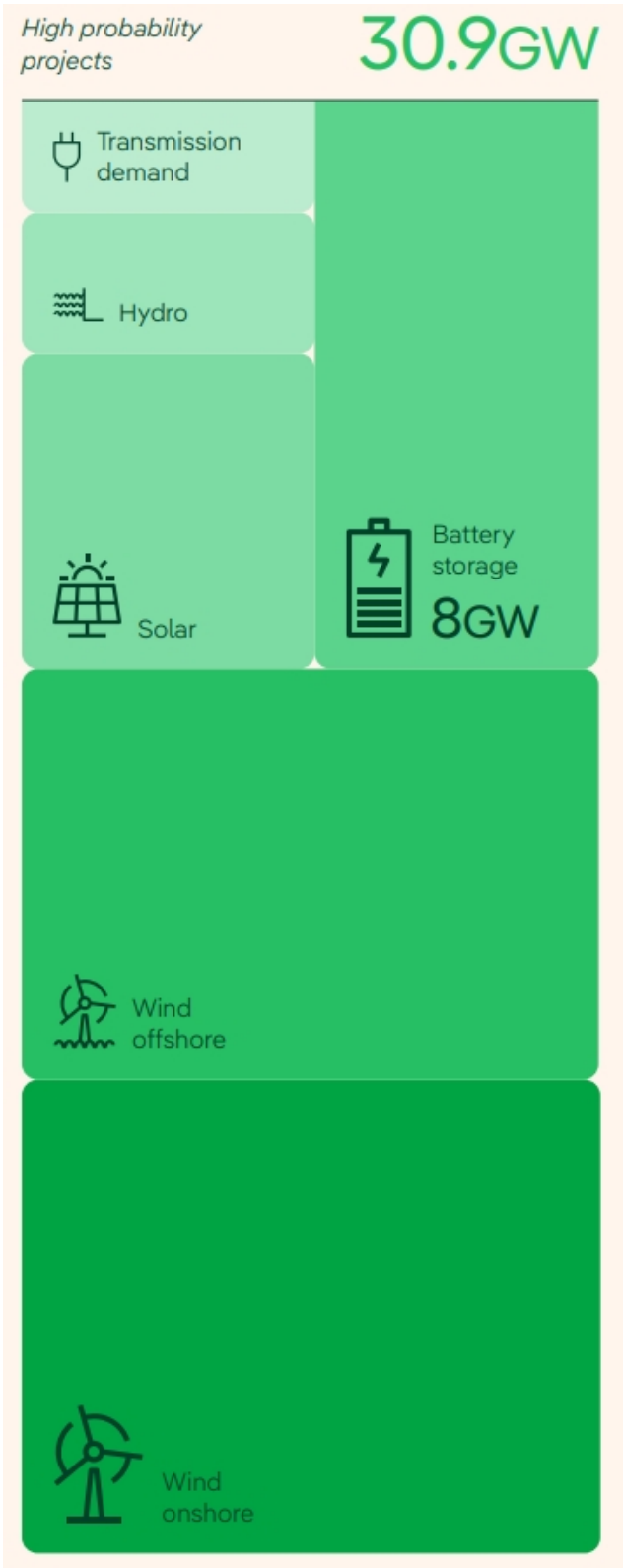
We set out the annual evolution of offshore wind capacity in the UK from 2011 to 2025 in Exhibit 40.

EXHIBIT 40: **Annual evolution of offshore wind capacity in the UK**

Source: DUKES, Bernstein analysis

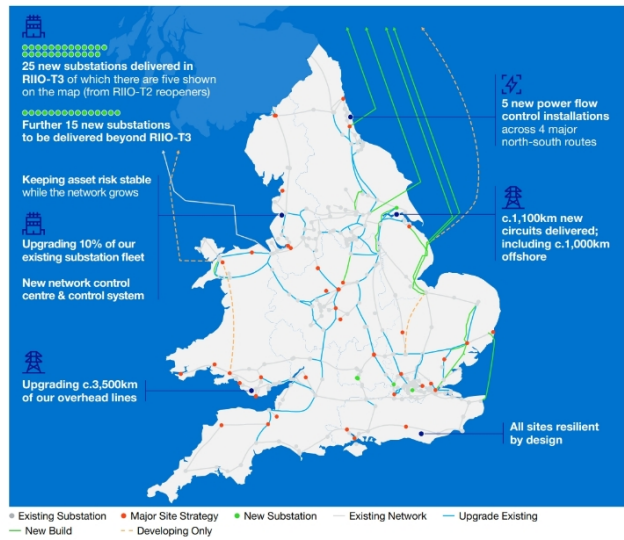
The impact of the ongoing renewable capacity roll-out in the UK on the electricity transmission companies' investment plans can be seen in Exhibit 41 and Exhibit 42. ScottishPower, for example, has identified ~31GW of high probability connection projects, with renewables the dominant driver of the investment case.

EXHIBIT 41: **ScottishPower’s RII0-T3 business plan** contemplates ~31GW of new high probability connections, dominated by renewables



Source: ScottishPower

EXHIBIT 42: **The offshore wind roll-out is an important driver of National Grid’s RII0-T3 investment plan**



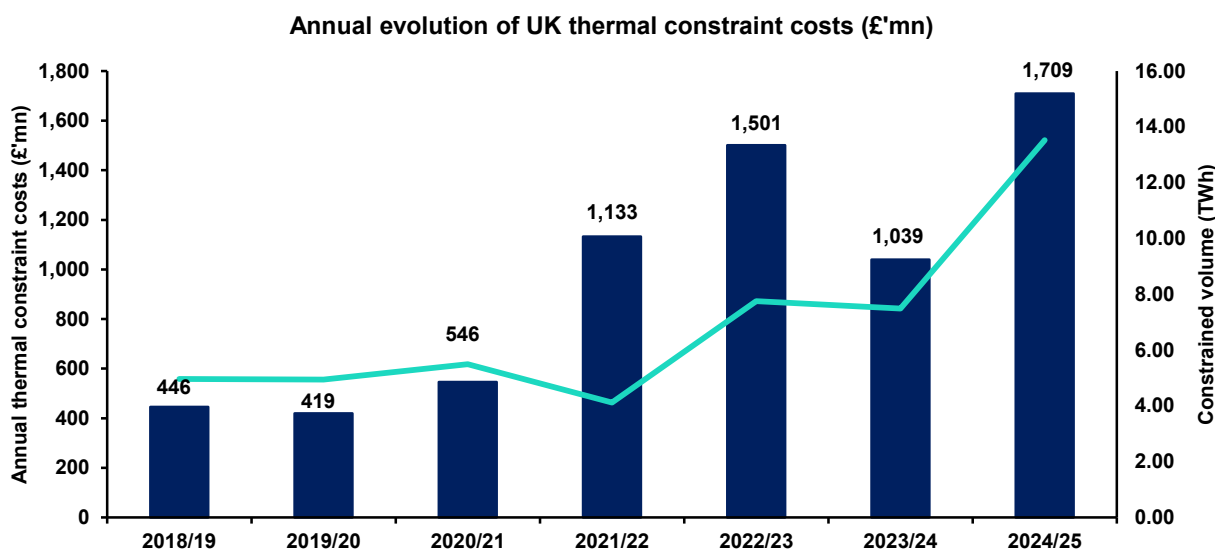
Source: National Grid

As with Germany, the need for grid capacity to catch up with previous renewable investment is a significant driver in the UK

As in Germany, there is a significant electricity network investment need in the UK from renewable investments which have already been made; the need to reduce payments for network constraints is one of the most commonly-cited justifications for the UK's significant ramp-up in electricity grid investment. The National Energy System Operator's [2025 Annual Balancing Costs Report](#) sets out that "investment in and timely optimization of the national electricity network is...the most impactful lever available to minimize balancing costs as we progress with the energy transition." Similarly, the NESO stated in its "[Beyond 2030 - Electricity Transmission Update](#)" report in June 2026 that "if Great Britain were to retain the existing transmission network into 2030, constraint costs in that year could reach around £12.7 billion. By building new network capacity, those costs onshore would be reduced by around 75%."

We set out the recent evolution of annual thermal constraint costs in the UK in Exhibit 43.

EXHIBIT 43: **Annual evolution of UK thermal constraint costs**



Source: NESO, Bernstein analysis

The decarbonisation push is also touching the distribution network

As highlight above, the decarbonization push in European countries is not restricted to the transmission network, and the role of the electricity distribution network in enabling the UK's renewable build-out is widely recognised. The UK's National Infrastructure Commission [explains](#) that "in the past, power moved from large, centralized generation sites on the transmission network down to distribution level at grid supply points. The location of generation is changing with the move to a highly renewable system, as renewable technologies and battery storage tend to be more dispersed. They are also more likely to be connected to the distribution network. Currently, 35% of generation is distribution-connected (also known as embedded generation), up from 15% in 2011, and this will continue to increase as more renewables are deployed."

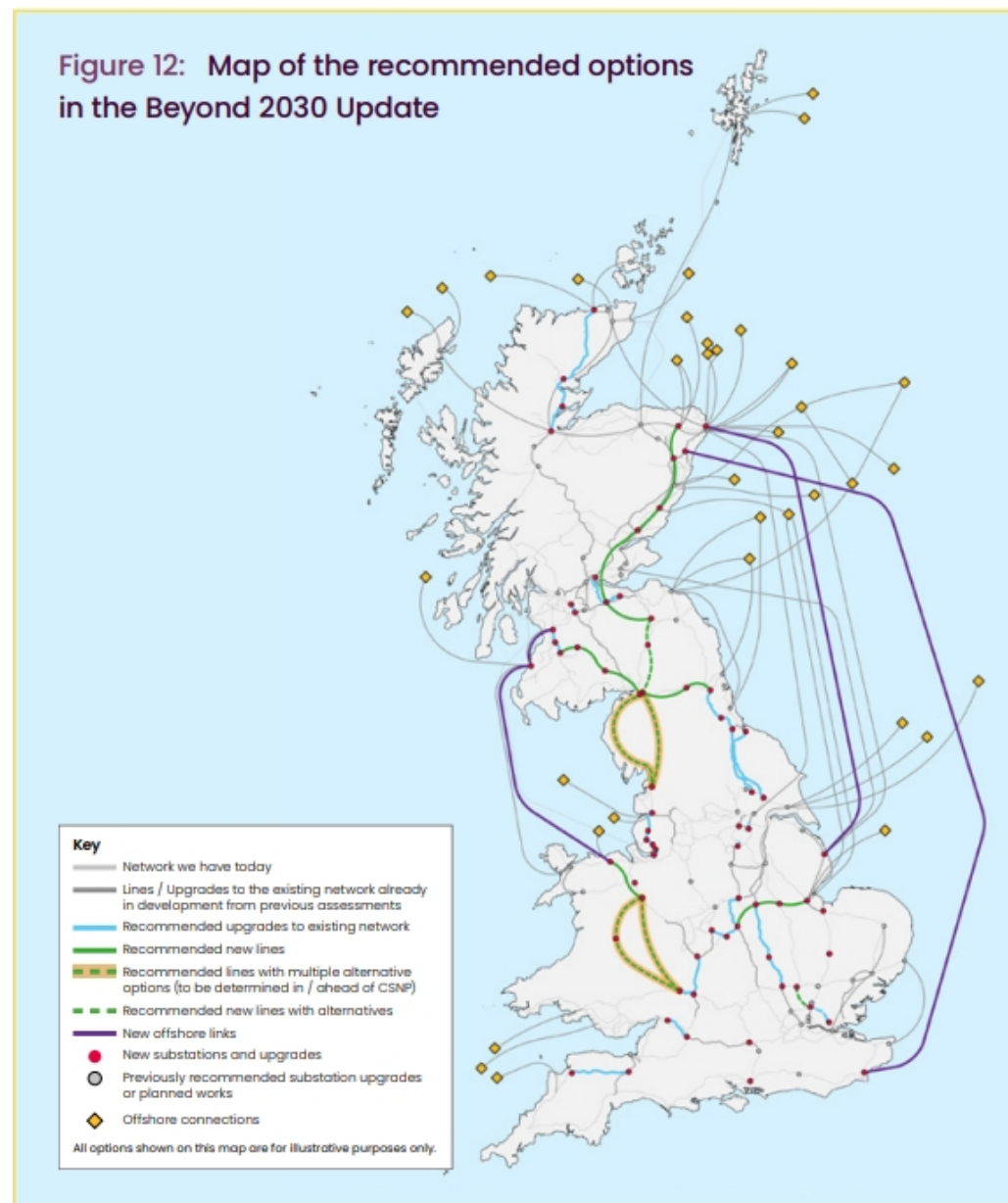
The UK has a relatively concentrated ownership base for its electricity distribution network, with 14 network licenses held by 6 different network operators. The [distribution grid connections queue](#) currently has 12GW of wind capacity, 73 GW of solar capacity and 101 GW of BESS capacity, vs 2030 Clean Power Plan targets of 29GW, 47GW and 27GW respectively.

The decarbonisation push will also drive an electricity network need in the UK significantly beyond 2030

The UK has clear visibility of the investment need in electricity transmission significantly beyond 2030. In June 2026 the UK National Energy System Operator published [Beyond 2030 - Electricity Transmission Update](#). Key findings included that "without [network] reinforcement the costs of balancing the system could see around a three-times increase between 2031 and 2035". The system operator recommends "a total of £89 billion of network investment beyond 2030 delivering a coordinated offshore and onshore network design, supported by targeted upgrades to local networks and power flow control. High capacity offshore

high voltage direct current (HVDC) links off the eastern and western coasts form the backbone of the system, enabling large volumes of renewable generation to move from North Eastern Scotland into Northern England, the English Midlands and further south to major demand centres. These are complemented by strategic onshore reinforcements across northern England, North Wales, the Midlands and Southern England, creating a more interconnected and flexible transmission system.”

EXHIBIT 44: **The UK's National Energy System Operator is advocating a significant further development of the UK's electricity networks beyond 2030**



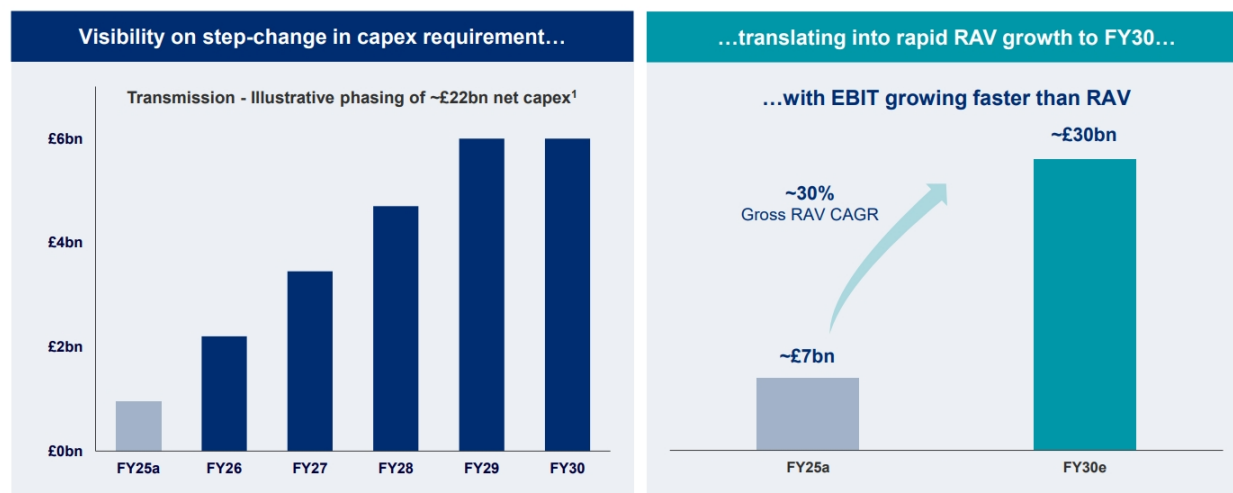
Source: NESO

The UK's investment needs translate into a strong growth outlook for its transmission owners

We would also highlight an abundance of regulated asset investment opportunities in the UK across electricity networks. At the most striking end of the spectrum, we would highlight SSE's ~30% RAV CAGR from March 2025 to March 2030, as shown in Exhibit 45. However, we would note strong growth across all 3 transmission system operators.

EXHIBIT 45: **SSE's ~30% RAV CAGR from FY25 to FY30 makes it one of the fastest-growing electricity networks globally**

Transmission: RAV to grow at ~30% CAGR, one of the fastest growing electricity networks in the world



¹ Capital investment net of SSEN Transmission's 25% Minority Interest

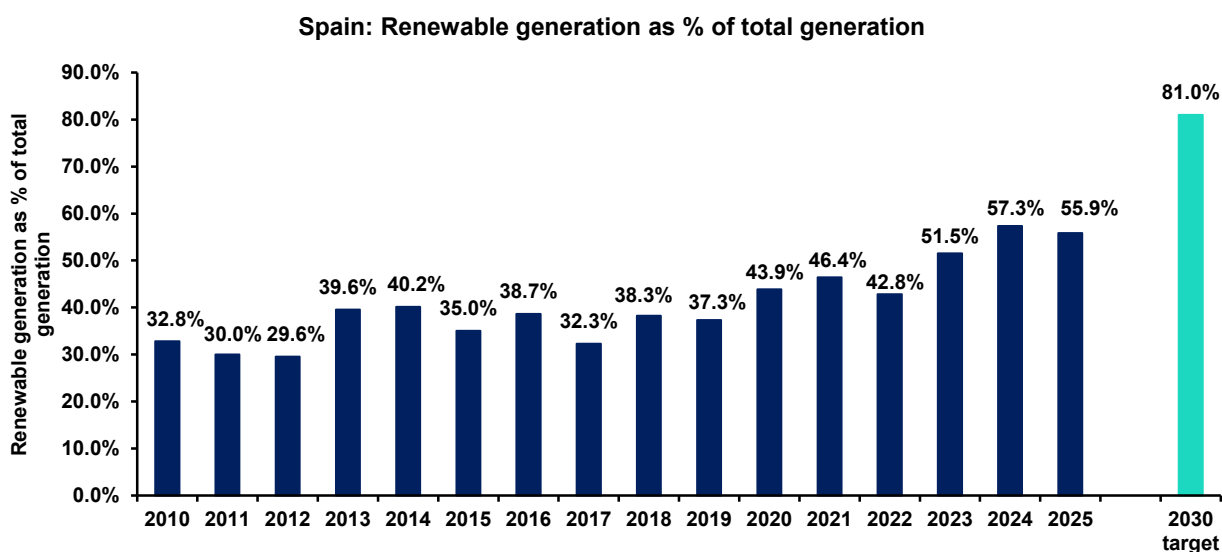


Source: SSE

C) SPAIN

Spain's [National Integrated Energy and Climate Plan](#) targets generation of 81% of the country's electricity from renewable sources by 2030, a goal that will require significant investment in grid infrastructure as a key enabler. This target is one of the most ambitious of the major European countries and requires a significant step-up from 2025 - see Exhibit 46.

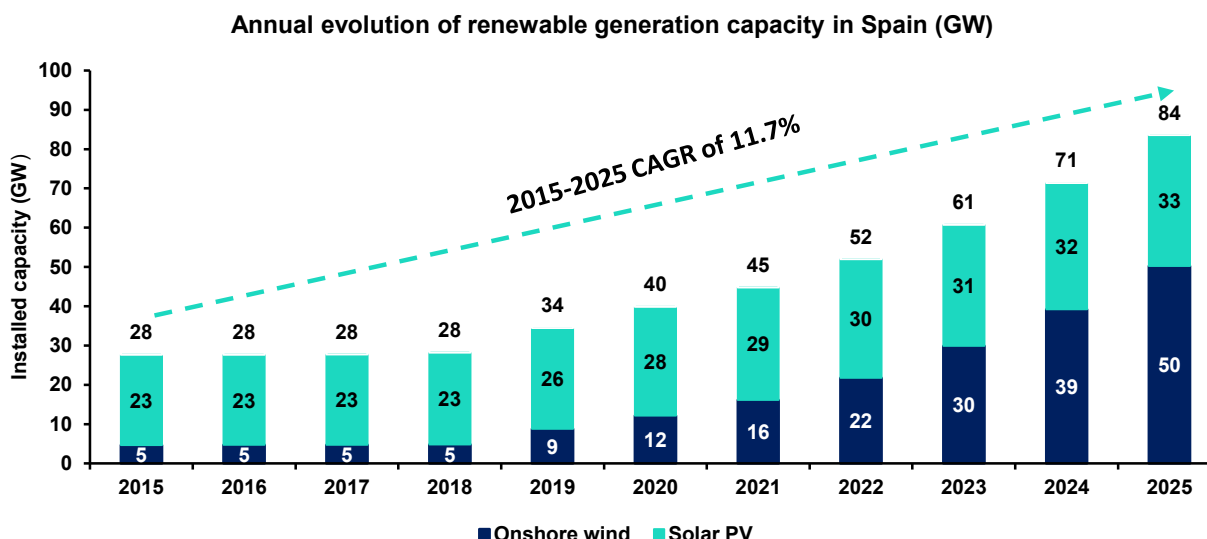
EXHIBIT 46: **Renewable generation as % of total generation in Spain**



Source: Ember, Spain Integrated National Energy and Climate Plan, Bernstein analysis

As set out in Exhibit 47, Spain has a strong track record of ramping up renewable generation capacity, which grew at a CAGR of ~12% from 2015 to 2025.

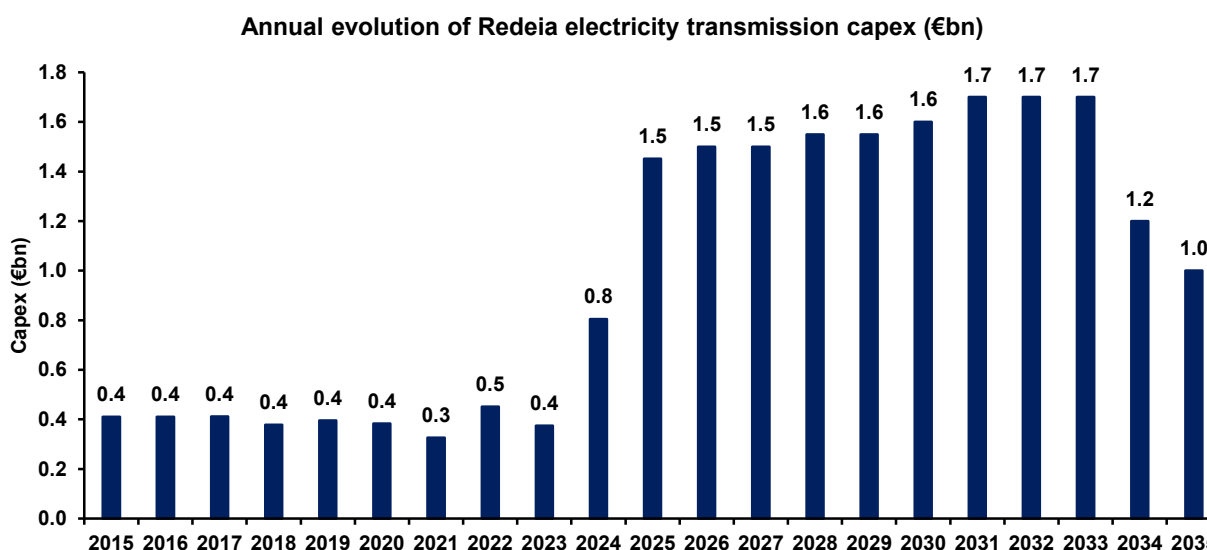
EXHIBIT 47: **Annual evolution of renewable electricity generation capacity in Spain**



Source: Red Elctrica Espana, Bernstein analysis

We set out our forecast annual evolution of Redeia's electricity transmission capex in Exhibit 48. Compared to the 2015-2023 norm, we see a significant step-up in investment for the period 2024-2035, peaking at ~€1.7bn per year in the early 2030s.

EXHIBIT 48: **Annual evolution of Redeia's electricity transmission capex (€bn)**



Source: Redeia, Bernstein analysis and estimates

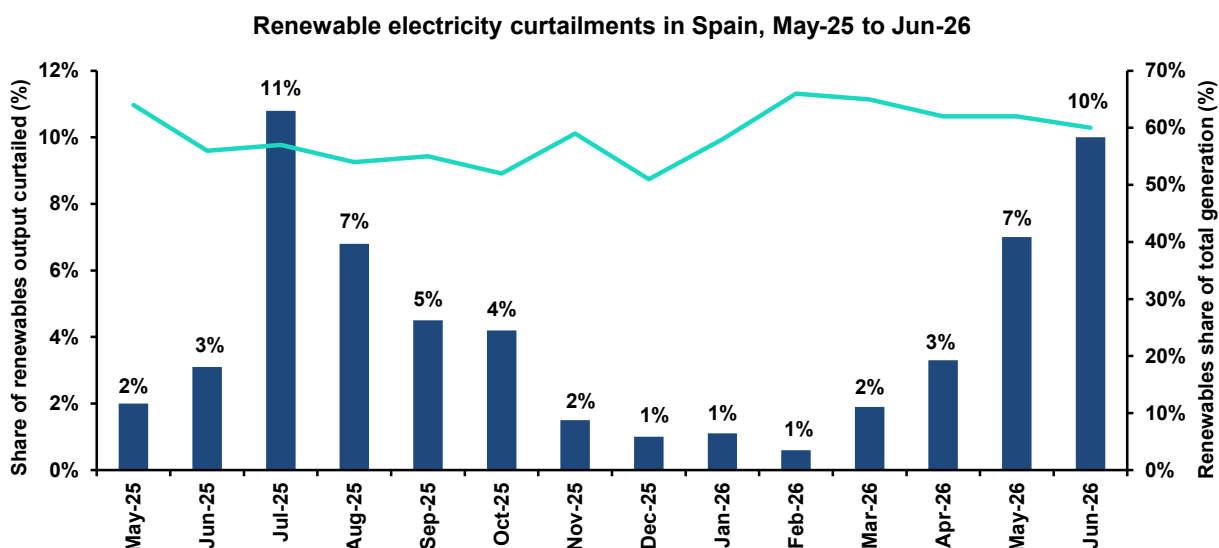
As with other European countries, alleviating significant renewable electricity constraint costs is an important investment driver in Spain

One idiosyncratic consideration in Spain is that the country has previously imposed caps on the level of annual investment in electricity grids, a legacy of 2013 reforms designed to alleviate a €30bn tariff deficit that had accumulated since 2001. Annual electricity transmission investment in Spain was previously capped at 0.065% of GDP and annual electricity distribution

investment was capped at 0.13%. These restrictions were temporarily relaxed in 2025 (increasing by >60%) in order to facilitate the energy transition, after contributing to a legacy of underinvestment. According to Bloomberg, during the 5 years from 2019-24, Spain had the lowest grid spending in Europe, investing €0.30 on grids for every €1 invested in renewable, significantly lower than the European average of €0.70. Following the removal of the investment caps, the current expectation of capex in the distribution grids would reach ~€8bn and ~€14bn for transmission grids by 2030.

The grid-vs-renewables investment shortfall in Spain has contributed to frequent output curtailment, as seen in Exhibit 49. Spain has seen curtailment of renewable output as high as 10-11% of the total in the summer months.

EXHIBIT 49: **Monthly evolution of renewable electricity curtailments in Spain, May-25 to Jun-26**

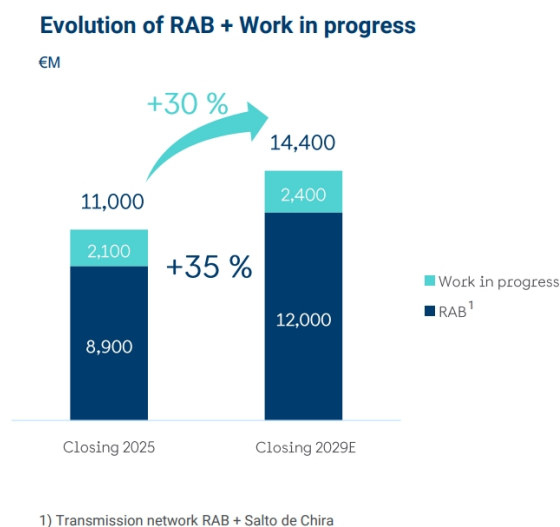


Source: Red Elctrica de Espana, Bernstein analysis

Spain likely sits mid-pack for electricity network investment

In our view, Spain likely sits mid-pack for investment prospects in European electricity grids. At its [2026-2029 Strategic Plan Update](#) in February 2026, Redeia set out expectations of a 2025-2029 RAB CAGR of ~7% - see Exhibit 50).

EXHIBIT 50: Redeia is contemplating ~30% growth in its Spanish electricity transmission RAB between 2025 and 2029, a CAGR of ~7%



Main projects commissioned by 2029

- Submarine **HVDC interconnection Bay of Biscay**.
- **New 400 kV alternating current (AC) 400 kV axis:**
 - Andalusia-Castilla la Mancha
 - Andalusia-Murcia-Castilla la Mancha-Aragón
 - Andalusia-Extremadura
 - Castilla la Mancha
 - Basque Country-Navarre
- **Synchronous compensators** in the Peninsula, Balearic and Canary Island systems.
- **Salto de Chira**.

Source: Redeia

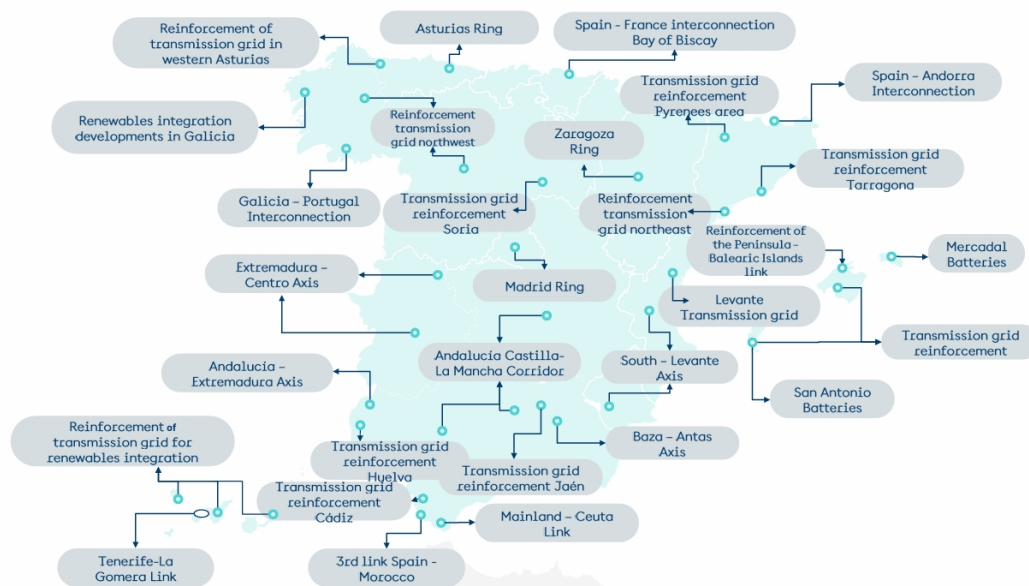
We set out the principal investment projects falling under Redeia's 2025-2030 capital plan in Exhibit 51.

EXHIBIT 51: Redeia's principal investment projects under its 2025-2030 capital plan

Proposed National Transmission Plan 2025-2030

Main actions

redeaia

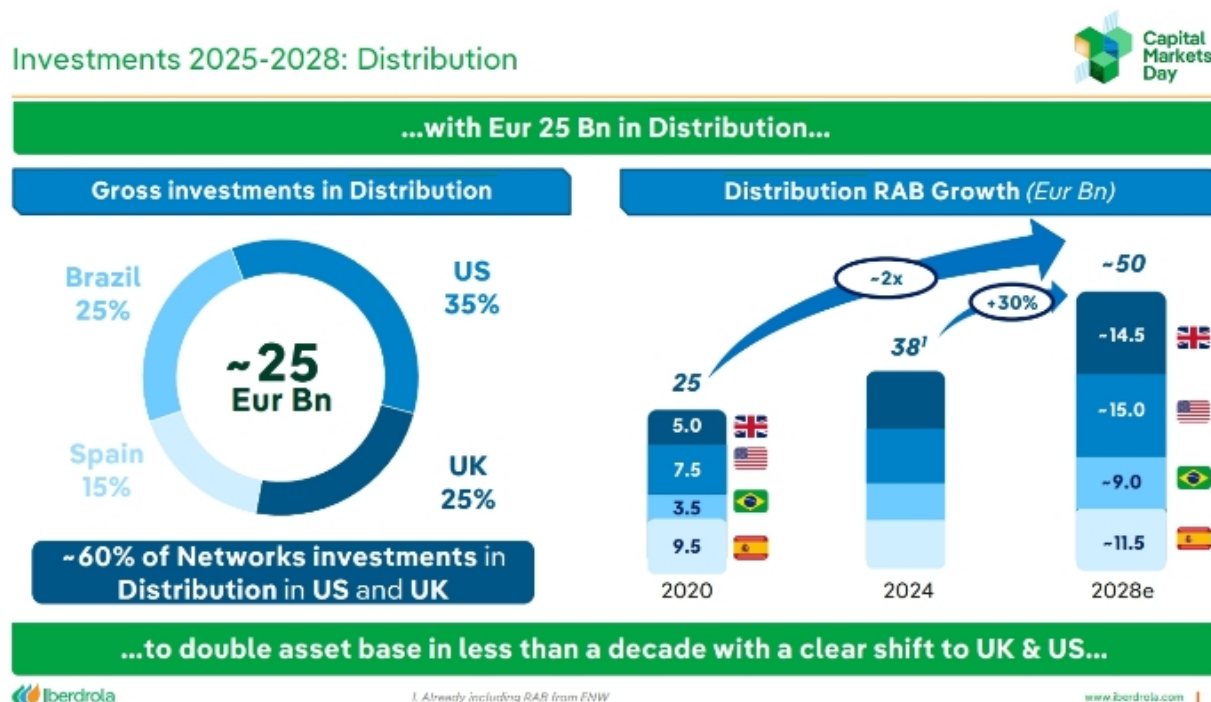


Source: Redeia

Distribution grids are also participating in growth, but on a lower trajectory

The Spanish distribution grid has a well-documented investment need. AELEC, the [Spanish Association of Electricity Companies](#), has argued that [more than 80%](#) of the nodes in the country's distribution grid are saturated, whilst Endesa reported that in 2025, [more than 50% of grid access requests were rejected](#) due to insufficient capacity. Nonetheless, investment expectations in Spanish electricity distribution are relatively modest. As seen in Exhibit 52, Iberdrola have [guided to](#) a ~2.5% electricity distribution RAB CAGR in Spain from 2020 to 2028, making the country the slowest-growing of its 4 major regions.

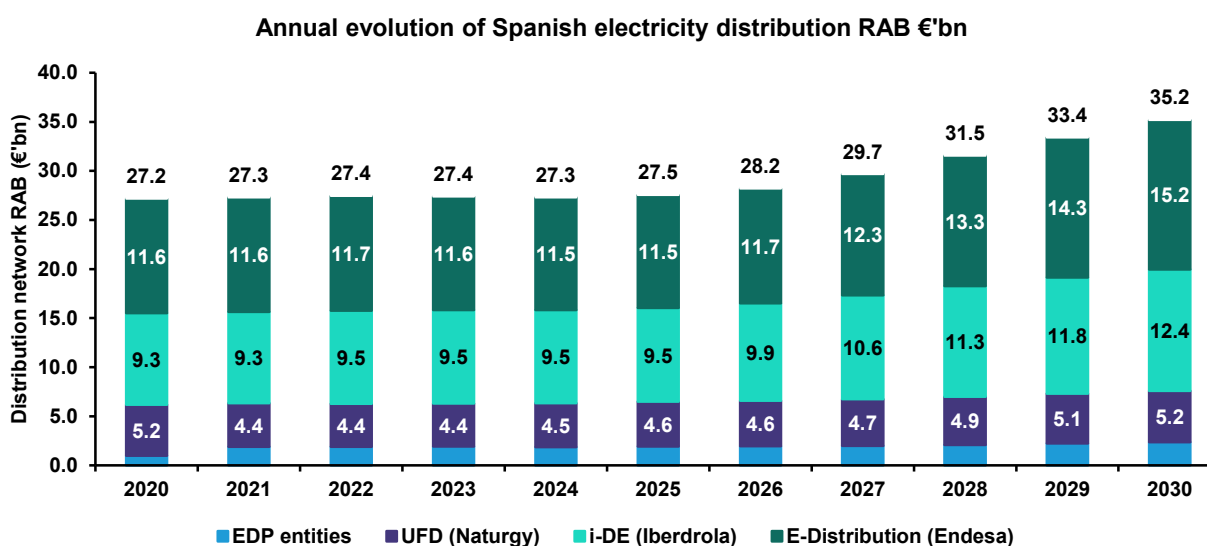
EXHIBIT 52: Iberdrola have guided to a ~2.5% electricity distribution RAB CAGR in Spain from 2020 to 2028



Source: Iberdrola

We set out our forecast evolution of the aggregate Spanish electricity distribution RAB across our coverage cohort in Exhibit 53.

EXHIBIT 53: Annual evolution of electricity distribution RAB in Spain



Source: EDP, Naturgy, Iberdrola, Endesa, Bernstein analysis and estimates

Like the UK, Spain also has a relatively concentrated ownership base within its electricity distribution network. The majority of customers are served by entities owned by 3 companies: Endesa, Naturgy and Iberdrola. We set out an indicative ownership map of the network in Exhibit 54.

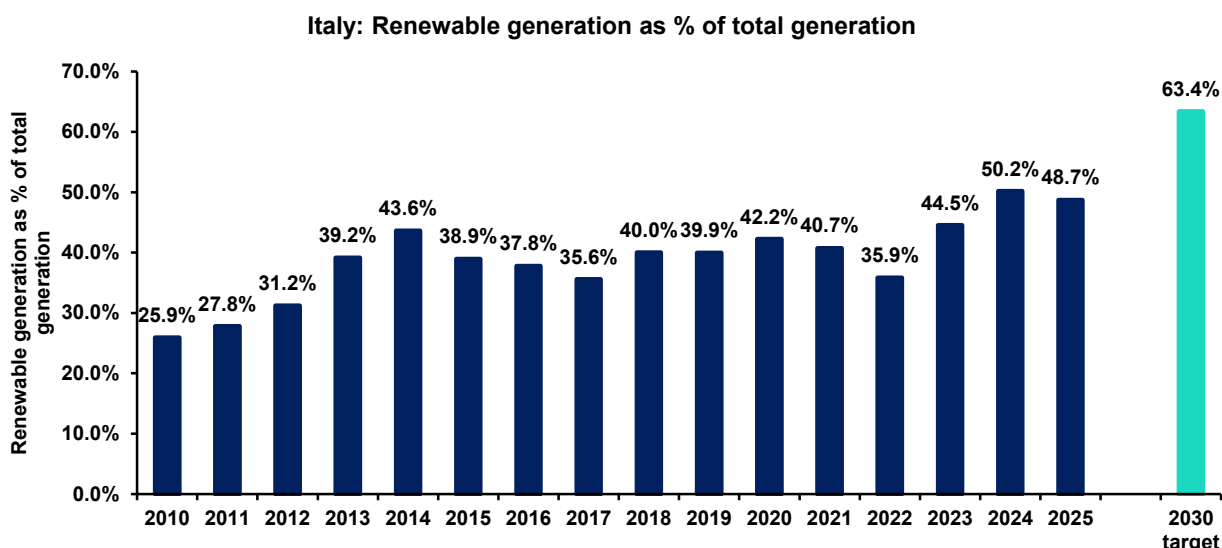
EXHIBIT 54: **Ownership map of Spanish electricity distribution companies**



Source: Podo, Bernstein analysis

D) ITALY

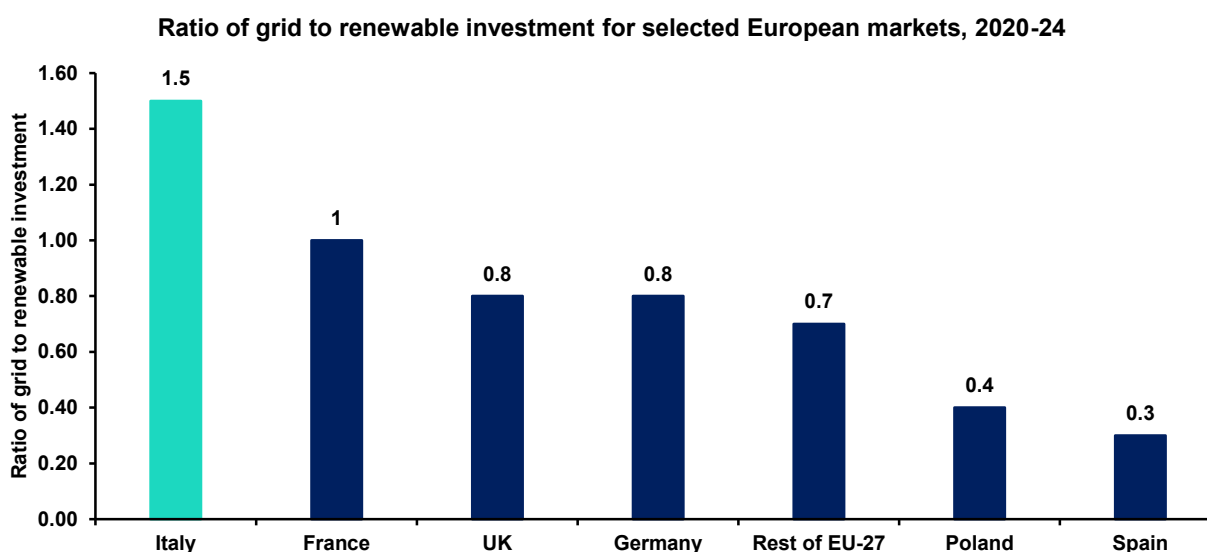
Italy is targeting a [63.4%](#) share of renewables in the electricity sector by 2030, up from ~49% in 2025 - see Exhibit 55. The country is pushing ahead with the required renewable roll-out; on 8 June 2026, the country [received approval](#) from the European Commission for €23bn of state funding for renewables. The funding is expected to support ~37GW of capacity under two-way CFDs. The expected new capacity is equivalent to ~48% of the current installed base.

EXHIBIT 55: Italy is targeting ~63% of electricity generation from renewable sources by 2030, vs ~49% in 2025

Source: Ember, Camera dei Deputati - Servizio Studi, Bernstein analysis

As with other countries, Italy needs grid investment to support the renewable roll out

Despite one of the highest ratios of grid to renewable investment in Europe over 2020-24 (see Exhibit 56), Italy still faces a significant investment need within its electricity grids as a result of its renewables push, reflecting the current significant mismatch between the locations of electricity demand and supply centres. This mismatch is currently manifested in significant network congestion and curtailed renewable generation volumes. As the IEEFA explain, “Northern Italy has a high concentration of industrial demand and historically hosted most of the thermal and reservoir hydro generation, while southern regions and islands now host a disproportionate share of renewable capacity, particularly solar and wind.”

EXHIBIT 56: Ratio of grid to renewable investment for selected European markets, 2020-24

Source: Bloomberg New Energy Finance, Bernstein analysis

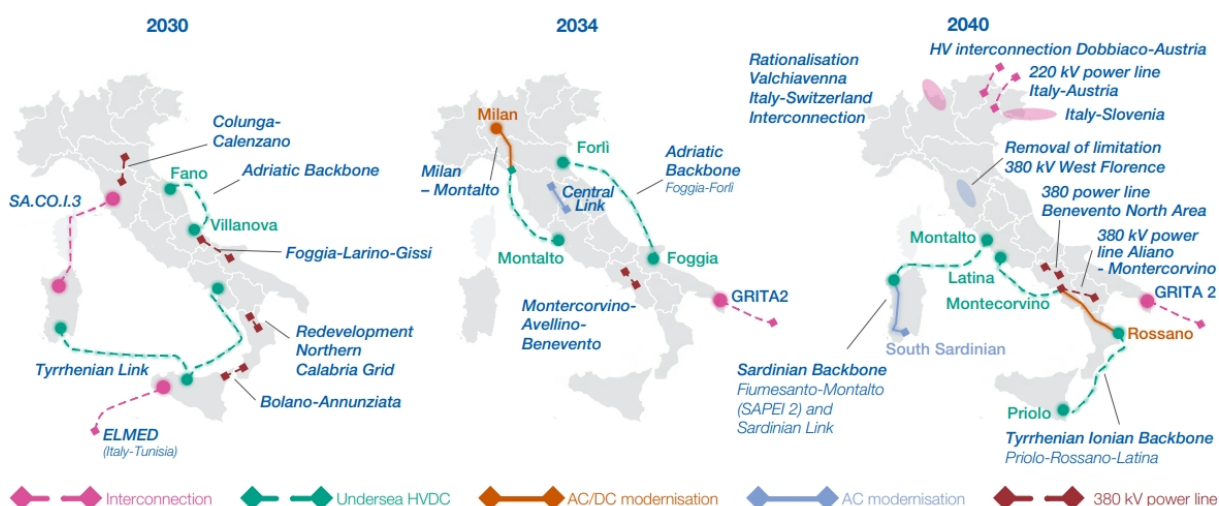
The Italian 10-year electricity transmission network development plan, published on 14 March 2025 (see our note here: [Terna: 10 year development plan supports long-term RAB growth](#)), outlines planned investments exceeding €23bn over 2025-2034. The total potential value of projects including those beyond 2034 is up to c.€40bn. This is up 10% versus the

previous 2023-2032 plan, which allocated €21bn, in turn up nearly 30% compared to the 2021 plan's €18.1bn. Prior to that, the 2020 plan anticipated investment of €14.4bn.

The 2025 Development Plan is aligned with the targets defined by the 2024 Integrated National Energy and Climate Plan (NECP), as detailed in the Terna-Snam 2024 Scenario Description Document, which foresees an increase of over 65GW in installed solar and wind capacity by 2030 and of 94GW by 2035, both versus installed capacity in 2023. Terna says that the plan will significantly enhance the capacity for energy exchange between market zones and aims to increase cross-border transfer capacity by c.40%. As of end-FY24, connection requests for renewable power plants totaled 348GW—comprising 152GW for solar, 110GW for onshore wind, and 86GW for offshore wind—while requests for storage system connections reached 277 GW, according to Terna data.

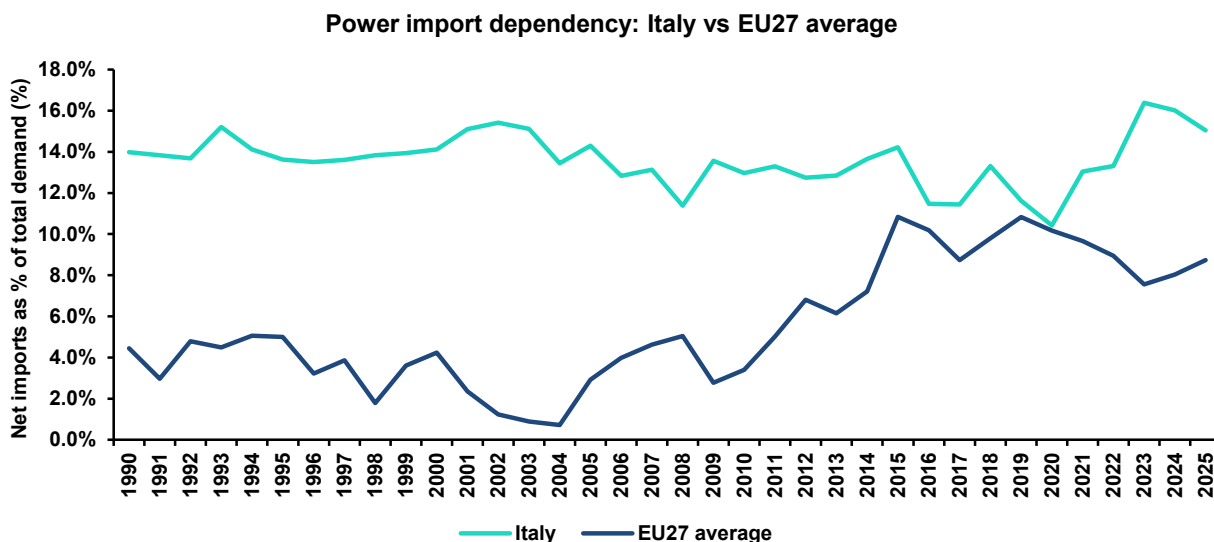
We set out the stages of the [Italian electricity transmission network development plan](#) in Exhibit 57.

EXHIBIT 57: Medium-term electricity transmission network planning in Italy



Source: Terna

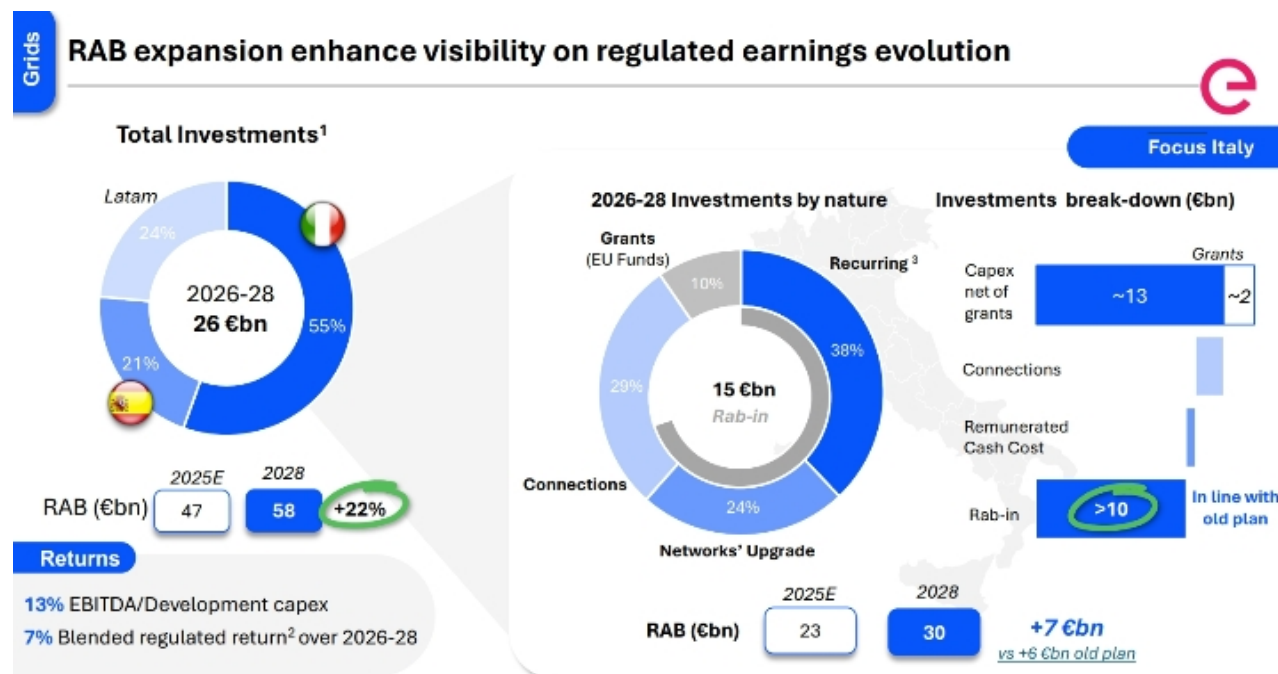
Alongside decarbonization ambitions, Italy's renewable electricity push also reflects the strategic imperative of achieving greater energy independence. Italy imported ~75% of its energy needs in 2023 and according to the Institute for Energy Economics and Financial Analysis, is the most import dependent country in Western Europe for electricity, importing ~14% of total domestic consumption in 2024.

EXHIBIT 58: **Italy is a long-standing significant net importer of electricity**

Source: Ember, Bernstein analysis

Italy likely sits mid-pack in Europe for electricity network investment potential

Italy sits in the middle of the pack in terms of its investment opportunity for regulated assets. In electricity transmission, system operator Terna [guided to](#) a ~9% RAB CAGR from 2024-28 in its March 2025 update, whilst in February 2026, electricity distribution grid operator Enel [indicated](#) that its distribution RAB would also grow at a ~9% CAGR from 2025 to 2028.

EXHIBIT 59: **Enel is guiding to an electricity distribution RAB CAGR of ~9% from 2025 to 2028**

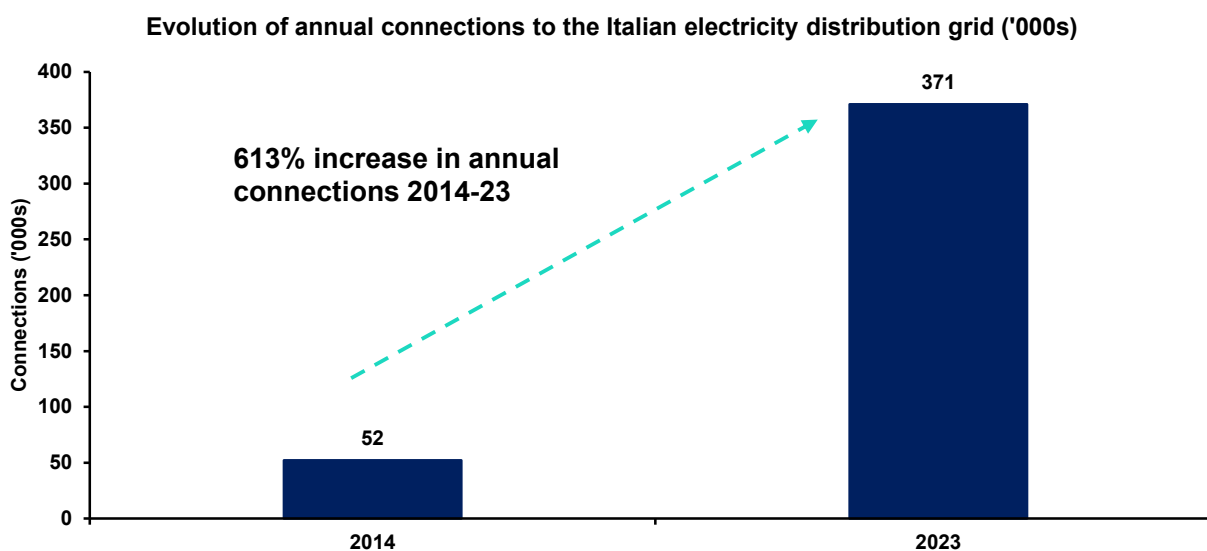
Source: Enel

Distribution is a key enabler

The distribution network in Italy is [operated by ~130 DSOs](#) but is dominated by E-Distribuzione, a subsidiary of the Italian integrated utility Enel, which serves ~85% of customers. The distribution network is currently acting as a partial inhibitor of renewable penetration in Italy, with ~53GW of solar capacity currently in the connection queue alongside ~9GW of battery capacity and a negligible quantity of wind capacity. These figures represent ~67%, ~50% and ~7% of the 2030 capacity target respectively.

Enel [explains](#) that “between 70% and 75% of additional RES [Renewable Energy Sources] capacity to be installed by 2030 in Italy will be connected to the distribution grid.” As we set out in Exhibit 60, this follows a 7-fold increase in annual new connections to the Italian distribution grid between 2014 and 2023.

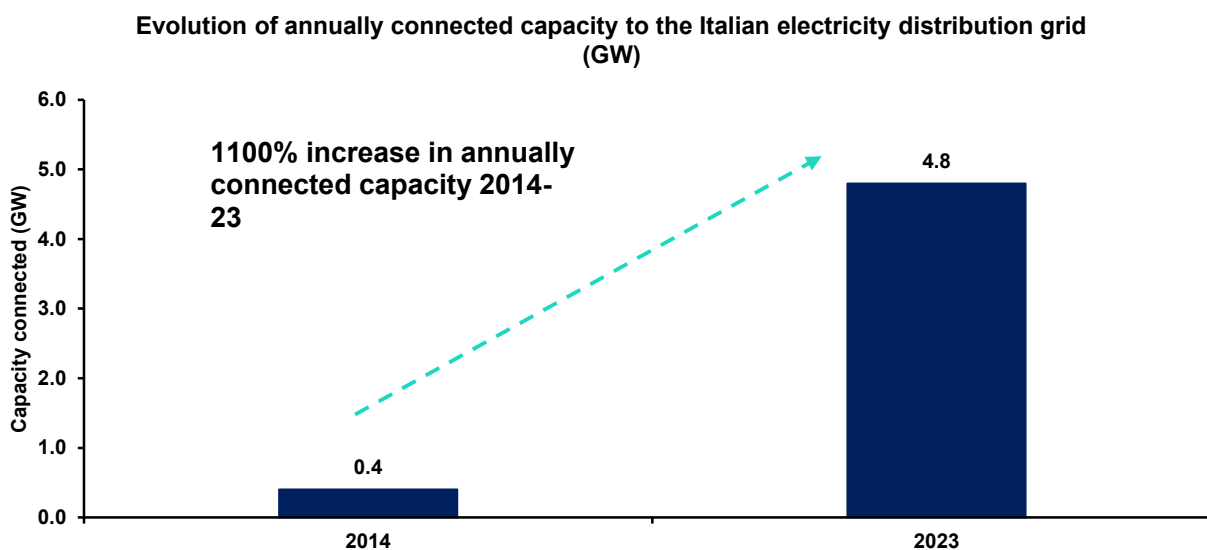
EXHIBIT 60: **Italy saw a 7-fold increase in annual new connections to the distribution grid between 2014 and 2023**



Source: Enel, TEHA Group on GSE "Rapporto Statistico 2023" data, Bernstein analysis

The period from 2014-2023 saw a 12-fold increase in annual new generation capacity connections to the distribution grid - see Exhibit 61.

EXHIBIT 61: This was accompanied by a 12-fold increase in annual new generation capacity connections to the distribution grid



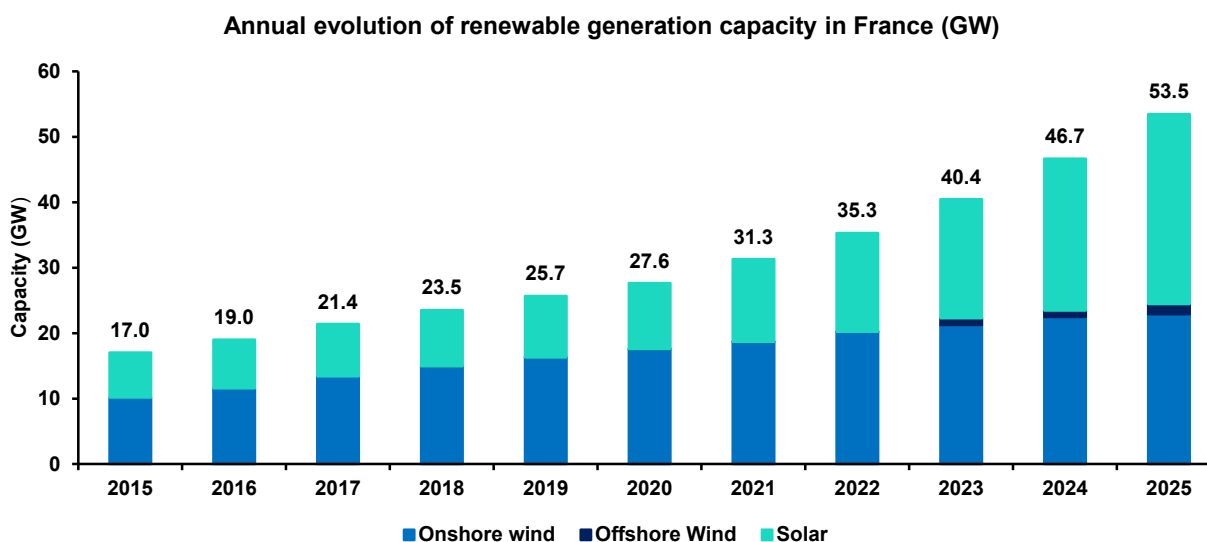
Source: Enel, TEHA Group on GSE "Rapporto Statistico 2023" data, Bernstein analysis

E) FRANCE

France is currently pursuing a multi-faceted energy agenda, as set out in the [3rd edition of its Multi-Year Energy Program](#) (PPE3/ Programmation Pluriannuelle de L'Energie 3). This places decarbonizing the energy mix in France alongside the other strategic priorities of energy sovereignty and affordability.

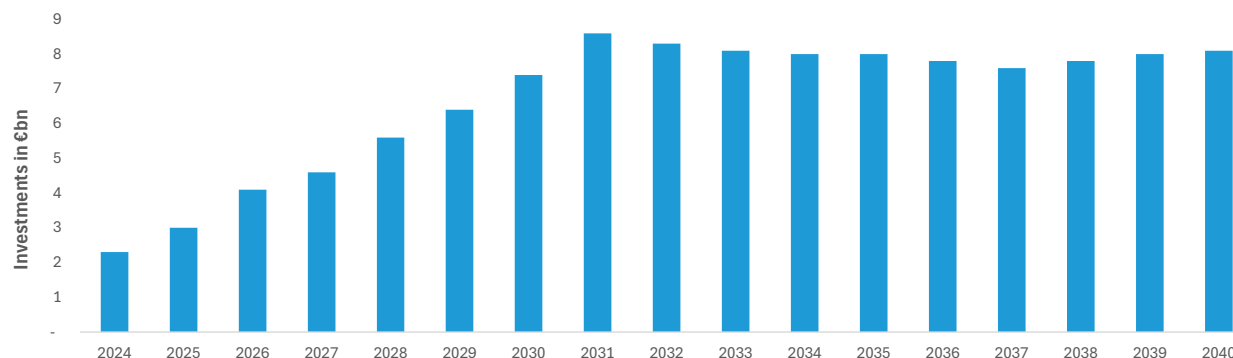
In service of these objectives, the French government is proposing some ambitious targets, including 6 new nuclear reactors, 15GW of offshore wind capacity by 2035, and a tripling of solar PV capacity by 2035. These ambitions clearly have implications for electricity networks investment.

France has already made significant progress in increasing renewable electricity generation capacity, as set out in Exhibit 62.

EXHIBIT 62: **Annual evolution of renewable generation capacity in France, 2015-2025**

Source: RTE, Bernstein analysis

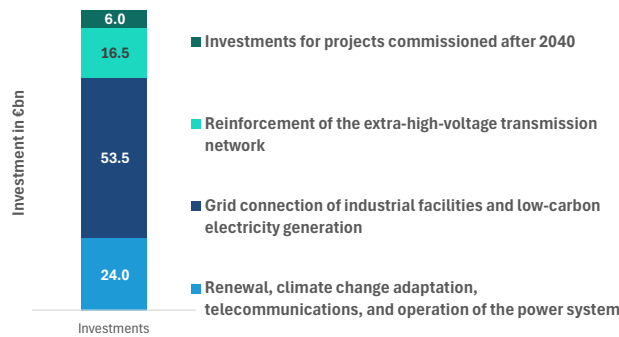
In 2025, French electricity transmission system operator (TSO) RTE [published](#) its Schéma Décennal de Développement du Réseau (SDDR), outlining approximately €100bn of investments over the next 15 years (through 2040) - see Exhibit 63.

EXHIBIT 63: **Investments in transmission grid**

Source: RTE

The plan includes €24bn for the renewal and climate adaptation of aging network infrastructure - including 23,500 km of transmission lines and 85,000 pylons across the country; €53bn for the grid connection of industrial facilities and low-carbon power generation: €16.5bn for strengthening the extra-high-voltage transmission network; and €6bn for projects scheduled to enter service beyond 2040. Two possible representations of the investment trajectory proposed in the SDDR are presented below.

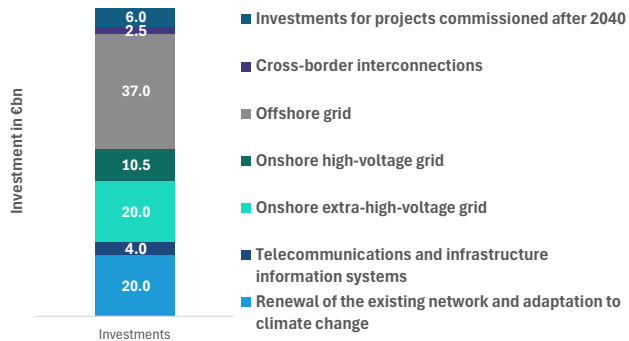
EXHIBIT 64: RTE investments by 2040



Source: RTE

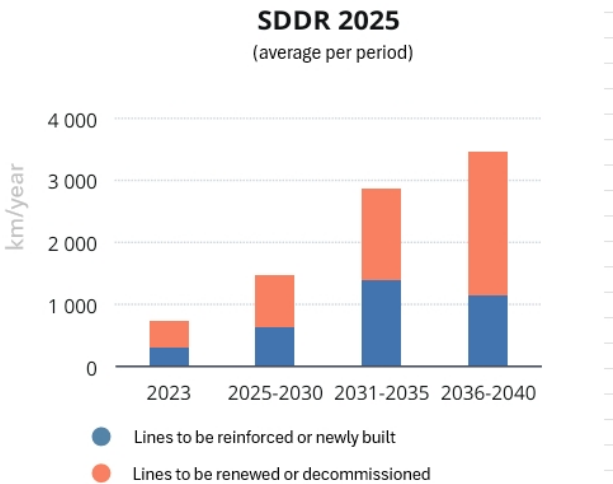
RTE estimates that approximately 40,000 km of transmission lines will need to be renewed, decommissioned, reinforced, or commissioned by 2040. Around 60% relates to the renewal or decommissioning of existing assets, while the remaining 40% reflects network reinforcement and expansion. RTE also anticipates a gradual ramp-up in activity, with annual volumes doubling from current levels over 2025–2030 and doubling again over 2031–2035.

EXHIBIT 65: RTE investments by 2040



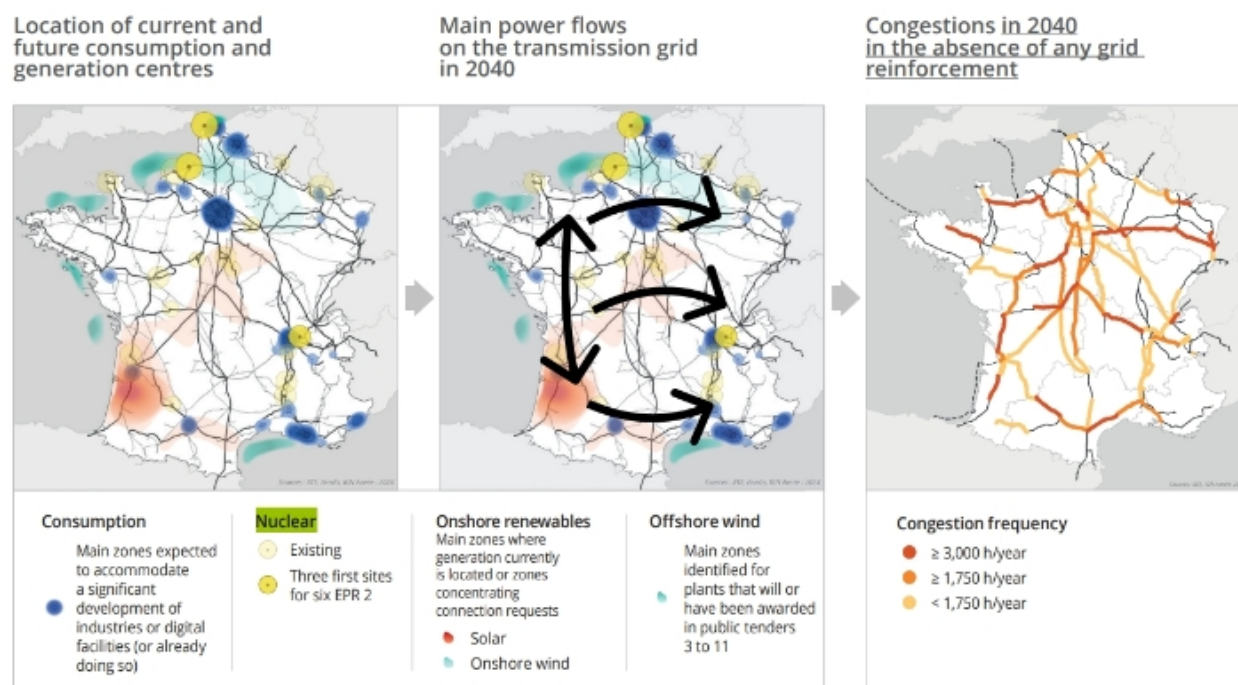
Source: RTE

EXHIBIT 66: Industrial needs in the SDDR reference scenario



Source: RTE

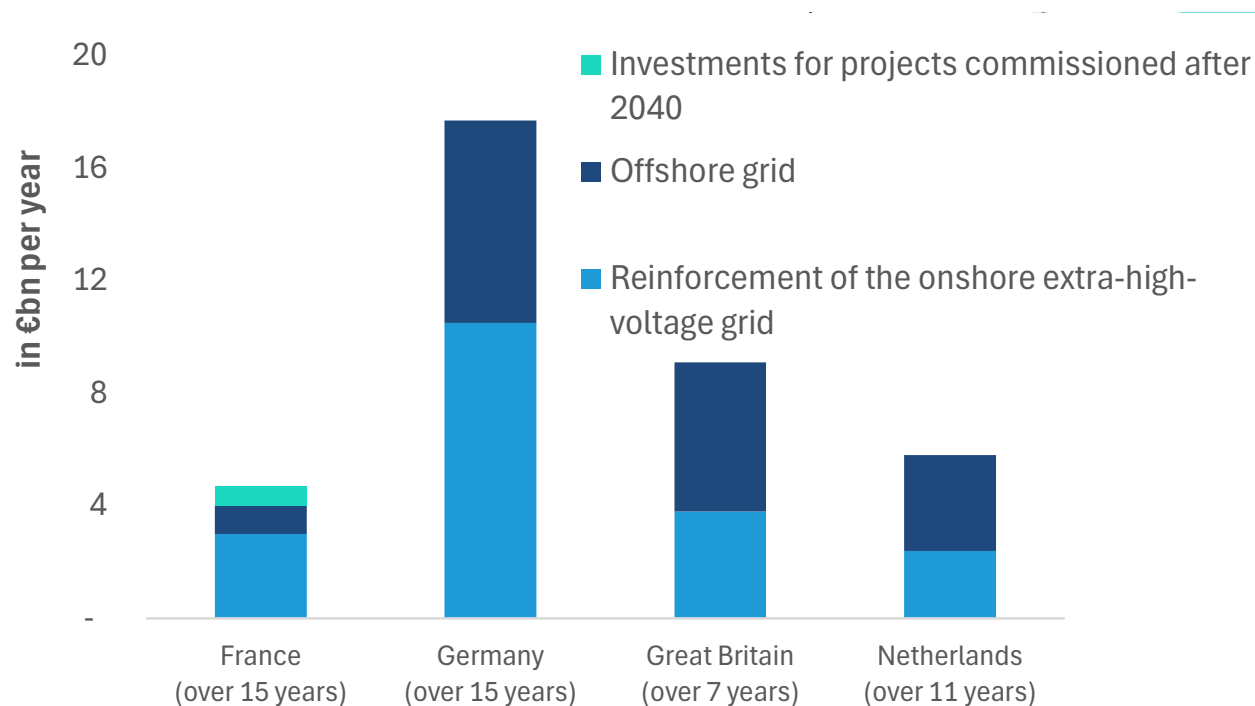
The SDDR states that “by 2040, the geography of the major drivers for the French power system (generation and consumption) will evolve significantly, modifying flows on the grid. In particular, new generation assets will be concentrated mainly in the west of the country and on the coast (EPR2 at Penly and Gravelines before 2040, offshore wind mainly off the Normandie, Bretagne and Charente coasts, solar already being deployed more rapidly in Nouvelle Aquitaine and Occitanie). New flow patterns will therefore be added to the current flows as a result of this new geography, and the additional generation will flow towards the major consumption centres in France and Eastern Europe...in France, without grid reinforcement, annual redispatching costs could reach €3bn/year by 2035 (compared with around €150m in 2024).

EXHIBIT 67: **France's changing electricity generation mix will place new demands on its transmission infrastructure**

Source: RTE

As set out in Exhibit 68, whilst clearly significant, France's proposed €100bn investment over 15 years is at the smaller end of a peer group including larger envelopes, e.g. Germany and the UK.

EXHIBIT 68: **Comparison of average annual investments in extra-high-voltage grid reinforcement across Europe (onshore and offshore)**



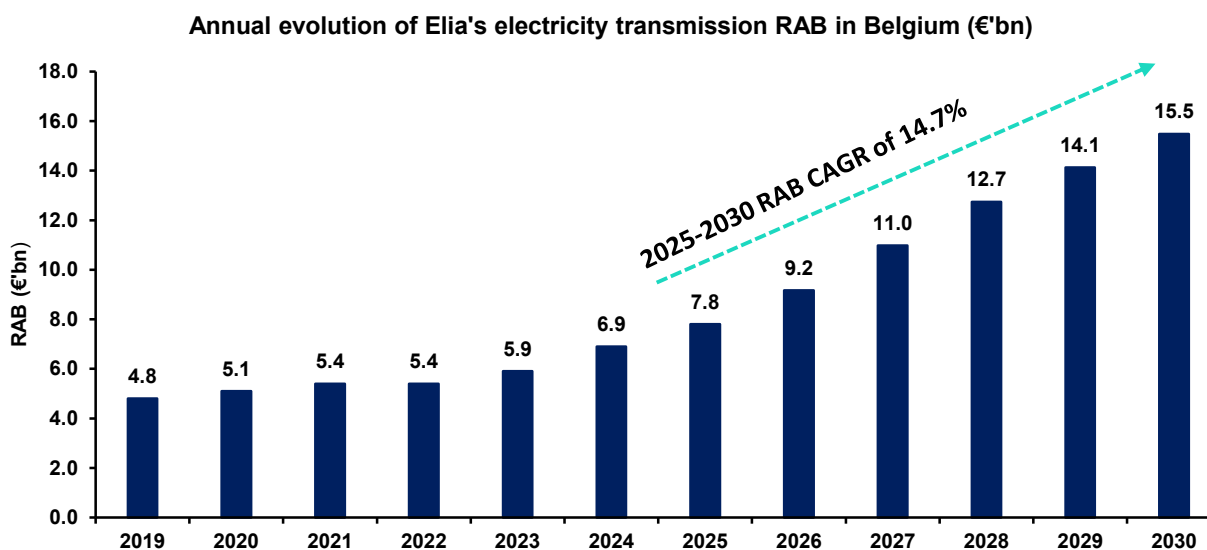
Source: RTE

On the distribution side of the network, French distribution operator Enedis [estimates](#) that its investments will increase from €5.7bn in 2025 to €6.8bn in 2030. In 2025 the company [publicly contemplated](#) a €96bn investment program over 2022-2040.

F) BELGIUM

We see a very strong growth outlook for electricity networks in Belgium, specifically in electricity transmission. The transmission system operator, Elia, published in 2023 its [Federal Development Plan](#) for the Belgian transmission system for the period 2024-2034, highlighting significant growth opportunities relating to electrification and renewable capacity integration. At its Capital Markets Day in the same year, Elia [set out expectations](#) for an ~18% RAB CAGR in Belgium from 2023-2028. We cover the significant growth opportunities available to Elia in Belgium in more detail in this note: [Elia: Growth, growth, growth](#), but in summary we see onshore wind capacity growing from 3.6GW in operation in 2026 to ~6GW by 2035, with offshore wind capacity growing from 2.3GW in 2026 to 5.8GW by 2030. This sits alongside significant interconnection needs, new battery connections and new gas plants as important growth drivers for Elia.

We set out our forecast evolution of Elia's electricity transmission RAB over the period 2019-2030 in Exhibit 69.

EXHIBIT 69: **Annual evolution of Elia's Belgian electricity transmission RAB, 2019-2030**

Source: Elia, Bernstein analysis and estimates

G) PORTUGAL

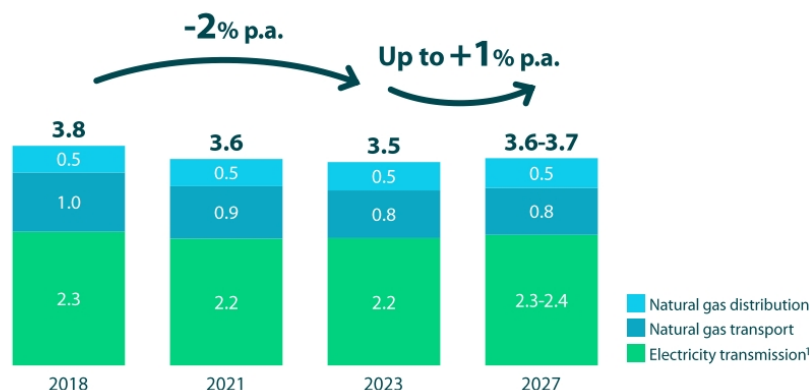
Overall, we see somewhat limited growth prospects for regulated utility networks in Portugal. At its 2024 [Capital Markets Day](#), Redes Energéticas Nacionais, the operator of the gas transmission, gas distribution and electricity transmission networks in Portugal, guided to low single digit RAB growth from 2023-27 - see Exhibit 70. EDP, which operates the electricity distribution network in Portugal, did however refer to a “significant increase in investments to support RAB growth” in its networks businesses in Spain and Portugal at its [November 2025 Capital Markets Day](#).

EXHIBIT 70: **REN guided to low single digit RAB growth in its regulated Portuguese assets at its 2024 Capital Markets Day**

CAPITAL
MARKETS
DAY 2024

We see our domestic RAB following the Energy Transition acceleration, with the 2024-27 cycle as an inflection point

Average RAB at reference costs, B€



Uptake in renewables capacity and impending green gases acceleration to drive significant investment growth in electricity and gas infrastructure, leading to up to +1% p.a. RAB growth during the 2023-27 period

1. Includes hydro land assets and Enonadas

REN

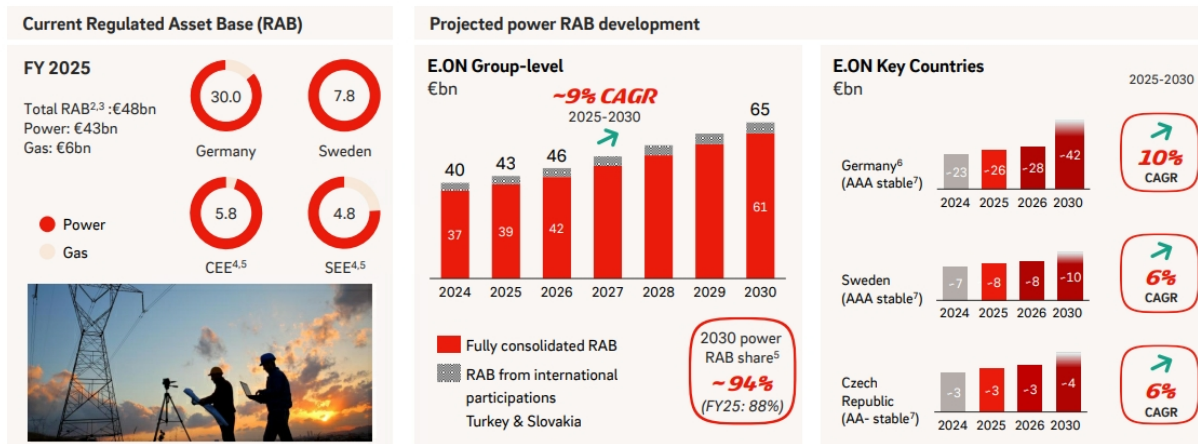
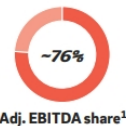
Source: REN

H) SWEDEN

The investment opportunity for public markets investors in Swedish electricity networks is restricted to electricity distribution, with operators in the remaining segments either state owned or privately held. We see a moderately attractive growth opportunity in electricity distribution, with E.ON guiding to a RAB CAGR of ~6% from 2025-2030.

EXHIBIT 71: **E.ON is guiding to an electricity distribution RAB CAGR of ~6% in Sweden from 2025-2030**

We offer an unprecedented regulated growth trajectory with a clear strategic focus on power infrastructure



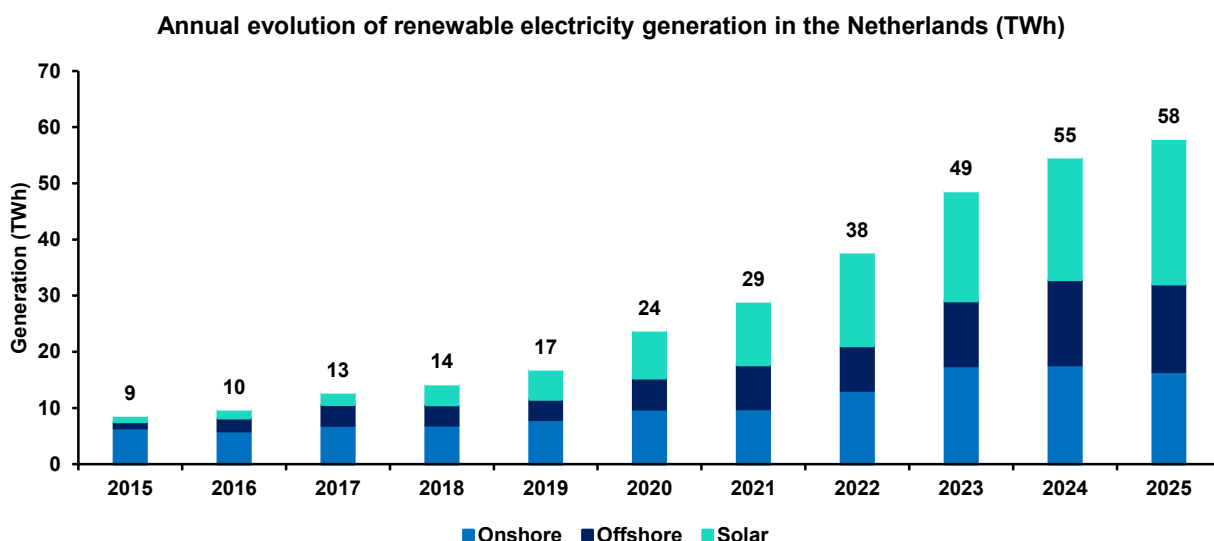
(1) Average Adj. EBITDA share over 2026-2030 plan excluding corporate functions. (2) RAB is the value of all distribution assets determined by the regulator. In general, RABs from different regulatory regimes are not directly comparable due to significant differences in regulatory calculation methods. These include for example different regulatory asset lifetimes, asset valuation methods or treatment of customer contributions for network connections. (3) As per 31-Dec-2025. (4) Central Eastern Europe including Czech Republic, Poland, Slovakia. South Eastern Europe including Hungary, Croatia, Romania, Turkey. (5) 100% view for Slovakia and Turkey. (6) Assuming constant number of network concessions. (7) Credit Rating from Standard & Poor's.

Capital Markets Story |

Source: E.ON

I) NETHERLANDS

The Netherlands have seen very significant growth in renewable electricity generation over the last decade, with combined output across onshore and offshore wind plus solar increasing almost 7-fold between 2015 and 2025 - see Exhibit 72.

EXHIBIT 72: **Annual evolution of renewable electricity generation in the Netherlands**

Source: StatLine, Bernstein analysis

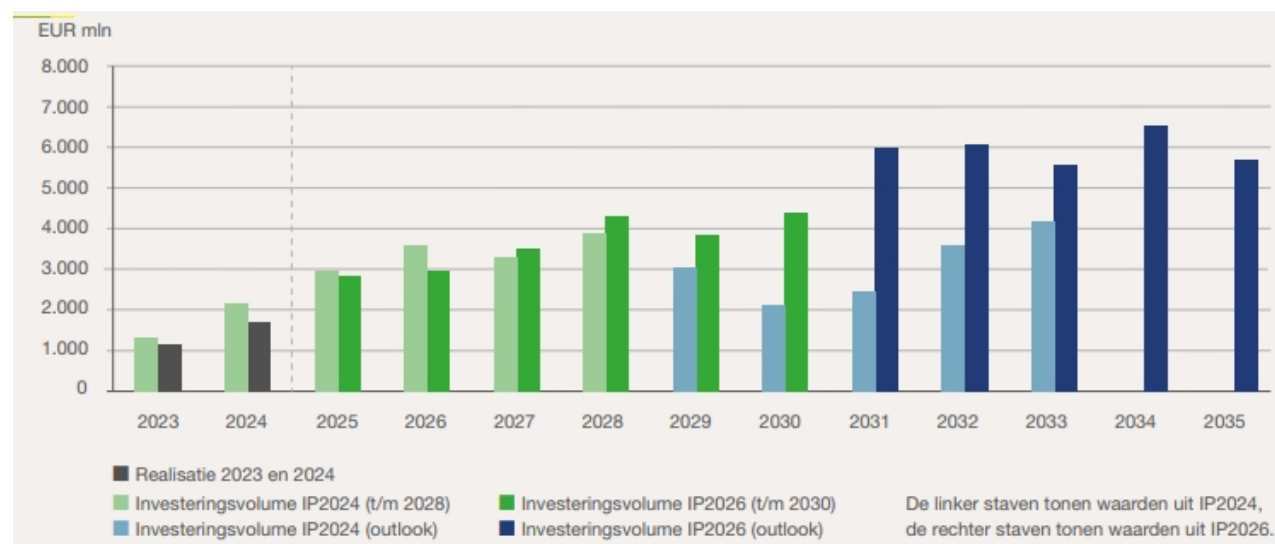
Amongst other factors, however, the renewable generation capacity roll-out has placed strain on Dutch grids, which now suffer from material congestion. The *Financial Times* reported in [July 2025](#) that access to electricity was being restricted in order to manage the stress placed on the grid, with a spokesman for Tennet Netherlands, the Dutch Transmission System Operator,

reportedly explaining that the country's historical reliance on natural gas had driven complacency in rolling out the necessary grid upgrades.

Tennet Netherlands published its [investment plan for the period 2026-2040](#) on 23 April 2026. For the onshore grid, the plan contemplates up to ~€50bn of investment over the period 2026 to 2035, as shown in Exhibit 73 and Exhibit 74.

There is also an acute need for more investment into electricity grids in the Netherlands. The Netherlands has rolled out significant renewable generation capacity over the past decade and now boasts, variously, 1) [the highest number of EV charging points per capita in Europe](#), 2) the highest number of solar panels per capita (almost 1/3 of Dutch homes have panels installed), and 3) a ~45% renewable generation share. However, the IEA [recently described](#) the Netherlands as "a striking example of a country where grid congestion has become a major bottleneck to the energy transition, challenging climate targets as well as energy security and affordability."

EXHIBIT 73: **Onshore grid investment by year**



"Realisatie 2023 en 2024" = "Actuals 2023 and 2024"

"Investeringsvolume IP2024 (t/m 2028)" = "Investment volume IP2024 (up to and including 2028)"

"Investeringsvolume IP2024 (outlook)" = "Investment volume IP2024 (outlook)"

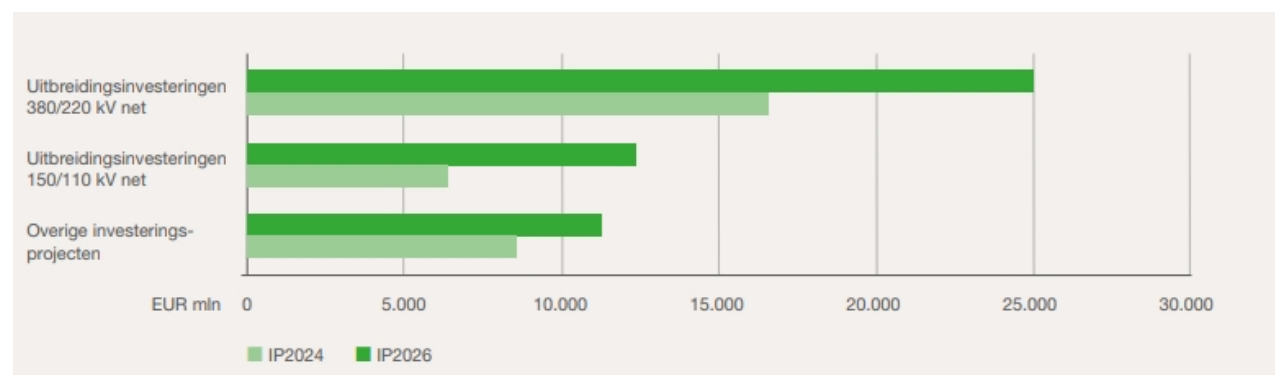
"Investeringsvolume IP2026 (t/m 2030)" = "Investment volume IP2026 (up to and including 2030)"

"Investeringsvolume IP2026 (outlook)" = "Investment volume IP2026 (outlook)"

"De linker staven tonen waarden uit IP2024, de rechter staven tonen waarden uit IP2026." = "The bars on the left show values from IP2024; the bars on the right show values from IP2026."

Source: Tennet Netherlands

We set out Tennet's investment categorisation by category in Exhibit 74.

EXHIBIT 74: **Onshore grid investment by category**

"Uitbreidingsinvesteringen 380/220 kV net" = "Expansion investments in the 380/220 kV grid"

"Uitbreidingsinvesteringen 150/110 kV net" = "Expansion investments in the 150/110 kV grid"

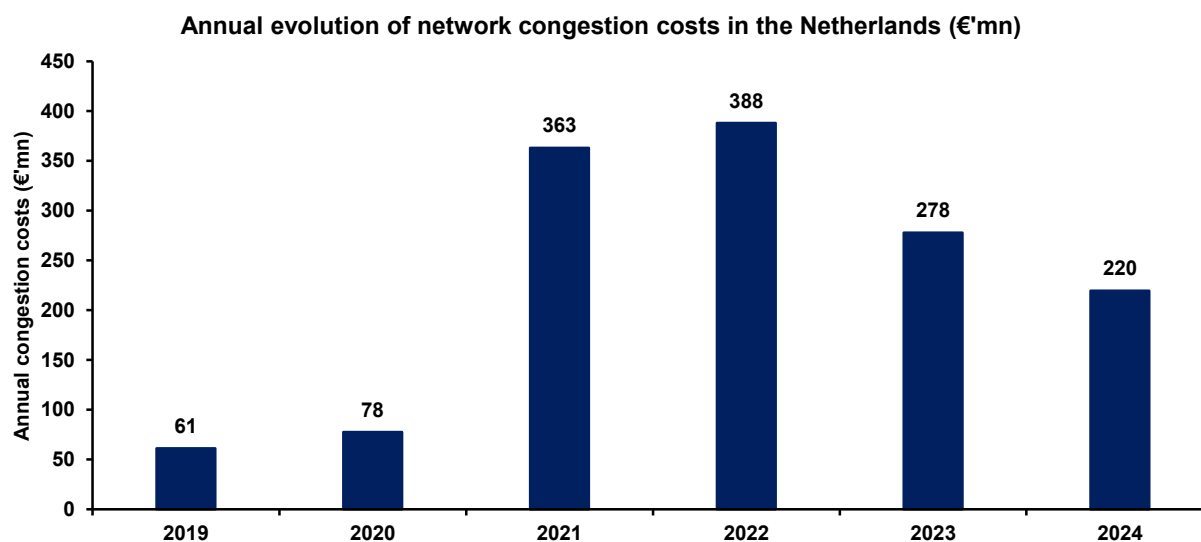
"Overige investeringsprojecten" = "Other investment projects"

"EUR mln" = "EUR million"

"IP2024" = "Investment Plan 2024"

"IP2026" = "Investment Plan 2026"

Source: TenneT Netherlands

EXHIBIT 75: **Annual evolution of electricity transmission network constraint costs in the Netherlands, 2019-24**

Source: TenneT, Bernstein analysis

3) ENERGY SOVEREIGNTY AND POWER MARKETS INTEGRITY

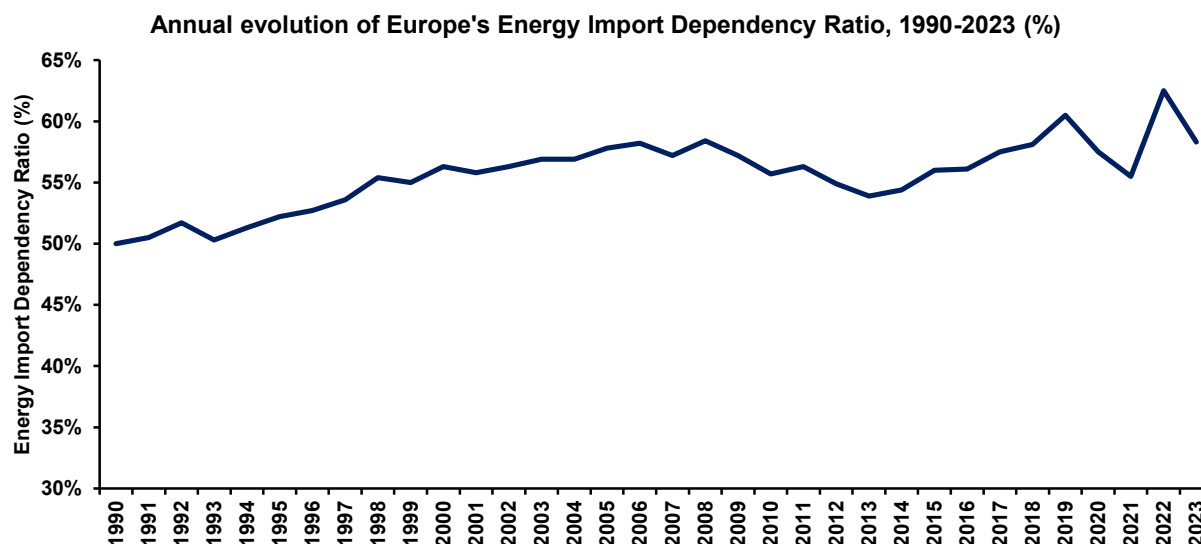
Energy sovereignty is a further key theme in the rise in European electricity grid investment. Europe is a significant net importer of energy; in 2024, net energy imports met [~57% of the EU's energy needs](#). Oil and petroleum products accounted for ~67% of energy imports and natural gas accounted for ~24%. As the continent seeks to reduce its import dependence, electrification is a key lever, contributing to the overall electricity grid investment supercycle.

A) ENERGY IMPORT DEPENDENCY IS A STRATEGIC WEAKNESS FOR EUROPE

Energy import dependence is frequently cited as a core strategic weakness for Europe. Frank Elderson, a member of the ECB's Executive Board, [explains](#) that “Europe’s energy dependence has become one of the critical vulnerabilities of our economy. Recent energy price shocks have transferred vast resources out of Europe, prompted emergency interventions and strained public finances. These costs are real, recurring and largely wasted.” Europe has chosen to address this strategic weakness principally by rolling out greater renewable and nuclear electricity generation capacity, thereby reducing reliance on imported fossil fuels. As previously explained, the planned renewables capacity roll-out drives a significant accompanying grid investment need.

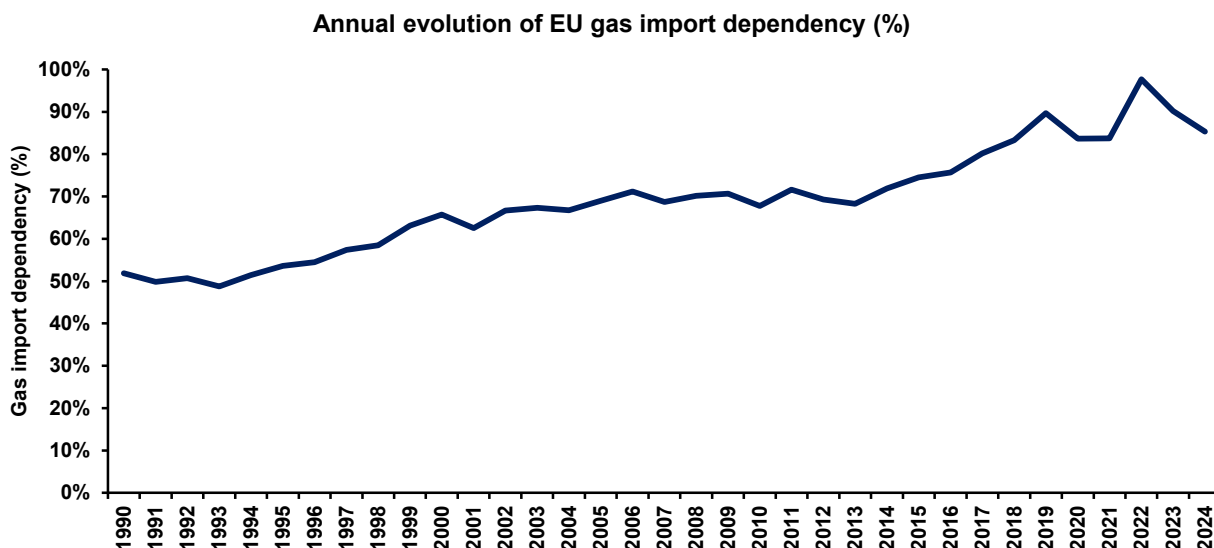
We set out the annual evolution of Europe’s energy import dependency ratio in Exhibit 76.

EXHIBIT 76: **Annual evolution of Europe’s Energy Import Dependency Ratio, 1990-2023**



Source: Eurostat via US Federal Reserve, Bernstein analysis

Europe’s energy import dependence is shown more starkly by its [gas import dependency data](#), as set out in Exhibit 77. In 2004, the EU imported ~67% of its gas, a figure which had risen to ~85% in 2024. Greater renewable penetration, enabled by a wider roll-out of electricity network capacity, is therefore seen as a key strategic priority for Europe.

EXHIBIT 77: **Annual evolution of EU gas import dependency (%)**

Source: Eurostat, Bernstein analysis

B) ENERGY INTERCONNECTION IS A KEY DRIVER OF IMPROVING EUROPEAN ENERGY RESILIENCE

As part of its push towards energy independence, the EU has [set an interconnection target of at least 15% by 2030](#), implying that each country must have electricity infrastructure in place that would allow it to import from neighboring EU countries the equivalent of at least 15% of the electricity production on its own territory. This is a potentially significant investment driver in its own right. As set out in Exhibit 78, the Spanish electricity transmission network operator Redeia anticipates that ~15% of its 2025-2030 National Transmission Plan will be driven by investment in international interconnectors.

EXHIBIT 78: **Redeia links ~15% of the >€13bn 2025-2030 National Transmission Plan to investment in international interconnectors**



Proposed National Transmission Plan 2025-2030
with investments over € 13 Bn

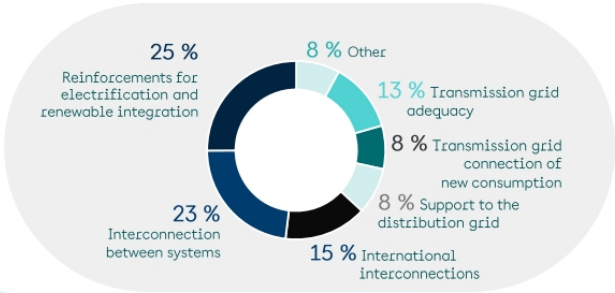
Defining TSO investments for the coming years, with an unprecedented volume



Guiding principles

- **Maximizing** the use of the **existing grid**.
- Deployment of new **renewable installations**.
- Connection of **new demands, electrification of the economy, island systems**, and reinforcement of **international interconnections**.
- Strengthening the **resilience** and **security** of the system.

Investment breakdown



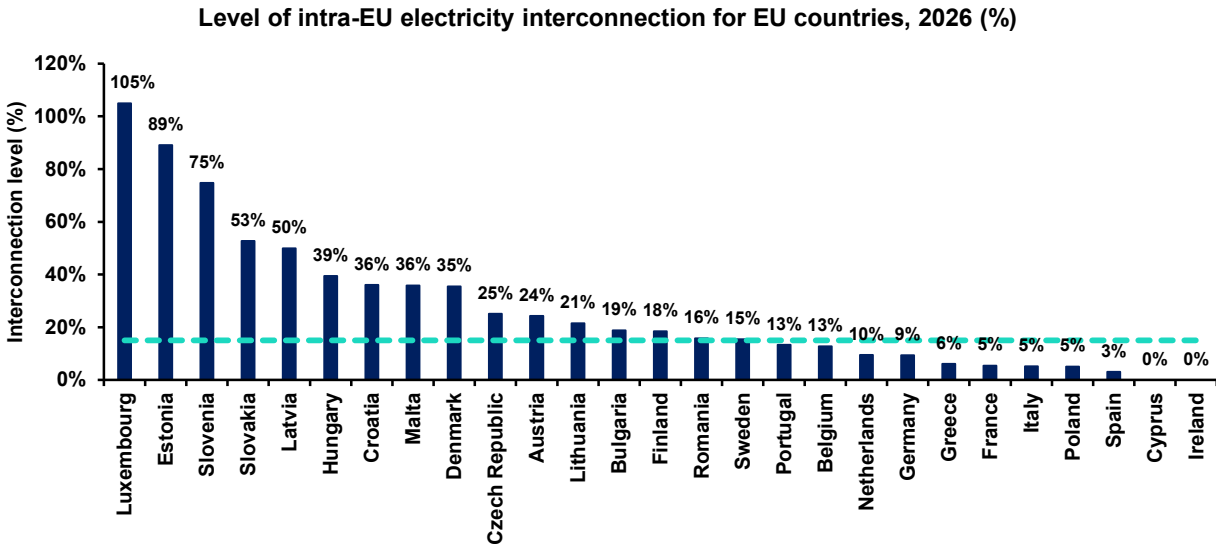
The transmission grid has capacity for new demand in 25% of its nodes

Source: Redeia

Several EU countries are currently lagging behind on interconnection targets

We set out the current levels of intra-EU electricity interconnection across EU countries in Exhibit 79.

EXHIBIT 79: **Levels of intra-EU electricity interconnection across EU countries, 2026 (%)**



Source: European Commission, Bernstein analysis

4) NETWORKS DIGITALISATION AND CYBERSECURITY

As demands rise on European electricity grids, digitalization and cyber-security investments are becoming increasingly important. Digitalisation is a key enabler of optimising the use of existing assets in electricity networks, enabling 1) real-time asset monitoring, 2) greater asset optimisation and 3) enhanced grid planning. In 2025, the IEA [reported](#) that Terna was “investing over 10% of their capital expenditure in grid digitalization”.

Networks digitalization is essential in an age of distributed generation

Digitalisation of electricity grids is likely to become increasingly essential as renewable capacity rises. Renewable capacity is inherently 1) distributed and 2) unpredictable on the very short time frames on which the grid needs to be balanced. As such, grid system operators are likely to become increasingly reliant on supporting digital infrastructure on a granular basis. According to recent reporting by the [Times](#) newspaper in the UK, a recent report for the electricity grid system operator found that growth of renewable capacity had “reduced visibility of the capacity which is supply and taking power from the GB network”, complicating the grid balancing process.

Illustrating the importance of enhanced monitoring of grid conditions, on 28th April 2025 at 12.33pm CEST, the Spanish and Portuguese electricity transmission systems suffered a major blackout, with both systems offline for over 10 hours (~12 hours in Portugal and ~16 hours in Spain). According to an expert panel [report](#), “the blackout occurred after an uncontrolled, rapid rise in system voltage and loss of voltage control on a day with multiple concurrent phenomena, accelerated by rapid generation output reductions and disconnections, leading to voltage instability and cascading generation disconnections in Spain.” Kristina Ruby, Secretary General at Eurelectric, Europe’s electricity industry association, described the 2025 Iberian Blackout as “a wake-up call”, showing that “the need to modernise and reinforce Europe’s electricity grid is urgent and unavoidable.” We would expect ongoing significant investment in grid digitalization in order to manage the successful integration of distributed renewable electricity generation into European grids.

EXHIBIT 80: The 2025 Iberian Blackout saw both the Spanish and Portuguese electricity transmission systems lose power for over 10 hours



Source: Voice of Renewables

Alongside the shorter term consequences, the Iberian Blackout has had the longer term impact of increased curtailment of renewable output in Spain as the transmission system operator has exercised increased caution in regard to intermittent power sources. According to [Aurora Energy Research](#), in the first half of 2026, uncompensated electricity curtailment reached 2.5 TWh in the first half of 2026, 50% higher than the 2024 total and more than 3x the 1H25 level. Separately, Ember [explains](#) that

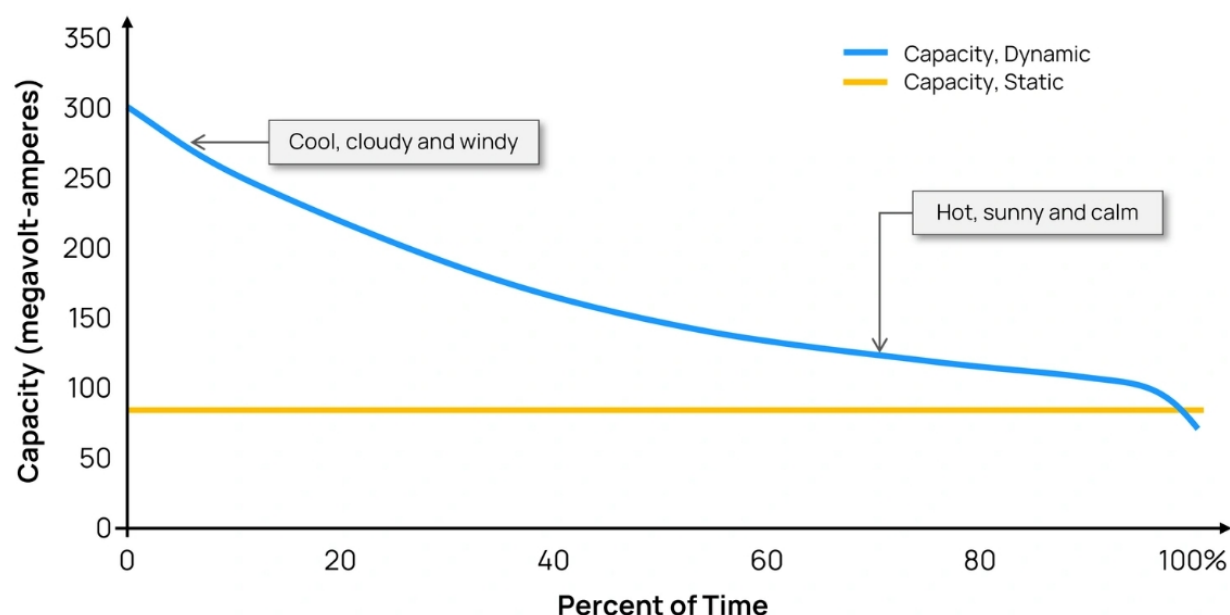
“curtailment of renewables has tripled since the blackout, from 1.8% in the last two years to 7.2% during May-July 2025.”

Digitalisation is also a key enabler of asset optimisation

Digitalisation is a key enabler of improvements in asset optimisation for electricity networks, for example via dynamic line rating, the practice of taking real-time data from overhead lines in order to assess the assets' thermal limits, adjusting load accordingly. The practice opens the possibility of temporarily increasing throughput through the assets when conditions are suitable, and thereby optimizing the network's performance and reducing the need for new physical asset investment. Dynamic line rating is one of several investment drivers underpinning a growing need for greater digitalisation in European electricity networks.

We set out an illustration of Dynamic Line Rating in Exhibit 81.

EXHIBIT 81: Dynamic Line Rating offers the potential to optimise the usage of physical network assets via digitalisation



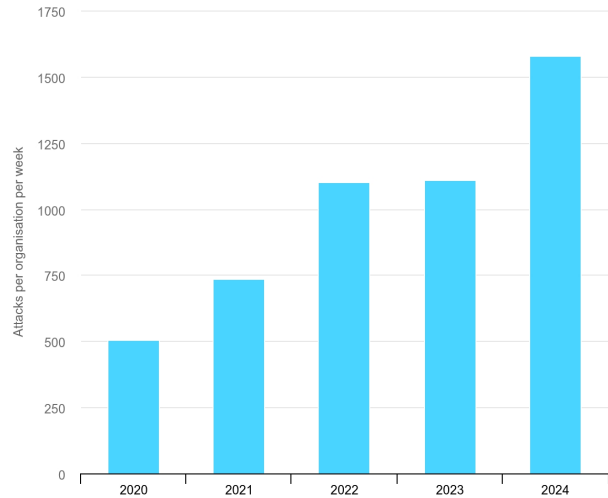
Source: Orennia

Cybersecurity is a non-negotiable

One corollary of the increasing digitalisation of electricity grids is the increasing importance of cybersecurity investments. The European Union Agency for Cybersecurity, ENISA, found in a [November 2024](#) report that the energy sector ranked 2nd in spending on cybersecurity, with only the banking sector spending more. Underlying the investment need, in a 2025 Report, [Energy and AI](#), the IEA cited analysis finding that a typical gas and electricity utility faced over 1,500 cyberattacks per week in 2024.

It is increasingly difficult to obtain granular analysis of individual company cybersecurity spending plans given the operational sensitivities. For its [RIIO-T3 business plan](#), covering the period from 1 April 2026 to 31 March 2031, National Grid set out plans to spend £1.7bn on network resilience, including against cyber attacks, but declined for security reasons to provide a detailed split. Given the trend in attack frequency set out in Exhibit 82, we would expect cybersecurity to be an increasingly material investment area.

EXHIBIT 82: IEA analysis of cyberattacks per week per energy organisation, 2020-2024



IEA (2025), Cyberattacks per week per energy organisation, 2020-2024, IEA, Paris <https://www.iea.org/data-and-statistics/charts/cyberattacks-per-week-per-energy-organisation-2020-2024>, Licence: CC BY 4.0
Source: IEA

EXHIBIT 83: The challenging geopolitical environment in the 2020s has raised the importance of cybersecurity investments

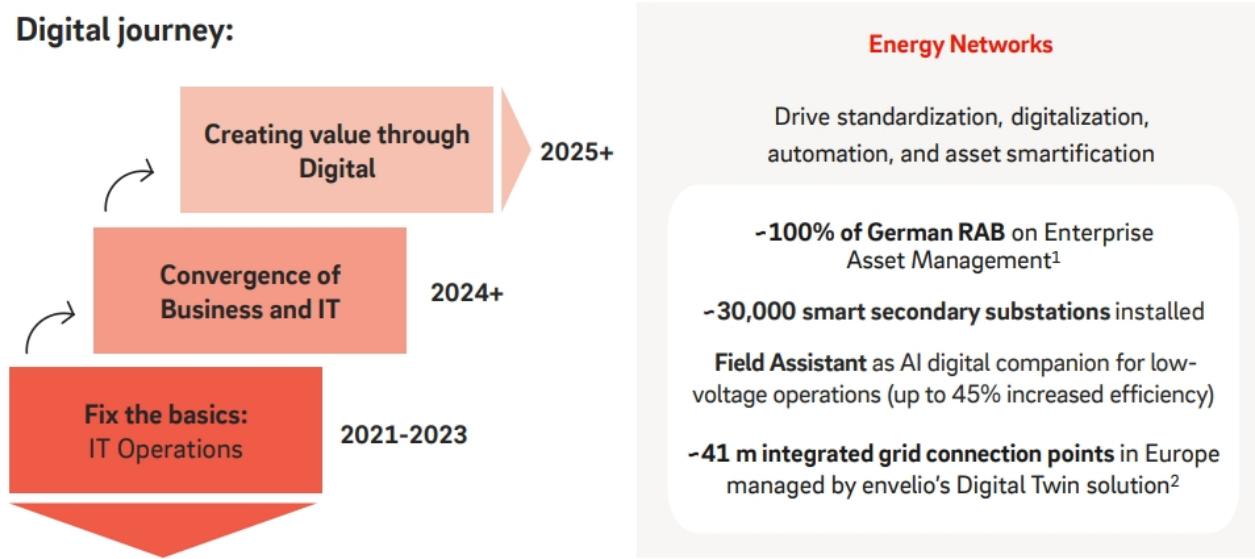


Source: Bloomberg

Digitalisation case study

We would highlight E.ON as a leader in the digitalisation of energy network assets. In September 2025, the company [announced](#) that it had completed a digital twinning exercise for its 700,000km of power grid, implying that 1/3 of the total German electricity distribution grid has now been mapped. The digital twinning capability means that E.ON can gather data from 55 million network components and more than 180,000 measuring devices.

EXHIBIT 84: E.ON Energy Networks digitalisation activity

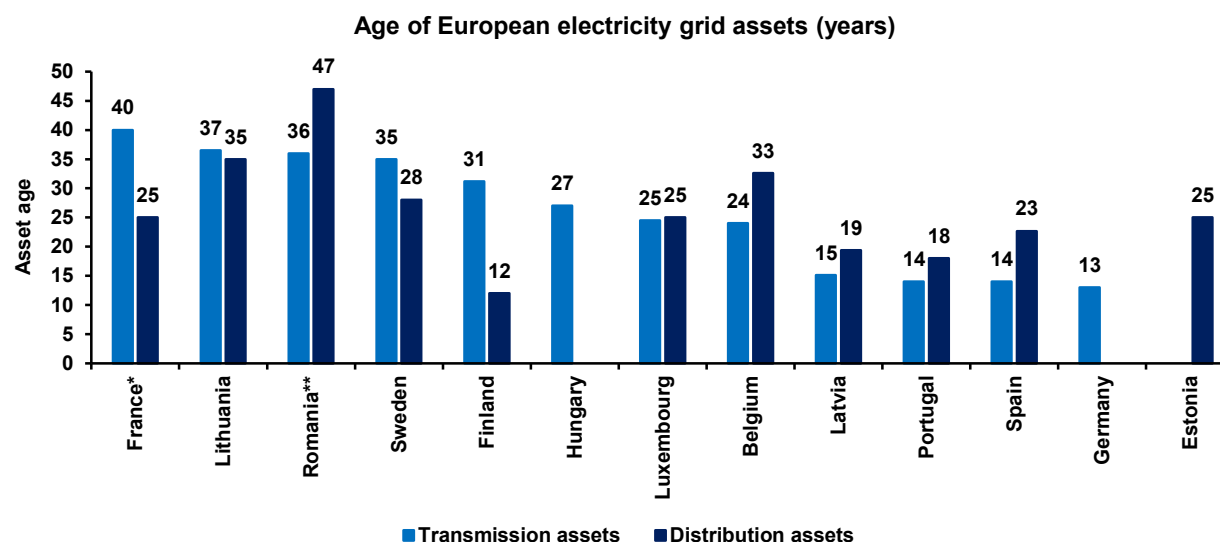


Source: E.ON

5) INFRASTRUCTURE RESILIENCE AND ENVIRONMENTAL STANDARDS

The final megatrend that we would highlight that underpins European electricity grids' investment outlook is the drive for higher levels of infrastructure resilience and compliance with more stringent environmental standards. European electricity grids are some of the oldest in the world (40% of European electricity distribution grids are [over 40 years old](#)), and electricity networks are vital pieces of infrastructure where unanticipated downtime can have severe consequences for economies and society. As such, ongoing maintenance expenditure is non-negotiable. We set out data from the European Court of Auditors on the average age of electricity grid assets in selected European countries in Exhibit 85.

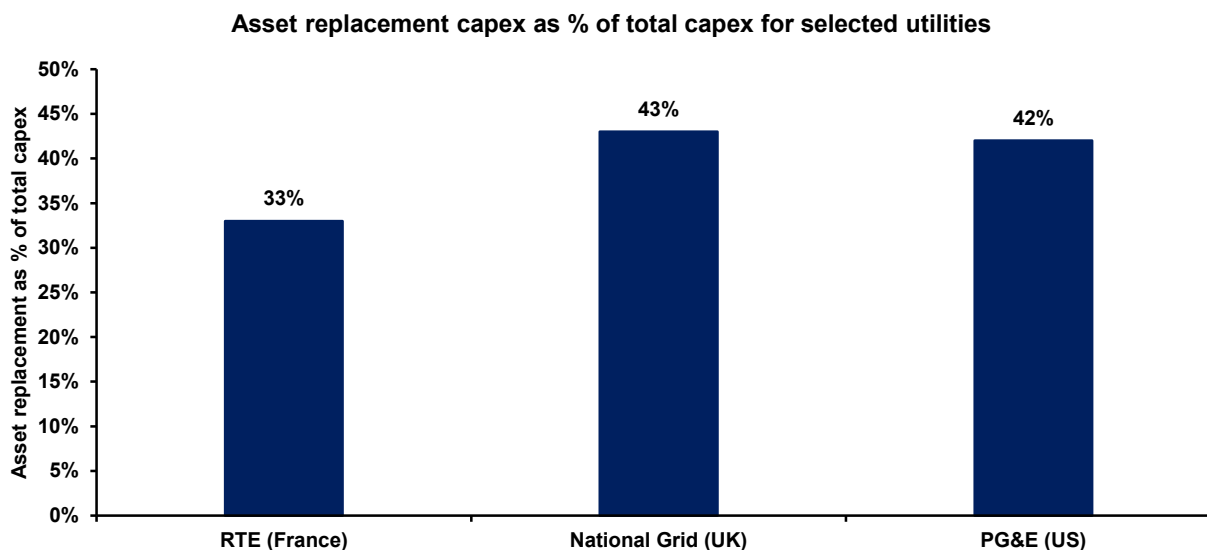
EXHIBIT 85: **Comparative ages of European electricity grid assets**



Source: European Court of Auditors, Bernstein analysis

In its [World Energy Investment 2026](#) publication, the IEA found that “in Europe, utilities are investing in grid resilience to make their networks more resistant to extreme events and other disruptions. Iberdrola allocated \$113m...following the 2024 floods, while Enedis is investing \$1.1bn annually and RTE is dedicating 40% of its yearly budget to renewing the transmission grid to adapt to climate change. Overall, utilities in Europe direct an estimated 25-40% of their annual capital expenditures to grid resilience and modernisation initiatives”.

We set out data for asset replacement capex as a percentage of total capex at selected electricity network operators in Exhibit 86.

EXHIBIT 86: **Asset replacement capex as a percentage of total capex at selected utilities**

Source: BNEF, Bernstein analysis

In the following sections we set out resilience considerations relevant for selected major individual European countries (France, the UK and Italy).

France

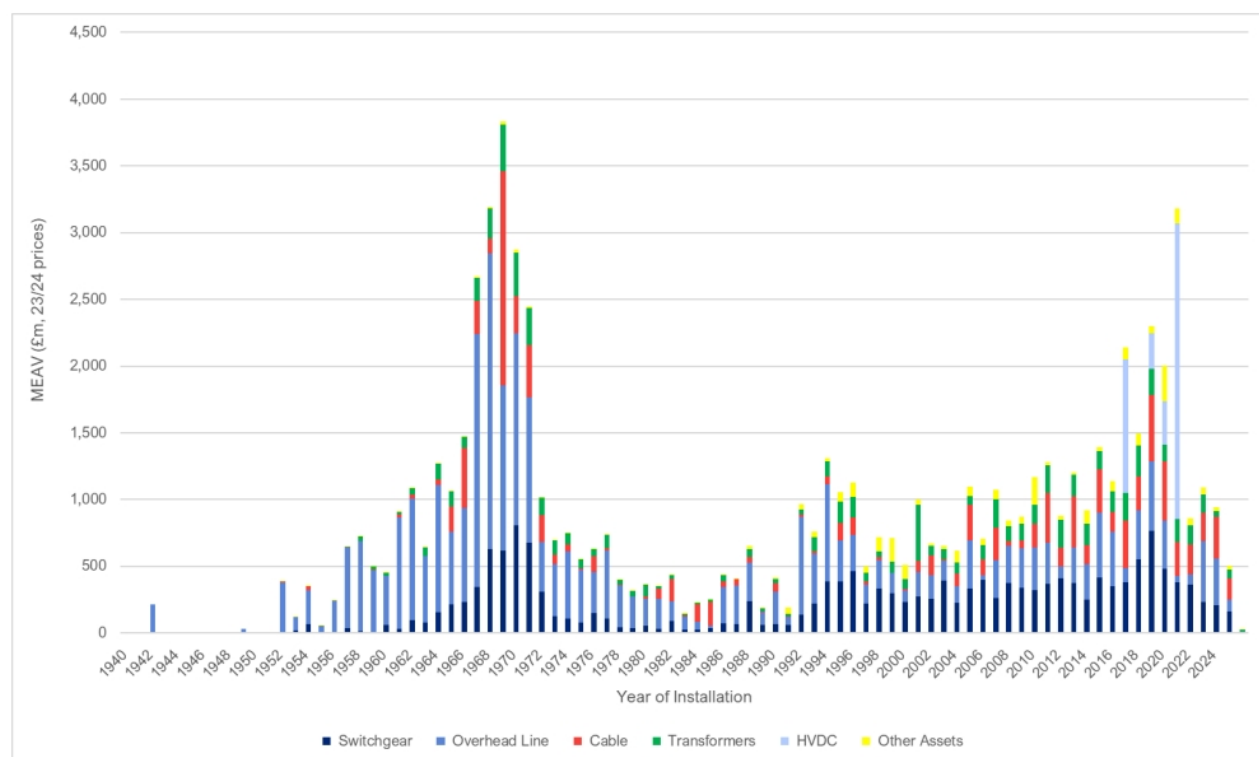
The French transmission network operator, RTE, recently published its [2025 Strategic development plan for the French transmission grid](#) (SDDR: Schema Decennal De Developpement du Reseau). The SDDR explains that “the age of the French grid makes it vulnerable to extreme heat and flooding: it was built in two waves (after the Second World War to rebuild the country, then between 1970 and 1990 to enable the development of nuclear power plants.” 27% of overhead electricity lines in France are now over 60 years old. RTE proposes to allocate €20bn to renewing the existing grid in France and adapting it to climate change.

UK

In the UK, a [June 2024 study](#) by Cambridge Economic Policy Associates found that “the average age of the electricity transmission networks range across the three network companies, from 29 to 37 years” and that “the weighted average for GB as a whole is 36 years.”

As part of the report, Cambridge Economic Policy Associates published the chart shown in Exhibit 87, illustrating the peak of electricity network investment in the UK in the 1950s and 1960s, followed by a multi-decade decline and then a gradual resurgence from the early 1990s onward. The age of the network underlines the need for additional maintenance investment in the UK.

EXHIBIT 87: Modern Equivalent Asset Value and year of installation for existing electricity transmission network assets



Source: Cambridge Economic Policy Associates

Italy

Granular data on the age and condition of the Italian electricity grids is not readily available, however we would again note the importance of the 1960s in the build-out, e.g. by [Enel](#) in Italy. As such, whilst more specific data is elusive, we would expect the ongoing capital maintenance requirements to be similar in Italy to those in France and the UK.

EXHIBIT 88: The Italian distribution grid, under Enel, saw significant growth in the 1960s and early 1970s

Source: Enel

WE THINK THAT THESE TRENDS WILL PROVE MORE DURABLE THAN THE MARKET CURRENTLY REALIZES, WITH THE NEXT >10 YEARS ALREADY SUBSTANTIALLY UNDERPINNED

Whilst the market understands the importance of these drivers to the growth outlook over the short and medium term, what is less well understood in our view is the durability of the investment outlook. In our view, an investment runway of 10-20 years is comfortably underpinned in the most attractive activities and jurisdictions, e.g. electricity transmission in Germany.

COUNTRIES ACROSS EUROPE HAVE A CAPEX RUNWAY LASTING AT LEAST A DECADE....

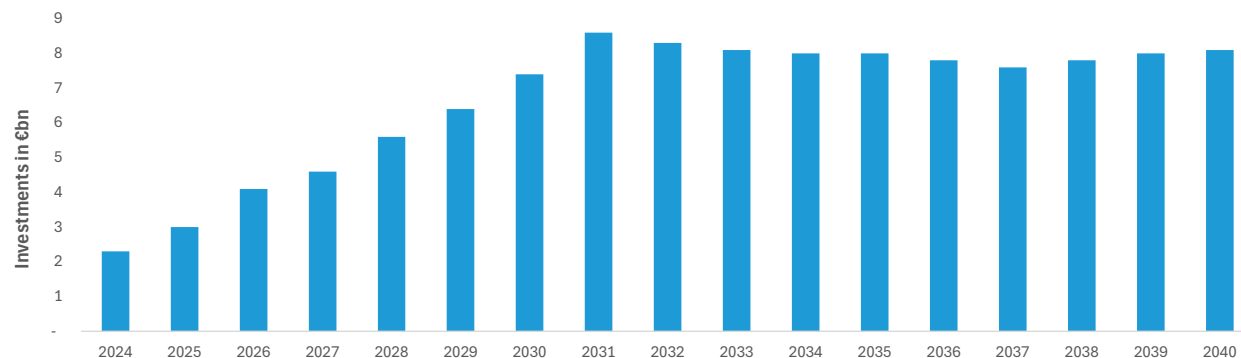
Based on our analysis, we see an investment runway across major European regulated utility subsectors that is underpinned for at least 1-2 decades in the most favorable cases. Visibility is typically higher in electricity transmission, where asset ownership tends to be concentrated in fewer entities than in electricity distribution, where ownership is more diffuse. Nonetheless, the 2 sub-sectors share common drivers and there is strong read-across from high visibility in a country's transmission network to its distribution network.

FRANCE

There is very high visibility of future investment runways in France across both electricity transmission and electricity distribution, driven by advanced grid planning processes.

Electricity transmission

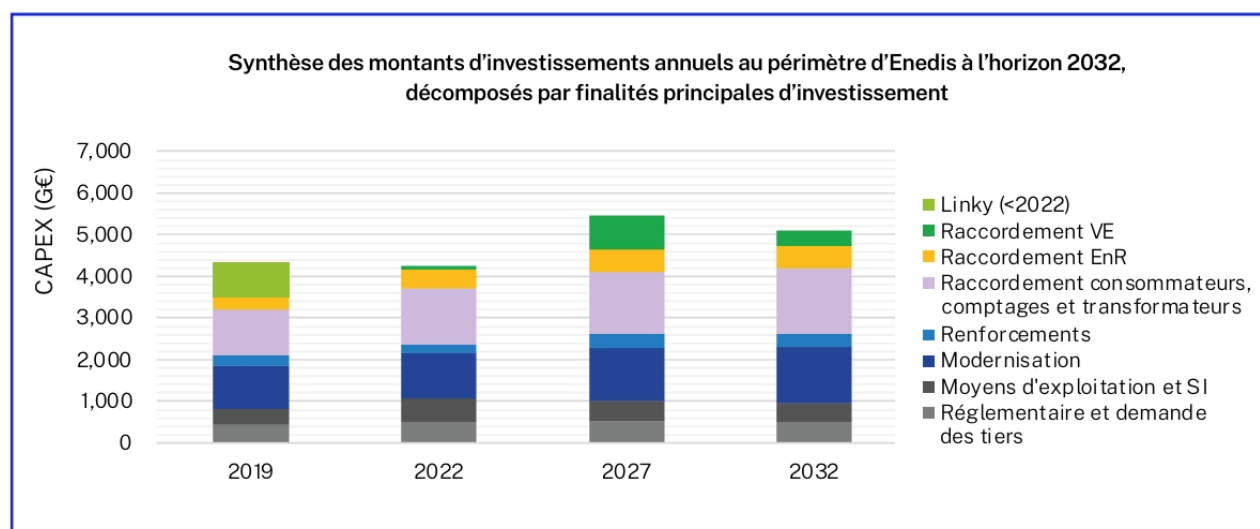
As noted in Exhibit 63 above and repeated in Exhibit 89 below, France has visibility on an investment cycle in electricity transmission lasting until at least 2040, as shown in RTE grid planning documents.

EXHIBIT 89: **Investments in transmission grid**

Source: RTE

Electricity distribution

We estimate that visibility of the investment runway in France extends to at least 2032, as shown in Enedis's grid planning documents.

EXHIBIT 90: **Investment drivers within Enedis' investment outlook, 2019-2032**

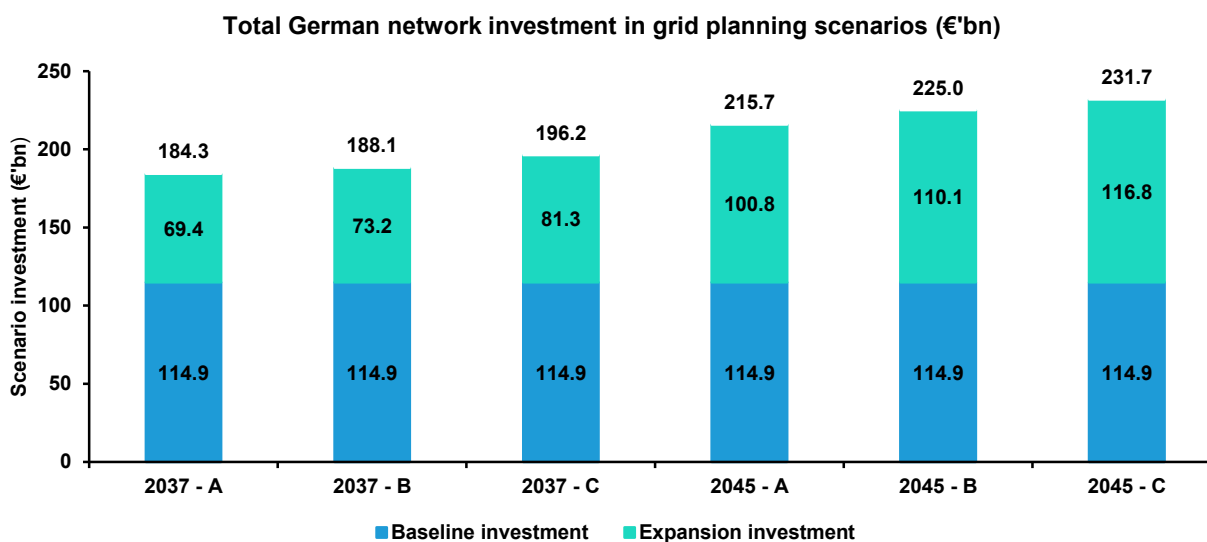
Source: Enedis

GERMANY

Germany has very high formal investment visibility in electricity transmission and well understood long-duration drivers of investment in electricity distribution,

Electricity transmission

As shown in Exhibit 91, investment visibility in electricity transmission in Germany is very high, with [current grid planning documentation](#) extending to 2045.

EXHIBIT 91: **Total German onshore electricity transmission network investment in grid planning scenarios**

Source: Netzentwicklungsplan Strom 2037 mit Ausblick 2045 (Version 2025). Bernstein analysis

Electricity distribution

Germany's electricity distribution network ownership is highly fragmented, with [more than 850](#) DSOs. A late 2025 study conducted by PWC for KfW, the state-owned development bank, found that electricity distribution network owners, alongside gas distribution network owners and grid-connected heating networks, would need to invest [~€535bn](#) in their assets by 2045, with ~2/3 of this total to be invested by 2035. With a post-2035 runway of >€150bn, we see strong visibility in German electricity distribution.

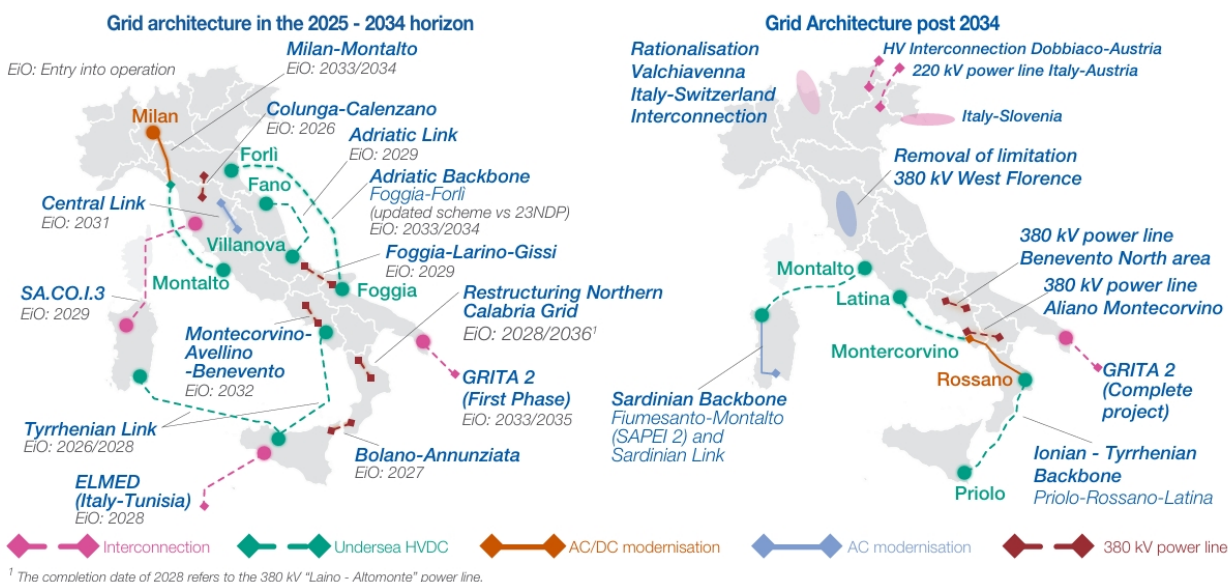
ITALY

Visibility of the investment need in electricity networks in Italy is high, with formal grid planning processes stretching to 2034 and additional visibility beyond that.

Electricity transmission

Investment visibility in electricity transmission in Italy is high. The 2025 [National Grid Development Plan](#) proposes > €23bn of investment from 2025-2034, with an additional up to ~€40bn of investment beyond 2034.

EXHIBIT 92: Terna's investment plan 2025-2034 and post 2034

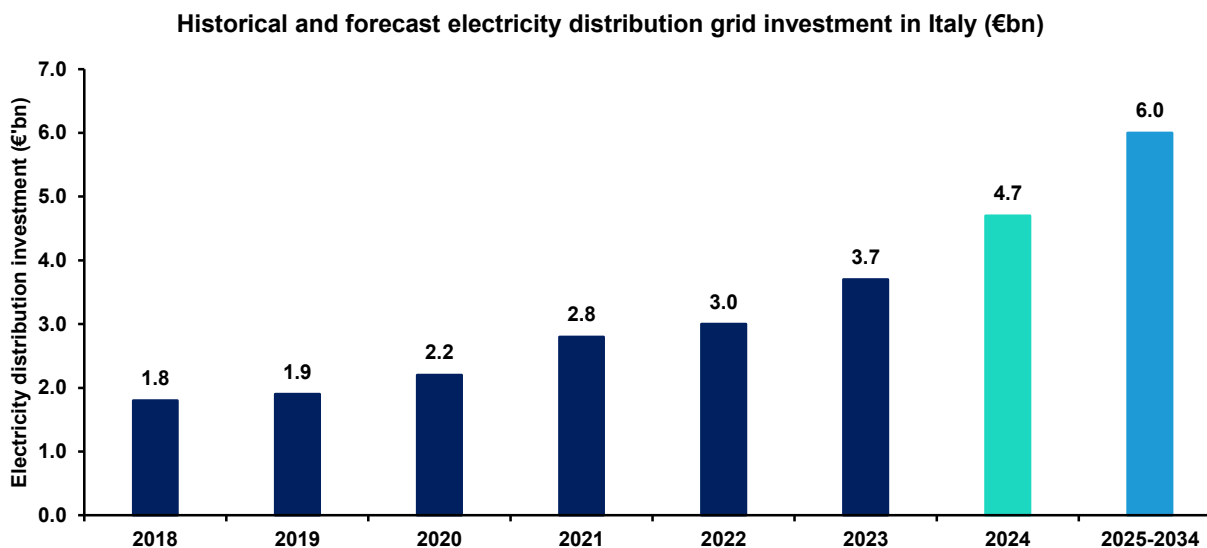


Source: Terna

Electricity distribution

Enel, via its subsidiary E-Distribuzione, dominates electricity distribution in Italy, with an >80% market share. Other players include A2A, Acea and Hera. In a September 2024 [report](#), Enel estimated that annual investments of ~€6bn per year would be required in electricity distribution between 2025 and 2034. Beyond 2034 the level of specific visibility appears reduced, but we would note commonality between Italy and other systems with a long growth runway.

EXHIBIT 93: Annual evolution of electricity distribution network investments in Italy, 2018-2034



Source: Enel, Bernstein analysis

SPAIN

Spain offers the lowest formal electricity network investment visibility of major European countries, with existing planning documentation (alongside the [suspension of the investment cap](#)) only extending to 2030.

Electricity transmission

The current governing document for electricity transmission in Spain is the [2025-2030 Network Development Plan](#), which as yet has not been formalised. New plans are presented every 4 years to cover a 6-year period. Formal visibility beyond the 2030 horizon is currently limited, although we note that the underlying investment need in Spain is no less acute than in other countries in Europe.

Electricity distribution

As with electricity transmission, we see limited investment formal visibility in electricity distribution beyond 2030, although as with transmission a significant investment need remains.

UK

Visibility on the electricity grid investment need in the UK is moderate, with a high degree of institutional sophistication across the National Energy System Operator; the energy regulator, Ofgem; and the Department for Energy Security and Net Zero, but a comparatively less ambitious grid planning horizon than in more ambitious jurisdictions such as Germany.

Electricity transmission

Visibility of the investment need beyond the current regulatory period in electricity transmission, RIIO-T3, which runs to 2031, is primarily provided by NESO documentation. In June 2026, the NESO [published](#) its "Beyond 2030 - Electricity Transmission Update", setting out recommendations for 43 network projects to be developed for delivery in the 2030s, with total investment reaching ~£64bn."

Electricity distribution

The UK is currently part-way through the so-called RIIO-ED2 price control, which runs from 1 April 2023 to 31 March 2028. The price control process for RIIO-ED3, which runs from 1 April 2029 to 31 March 2033, remains at a comparatively early stage, with business plans not expected to be submitted until Q4 2026. We would note, however, the diversity and quality of potential investment drivers across EVs, heat pumps and distributed generation, all of which are supportive for the future investment runway.

...WHILE GOVERNMENTS AND OTHER PARTIES ARE PUSHING HARD TO UNLOCK INVESTMENT

Historically there have been significant barriers against the realization of the sector's growth potential in the form of 1) planning and permitting inefficiencies and delays and 2) supply chain restrictions. In this section of the note we explore the progress that has been made on these fronts in recent years, noting significant improvements in planning in some countries, whilst supply chain availability remains a generally-applicable hindrance.

PLANNING AND PERMITTING

The most commonly-cited barrier to European utility grid growth is permitting. For example, the European Union has [estimated](#) that a typical electricity transmission grid project can take ~10 years for completion, of which the permitting process accounts for more than half.

In recognition of the restrictive impact of slow permitting procedures on electricity grid expansion, on 10th December 2025 the European Commission published its [European Grids Package](#). Amongst other proposals (see Exhibit 94), the package contemplates 1) better co-ordination at the EU level to map and plan required grid infrastructure, and 2) a streamlining of permitting processes for electricity grids alongside renewable generation, storage and recharging station projects.

EXHIBIT 94: **The European Commission is seeking a pan-European approach to energy grid planning****A common European project for the common European good**

Greater direction and coordination at EU level to map and plan the grids infrastructure.



Enhance security and resilience of physical energy infrastructure.



Speed up and streamline permitting processes for renewable projects, while ensuring public acceptance and benefit-sharing.



Make existing infrastructure more efficient, reinforced by new technology, flexibility, and storage capacity.

Source: European Commission

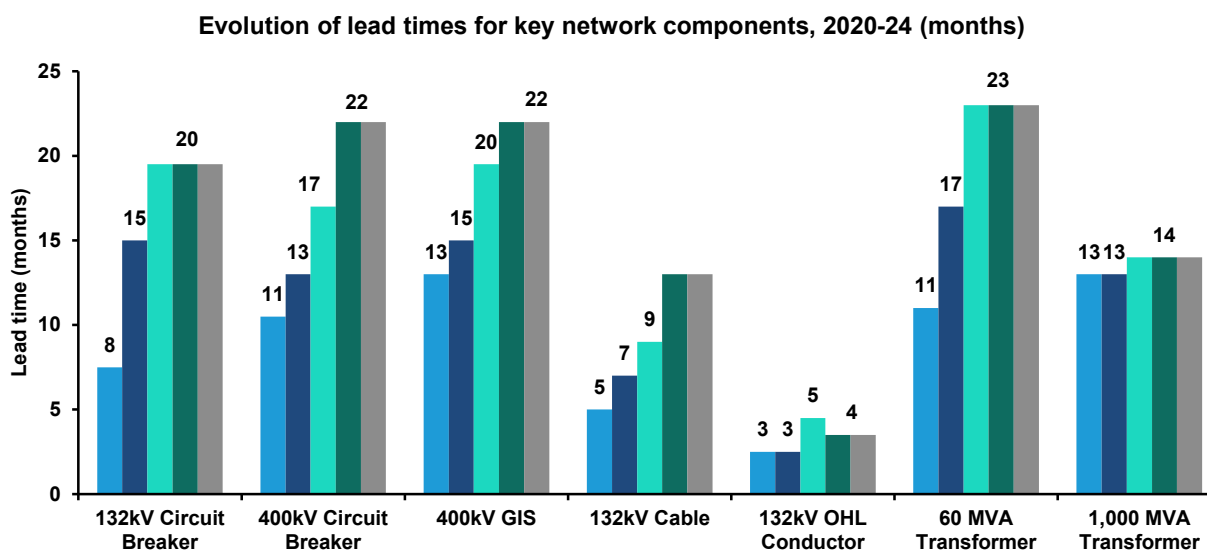
Whilst the effectiveness of the European Grids Package and other similar measures remains to be seen, we note the [impact of permitting reforms in Germany made following the 2022 energy crisis](#). These reforms led to more than a tripling of permits for new onshore wind projects and a saving of 12 months to 3 years in the permitting time for grid projects.

SUPPLY CHAIN

The second key potential barrier to realisation of Europe's infrastructure growth ambitions is **the level of supply chain capacity**. The strain on the supply chain is particularly noteworthy given the global context, given that the investment need is not confined to Europe; a recent IEA report [noted](#) for example that countries all around the world are ramping up investments in grid infrastructure on similar time frames. In their RIIIO-T3 [business plan](#), which addresses electricity transmission in the south of Scotland from 1st April 2026 to 31st March 2031, Iberdrola's ScottishPower wrote that "we have engaged with a wide range of suppliers for equipment and services and closely analysed the available data to build a clear understanding of the supply chain opportunities and constraints in the global supply chain for transmission equipment and services we need to access. The lead time for critical network components...is continuing to increase due to imbalanced global supply and demand. Since 2020, we are seeing lead times increasing by between 50% to 300%. For some equipment this is a delay of more than a year. Insights provided by Original Equipment Manufacturers suggest that these times will be moving out further in the next 3-4 years due to the quantity of orders and factory capacity being reserved by transmission companies in other jurisdictions and countries who have similar expansion plans to facilitate a transition to renewable energy and away from fossil fuels."

As shown in Exhibit 95, ScottishPower disclosed that lead times for key components increased by 8-160% between 2020 and 2024.

EXHIBIT 95: Evolution of lead times for key network components, 2020-24

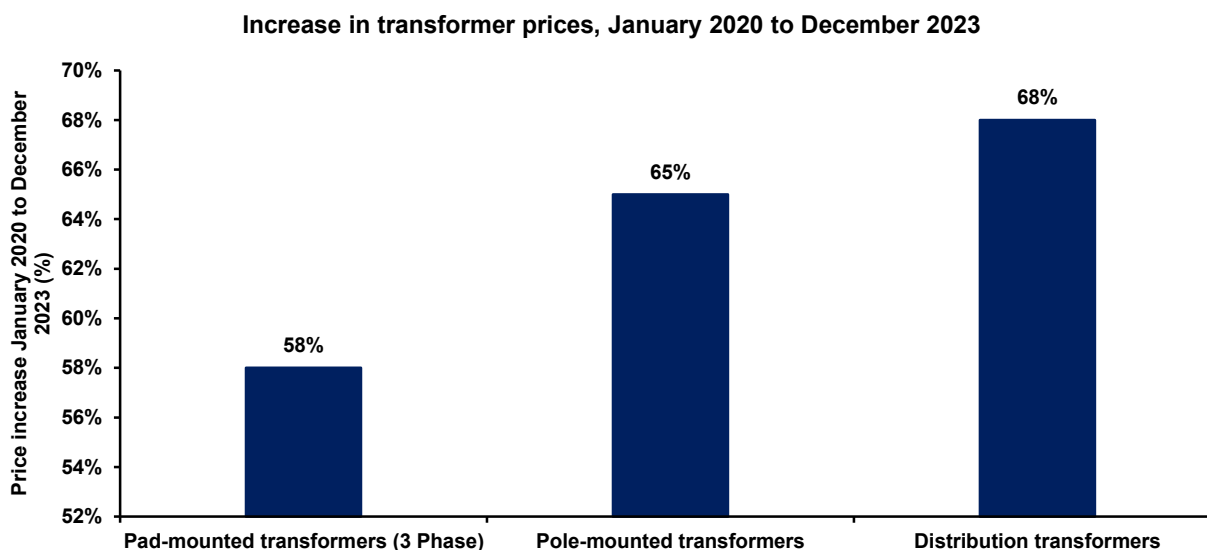


Source: ScottishPower

Similarly, a 2024 Eurelectric report [found](#) that “transformer prices, dependent on size and applications, have risen 60%-70% on average since January 2020...due to supply chain issues, manufacturing shortfalls and increased commodity prices.”

We set out the Eurelectric data on transformer price increases in Exhibit 96.

EXHIBIT 96: Increases to transformer prices, January 2020 to December 2023



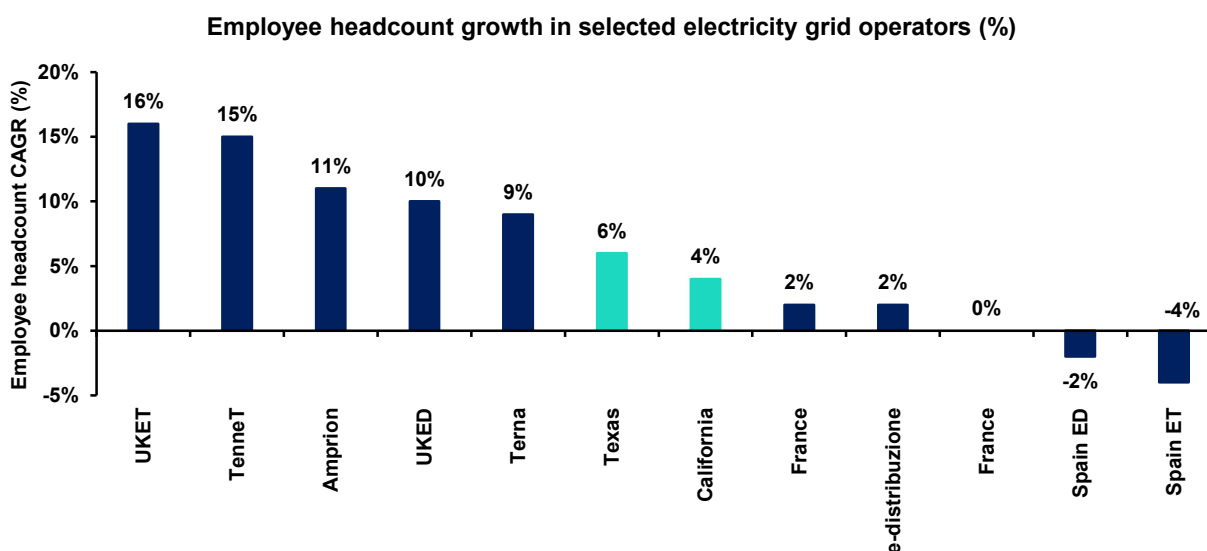
Source: Eurelectric, Bernstein analysis

There is a wide variety of other evidence of cost increases and supply chain delays in electricity grids. The IEA in its 2026 [World Energy Investment](#) publication noted rising grid equipment prices outpacing inflation, whilst a May 2025 [briefing to the European Parliament](#) observed that even the most urgent grid projects were suffering delays in sourcing key components. The supply of transformers [appears to be particularly challenging](#), with lead times for large power transformers seeing lead time increases of close to 100% since 2019. A [2025 IEA survey](#) of leading players in the electricity network supply chain found that “it now takes 2 to 3 years to procure cables and up to four years to secure large power transformers” and that “some specialized components are even more difficult to source: waiting times for direct current cables - often preferred for long-

distance transmission lines - extend beyond five years.” Finally, a report [published](#) in September 2025 has found that the current German Grid [Development Plan 2037/2045 \(2023\)](#) (a.k.a. NDP 2023 - 2037/2045), which was developed between January 2022 and September 2023, would cost **c.€440bn if realized today, vs c.€323bn previously estimated.**

Alongside components, **labour is another key bottleneck.** [Around 8 million people](#) currently work in roles associated with the construction, maintenance and operation of electricity grids, a figure that the IEA estimates will need to rise by ~20% by 2030 even assuming no further acceleration in policy ambition. We set out in Exhibit 97 the growth in employee headcount at selected electricity grid operators between 2019 and 2024, showing that UK electricity transmission companies have been most successful in recruiting, whilst Spanish network companies have lagged.

EXHIBIT 97: **Employee headcount growth in selected electricity grid operators, 2019-24**



Source: BNEF, Bernstein analysis

IT WILL BE IMPORTANT FOR INVESTORS TO MONITOR THE CAPEX OUTLOOK ON AN ONGOING BASIS

Notwithstanding our conviction that the current investment supercycle in electricity grids will persist for longer than the market currently appreciates, we nevertheless concede that once it has run its course, historical precedent points to a significant and perhaps prolonged downturn in growth prospects. In our view the key uncertainty regarding the timing of this slowdown relates to the future path of power demand; in a scenario where power demand continues to grow in the late 2030s and beyond, driven by ongoing uptake and roll-out of technologies such as electric vehicles, heat pumps and data centers, we see potential for a second leg of growth. As such, it will be important for investors to monitor the trajectory of power demand on an ongoing basis.

ALL GOOD THINGS COME TO AN END, INCLUDING CAPEX SUPERCYCLES...

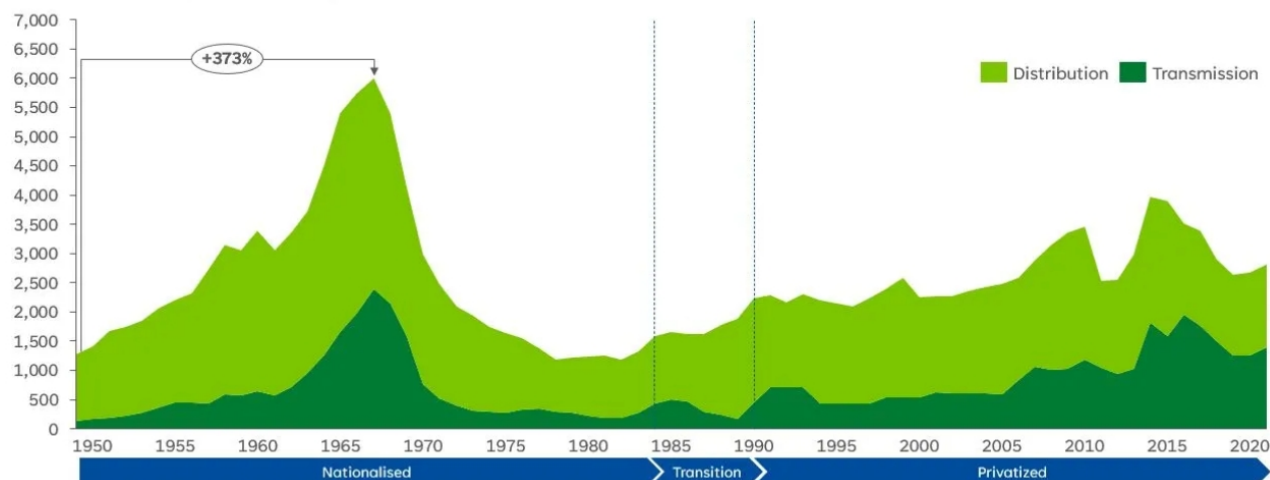
In our view, **the current investment supercycle can be characterized as primarily (although not exclusively) decarbonisation-driven**, with the roll-out of new, decentralized renewable generation capacity to replace outgoing centralised fossil-powered capacity being the most important of the range of investment drivers previously discussed. Despite our optimism, we nonetheless acknowledge that this process has a finite horizon, and that the historical cases studies we review below point to the potential for a sharp slowdown after the peak.

UK case study

Between the 1950s and the 1970s the UK built out what was then known as the “[supergrid](#)”, a 275kV network (subsequently accompanied by 400kV lines) to replace the original 132kV network. The first 400kV line in the UK was [inaugurated in 1965](#) after the 1961 formation of a specific project taskforce. As set out in Exhibit 98, the supercycle lasted for ~10 years, with capex

reverting to close to pre-cycle levels by ~1970, at which point ~1,300 miles of 400kV line were [reportedly in operation](#).

EXHIBIT 98: Annual evolution of UK electricity transmission and distribution capital investment

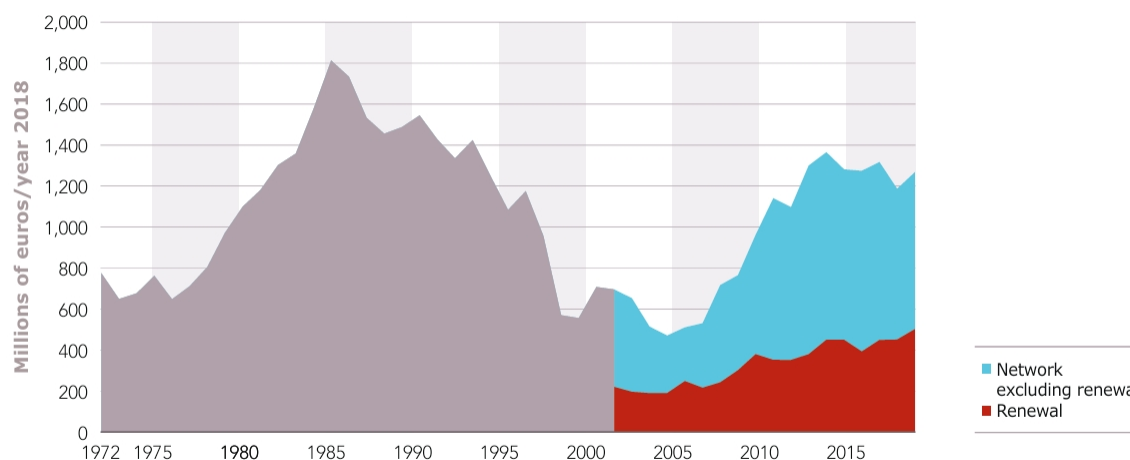


Source: Arthur Downing via Substack

France case study

Similarly, following the 1973 oil crisis, France launched the [Messmer Plan](#) in 1974, aiming to build 13 900MW nuclear plants over 2 years in order to insulate the country's electricity supply against fossil fuel price volatility. France subsequently built [56 nuclear reactors](#) between 1974 and 1989. Alongside the reactor build out, France also saw a very significant increase in electricity network investment, [peaking in the mid-1980s](#) - see Exhibit 99.

EXHIBIT 99: Annual evolution of electricity transmission network investment in France



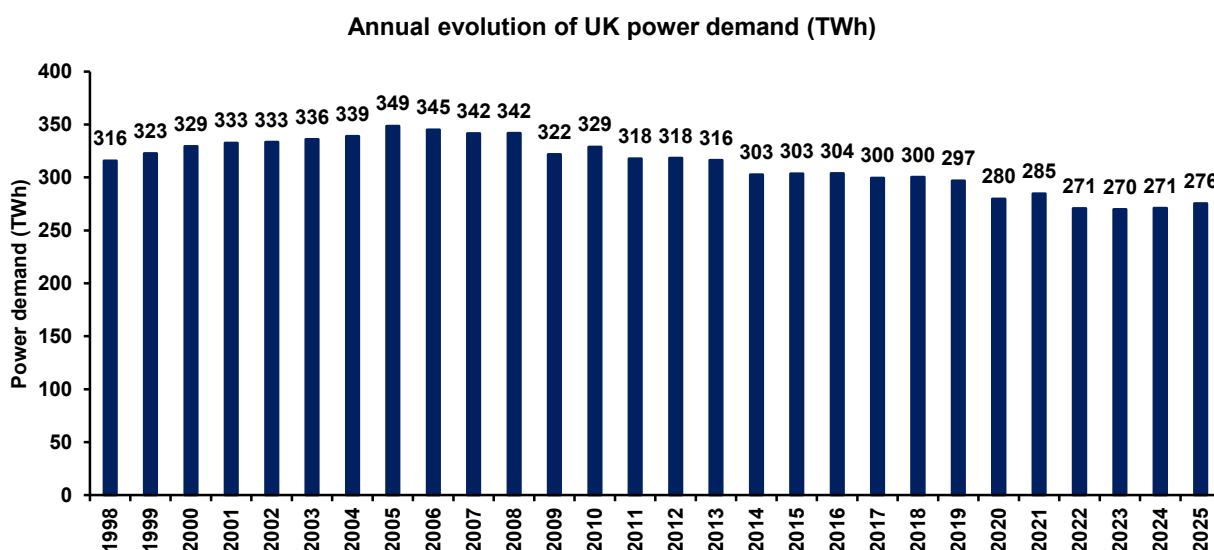
Source: RTE, Bernstein analysis

....BUT THIS TIME THERE IS THE POTENTIAL FOR GROWING POWER DEMAND TO TAKE ON THE BATON FROM ELECTRIFICATION AND DECARBONISATION

In the event that key drivers such as EV and heat pump uptake, data centers and even green hydrogen play out as anticipated, power demand will grow significantly during the 2030s, **adding another leg to the grids' investment runway** as new generation capacity is added to serve the additional demand, requiring accompanying network investment. Whilst (as set out in

Exhibit 100) this would represent a reversal of the current sluggish trends in power demand growth, it would nonetheless (as previously noted) be entirely in line with the direction of policy in Europe.

EXHIBIT 100: A further leg of networks growth is conditional on reversing recent sluggish power demand trends in markets such as the UK



Source: Digest of UK Energy Statistics (2026), Bernstein analysis

DURABILITY OF GROWTH IS HIGHLY SUPPORTIVE FOR VALUATIONS GIVEN THE IMPORTANCE OF RAB GROWTH IN DRIVING THE TERMINAL RAB PREMIUM

The single most important parameter in our regulated utility valuation models is the terminal premium to RAB assumption, which in turn primarily reflects 1) the level of outperformance expected across both financial and operational metrics, and 2) the growth outlook. Our findings in this note are highly supportive for the second element.

We would highlight Germany and the UK in particular as markets with strong growth outlooks, the former driven by a pressing need for north-south interconnections in electricity transmission, as well as a more generalised renewables catch-up push. In the UK, successive governments have set ambitious decarbonisation targets

HIGH VISIBILITY ON SECTOR GROWTH IS SUPPORTIVE FOR VALUATIONS

The level of visibility in a regulated utility's growth outlook is a [key driver of valuation](#). Where there is a long runway of asset growth ahead, investors tend to be willing to pay higher premia to the current RAB. Uncontroversially, we see the most significant regulated network growth opportunities over the next decade in electricity networks, and hence, all else equal, would expect valuations to be richer in these sectors. As previously discussed elsewhere in the primer series, however (see [Regulated Networks Primer series: Part 1: How do regulated return frameworks compare across jurisdictions?](#) and [Regulated Networks Primer series: Part 2: A deep-dive into valuation](#)), the attractiveness of growth is inextricably linked to the returns associated with it.

As we set out in Exhibit 101, for a given level of returns spread, the RAB growth outlook is a very significant driver of valuation.

EXHIBIT 101: Theoretical RAB premium under a range of growth and return (relative to cost of capital) scenarios

		Return/WACC-1 (WACC = 5%)						
		0%	5%	10%	15%	20%	25%	30%
10 year RAB CAGR*	0%	0%	5%	10%	15%	20%	25%	30%
	5%	0%	9%	17%	24%	32%	39%	47%
	10%	0%	15%	26%	38%	50%	61%	73%
	15%	0%	24%	41%	59%	77%	94%	112%

*Assumes 0% RAB growth after 10 years

Source: Bernstein analysis and estimates

THE MISSION-CRITICAL NATURE OF THE SPEND IS POSITIVE FOR RETURNS

Regulators of utility networks are faced with 2 pressing challenges during the regulatory price setting process: 1) the unavoidable asymmetry of information between the regulator and the regulated entity; and 2) the strategic importance of the regulated infrastructure. In our view, there is evidence to suggest that in circumstances where there is a need to fund strong capacity growth, regulators may be inclined to err towards over-remuneration given the adverse consequences of failing to secure the necessary investment. In a June 2025 paper, [PR6 Cost of Capital Estimation](#), for example, Cambridge Economic Policy Associates explain that “regulators may deal with uncertainty in the cost of capital by weighing up the relative risks of over/under-estimating the [allowed] WACC. There are risks with setting a WACC that is both too high and too low, but many regulators (and appeal bodies) have tended to emphasise the asymmetry of the risks, in that setting a WACC that is too low is likely to be more harmful for consumers than one that is too high.” This implies that a strong growth outlook is supportive of returns as well as growth.

INVESTORS CAN HAVE VERY HIGH CONFIDENCE THAT THE GROWTH WILL ACCRUE TO INCUMBENTS

One of the principal potential risks for investors contemplating a long investment runway for a regulated utility is the risk of regulators pursuing competitive delivery models rather than awarding the project (and the accompanying RAB and earnings growth) to the incumbent. To date and in Europe, this risk appears restricted to the UK. Given the strategically important nature of much of the planned investment, as well as the significantly greater operational complexity of competitive delivery models, we see limited risk in this area across the continent.

THE UK IS AN OUTLIER IN CONTEMPLATING COMPETITIVE DELIVERY

The UK is an outlier in Europe in exploring competitive delivery models. In December 2025, the UK’s energy regulator, Ofgem, [published](#) a stakeholder update in which it stated that “introducing competition into the delivery of onshore electricity transmission projects has been a long-term ambition of both Ofgem and government, which we believe has the potential to foster innovation, drive performance in the sector, attract investment into Britain’s energy networks and help deliver Net Zero at the lowest possible cost to consumers. We are continuing to progress work to finalise the new framework for onshore early competition in electricity transmission ahead of identifying projects that will be competitively tendered...the legislative framework to allow for onshore competition is now in place.”

The vehicle that the UK is exploring to enable competitive delivery of electricity transmission infrastructure is the [Competitively Appointed Transmission Operator](#) regime. Ofgem first consulted on proposals for competitively appointed transmission operators in 2021, and the 2023 Energy Act established the necessary legal framework.

COUNTRY-BY-COUNTRY NUANCES MEAN THAT SPECIFIC NATIONAL ANALYSIS IS STILL IMPORTANT

As highlighted in the [first edition](#) of our primary series, it is vitally important for investors to take a holistic view of the regulatory frameworks in which companies operate. We republish our overall framework ranking in Exhibit 102 below. We would urge investors to combine our commentary on the growth outlooks for individual countries with our assessment of the overall

attractiveness of the returns offered.

EXHIBIT 102: **Bernstein European utilities regulatory framework ranking**

Jurisdiction	Headline Returns on new investments	Inflation Protection	Returns Protection	Treatment of WIP	Operational incentive opportunities	Overall attractiveness	Rank
Italy	5.0	5.0	4.0	3.0	4.5	4.3	1
UK	3.5	4.0	4.5	5.0	4.0	4.2	2
Belgium	2.0	3.0	5.0	4.0	5.0	3.8	3
Portugal	3.5	3.0	4.5	3.0	3.0	3.4	4
Germany	2.0	3.0	3.0	4.0	2.0	2.8	5
Sweden	4.0	3.0	1.0	1.0	2.0	2.2	6
Spain	3.0	1.0	1.0	2.0	3.5	2.1	7
France	2.0	3.0	1.0	2.0	2.0	2.0	8
Average	3.1	3.1	3.0	3.0	3.3	3.1	

Source: Regulatory data, Bernstein analysis

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